UNIVERSITÄT BONN

Chapter 4:
Color and Perception

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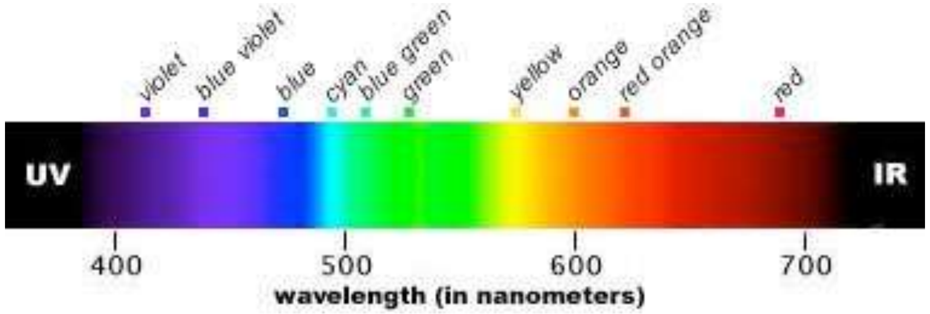
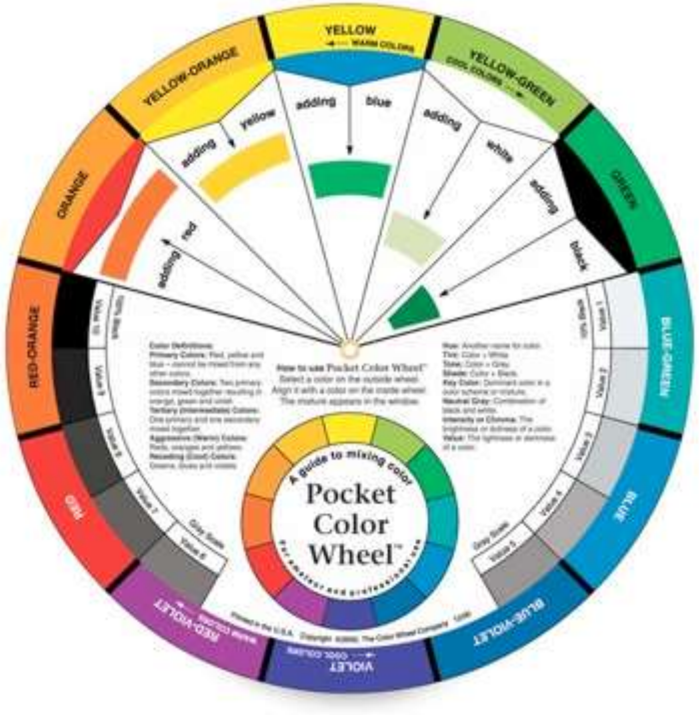
b-it Bonn-Aachen International Center for Information Technology

Section 4.1:
Human Visual Perception

What is Color?





- **Physicist's view:** *Linear* range of wavelengths of visible EM radiation

- **Artist's view:**
  Circle of hues, with mixing of white or black

Electromagnetic Spectrum

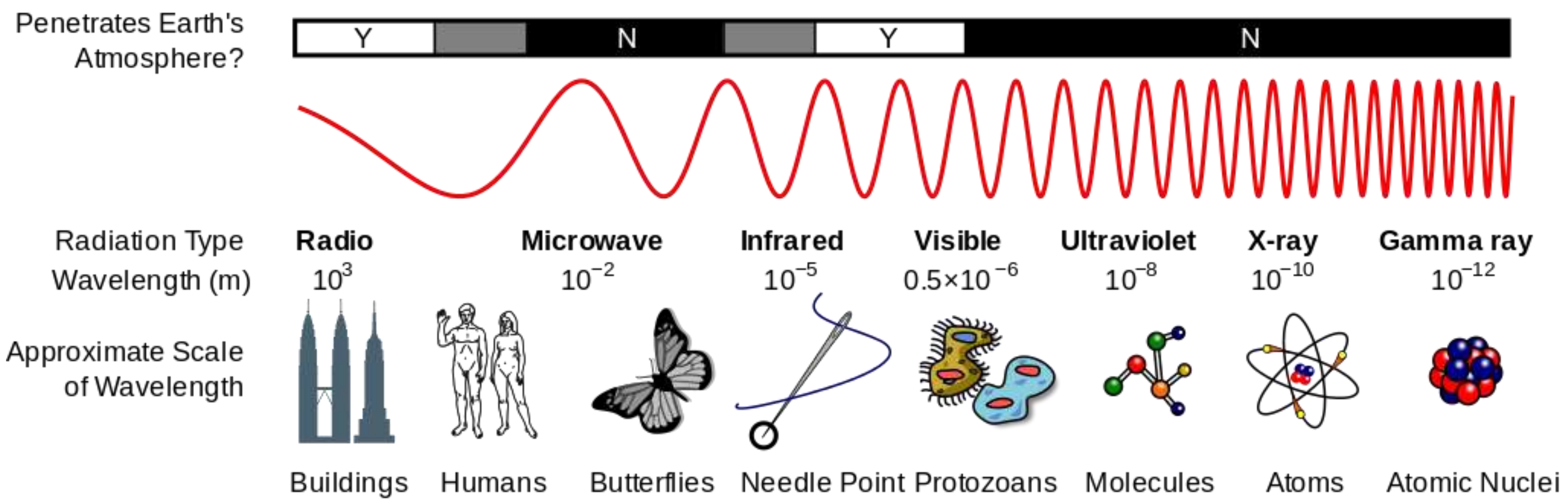
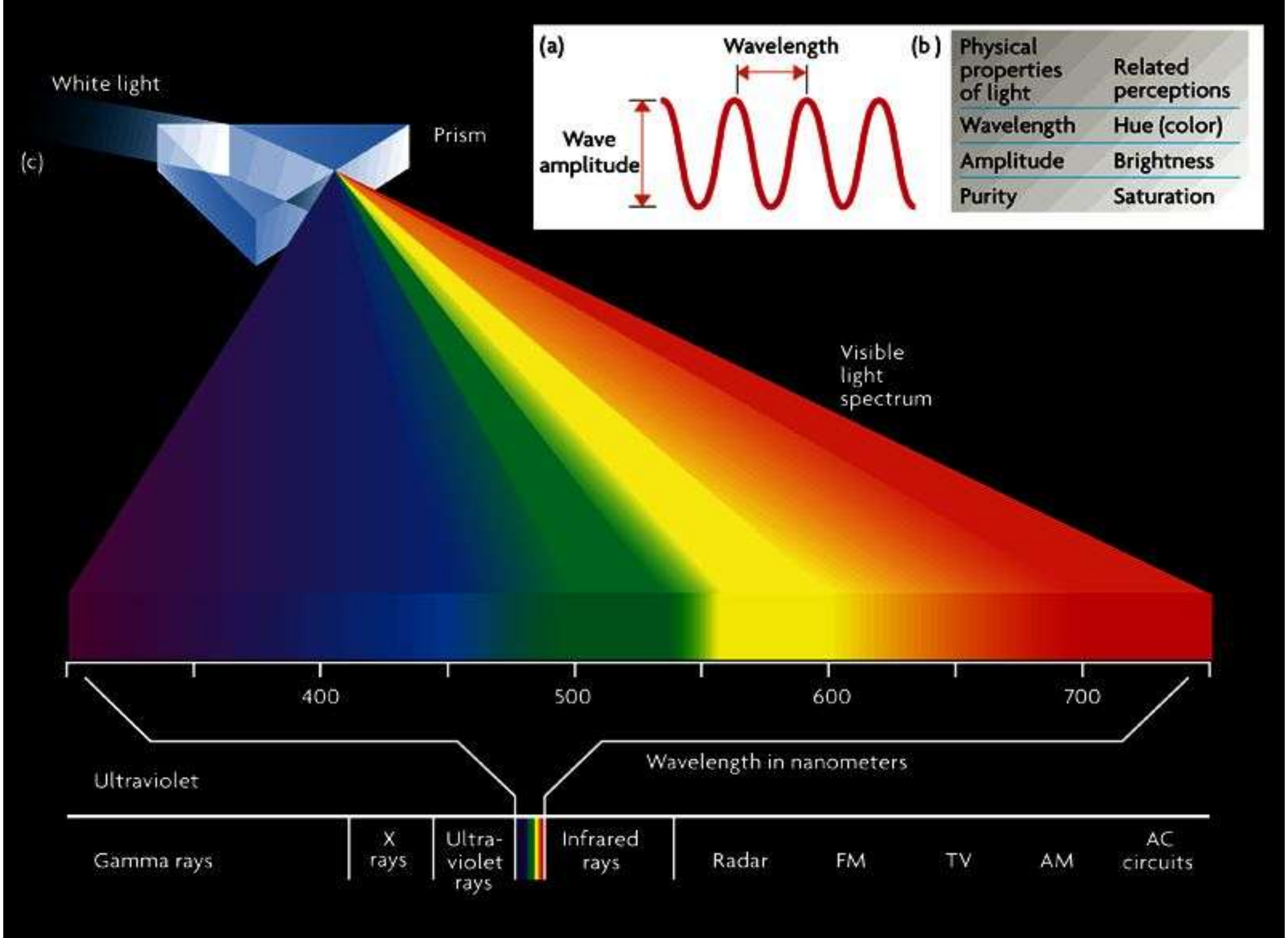
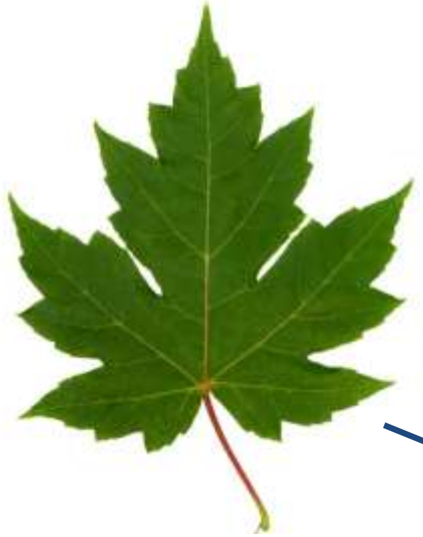


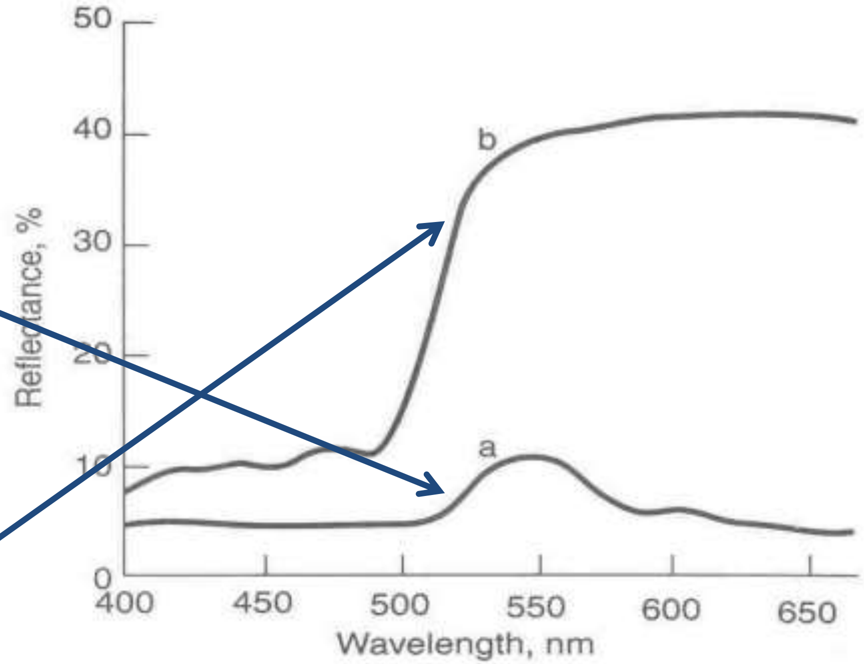
Image from Wikimedia commons, user inductiveload



(a)
Wavelength
Wave amplitude
(b)
Physical properties of light
Related perceptions
Wavelength
Hue (color)
Amplitude
Brightness
Purity
Saturation
White light
Prism
(c)
Visible light spectrum
400
500
600
700
Wavelength in nanometers
Ultraviolet
Gamma rays
X rays
Ultra-violet rays
Infrared rays
Radar
FM
TV
AM
AC circuits

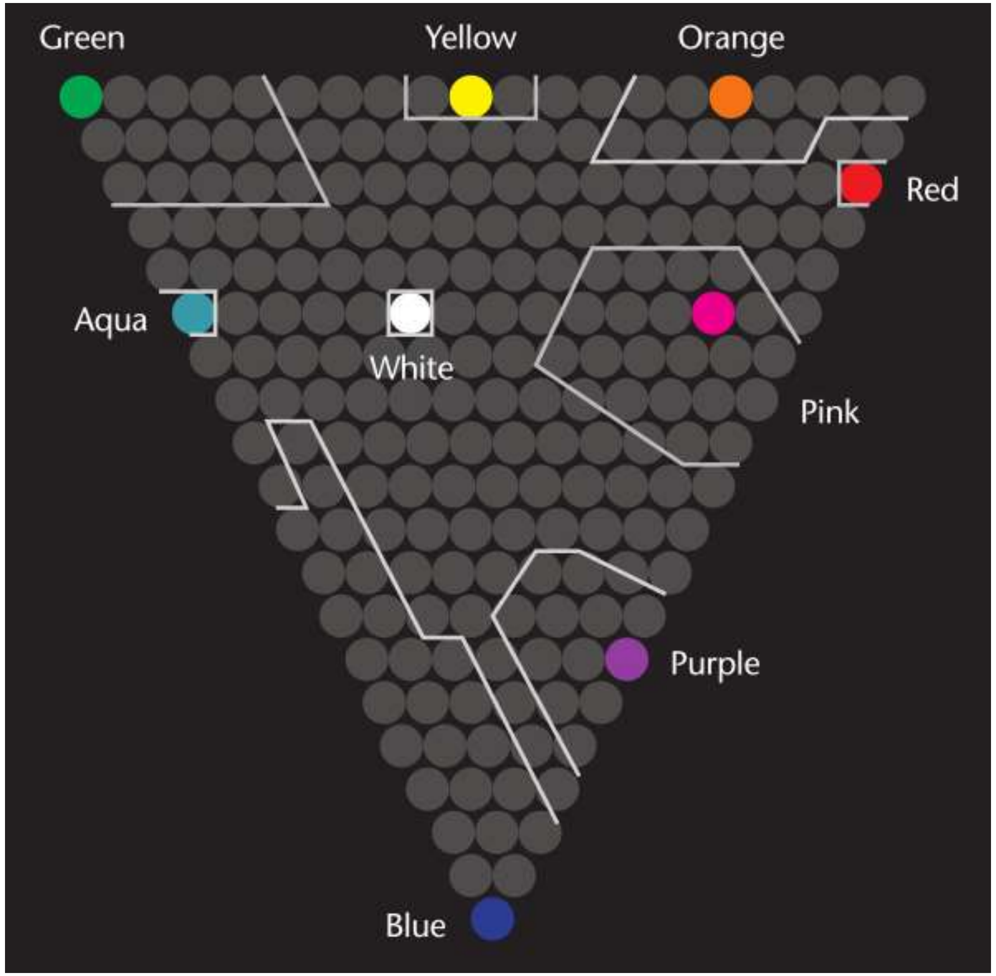
Color vs. Wavelength





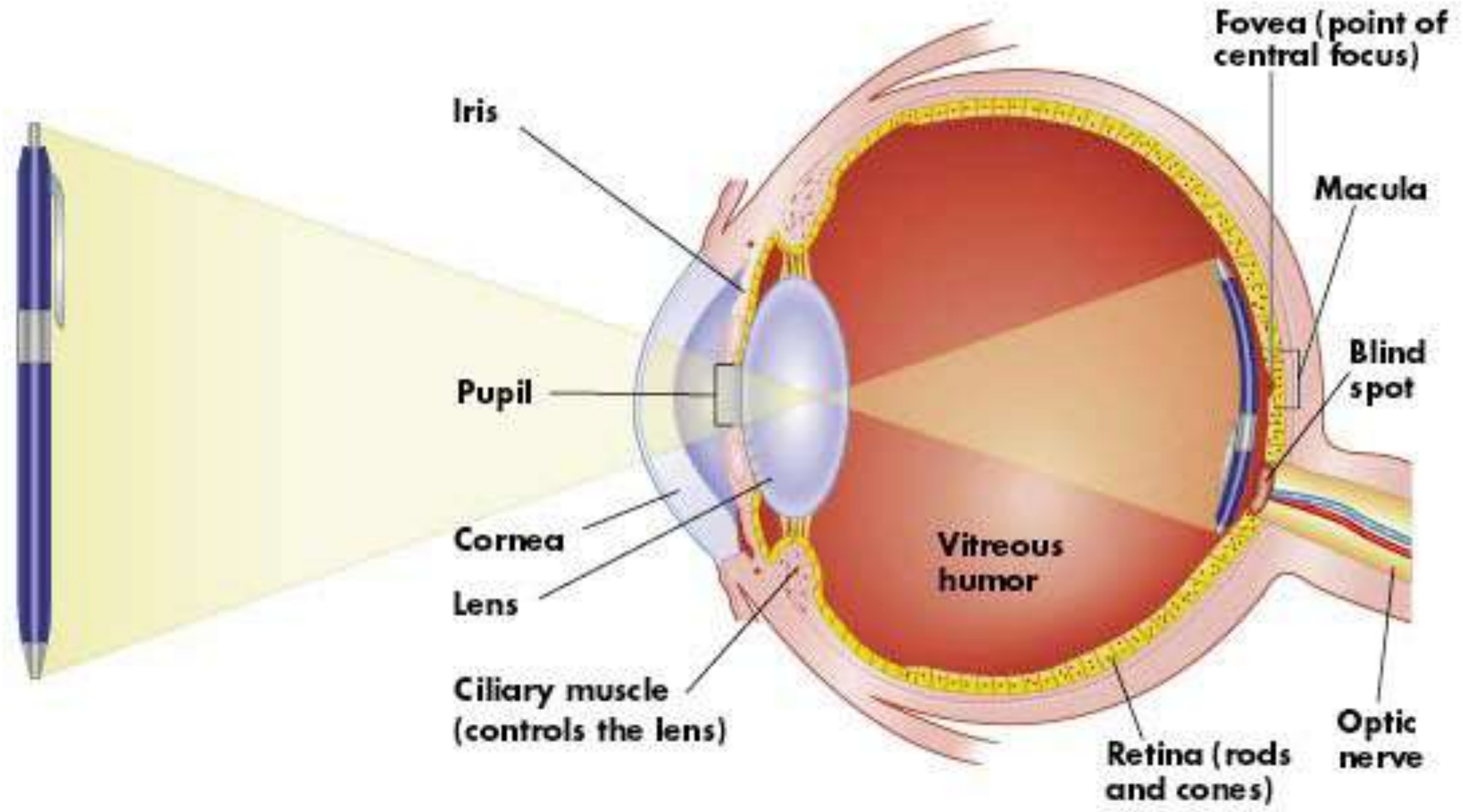
Naming Color

- **Color is a perception**
  - Out of 210 colors, 8 names were consistently used by at least 75% of subjects
  - Most people called monitor red "orange"
  - Most set "pure green" around 514nm, about 30% of the population set it around 525nm
  - Subjects typically agree on "pure yellow" within 2nm



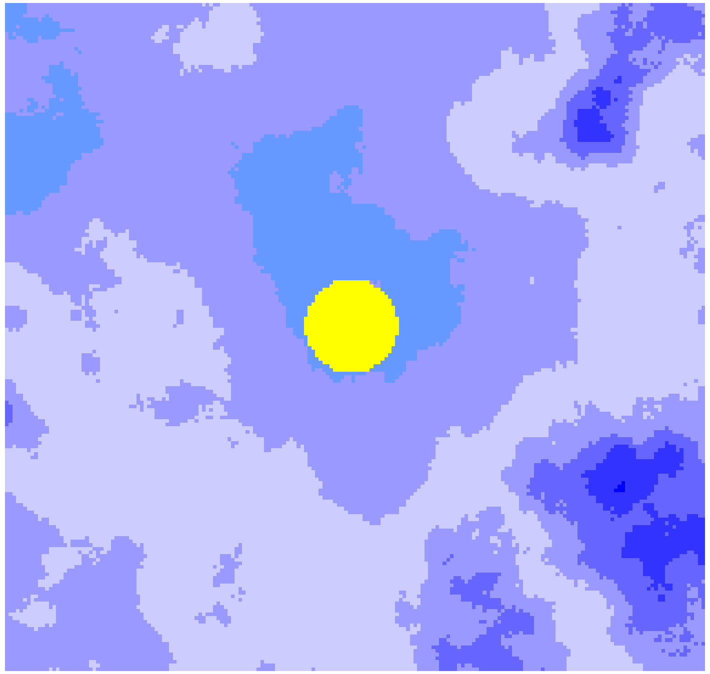
[Post and Greene 1986]

Anatomy of the Eye



The Blind Spot

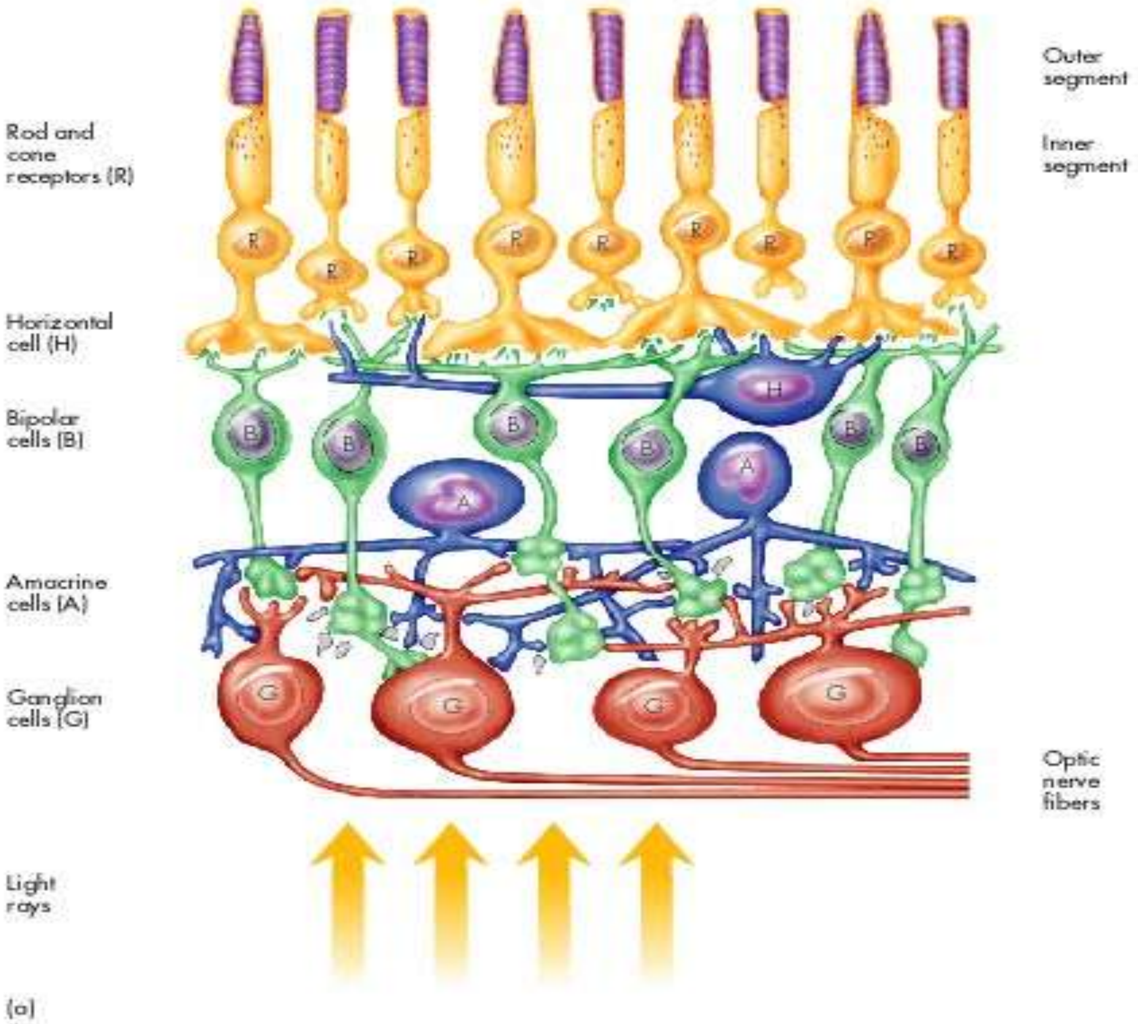
- Close your **right** eye!



1 2 3 4 5 6

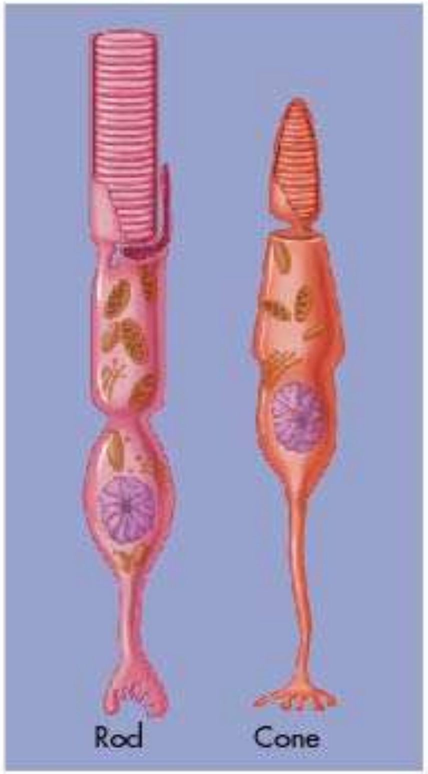
Anatomy of the Retina

- The **retina**
  - Transduces light energy into electrical impulses
  - Performs initial encoding / processing
  - Is counted as part of the brain

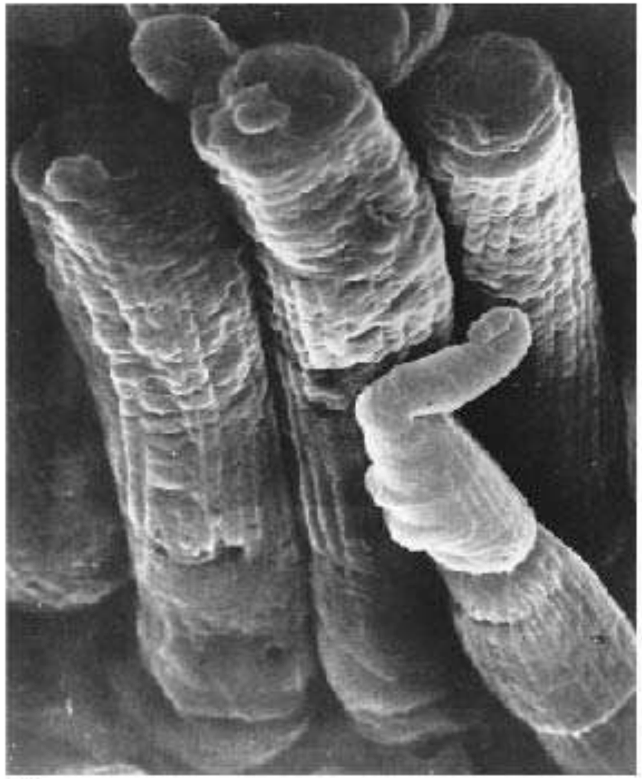


Rods vs. Cones

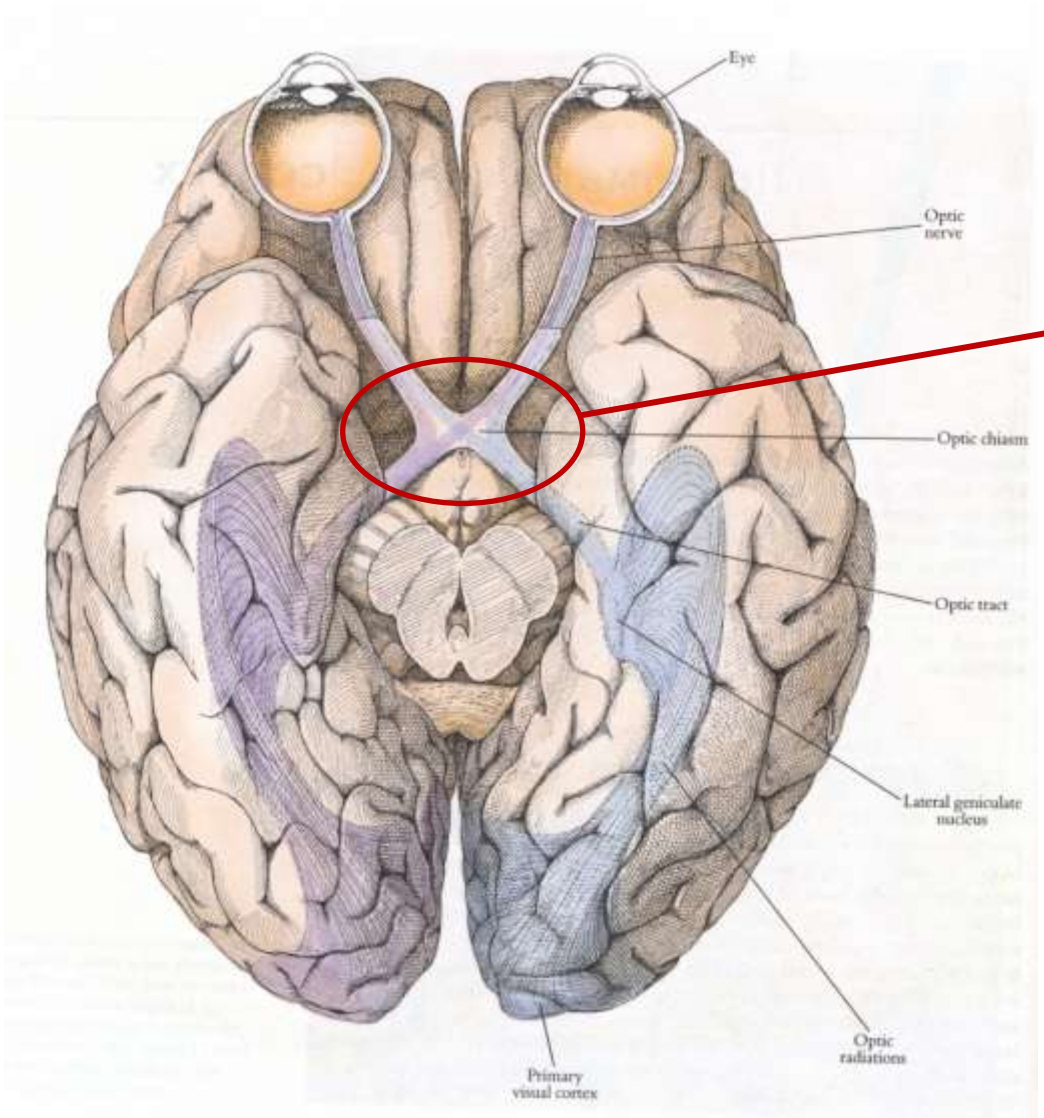
- The retina has approx. 120 million rods and 5 million cones
- **Rods:**
  - Used for night (scotopic) vision
  - Achromatic
  - Peripheral vision
  - Less acuity
- **Cones:**
  - Require strong illumination (photopic vision)
  - Chromatic
  - Central vision
  - Higher acuity



(a)


(b)

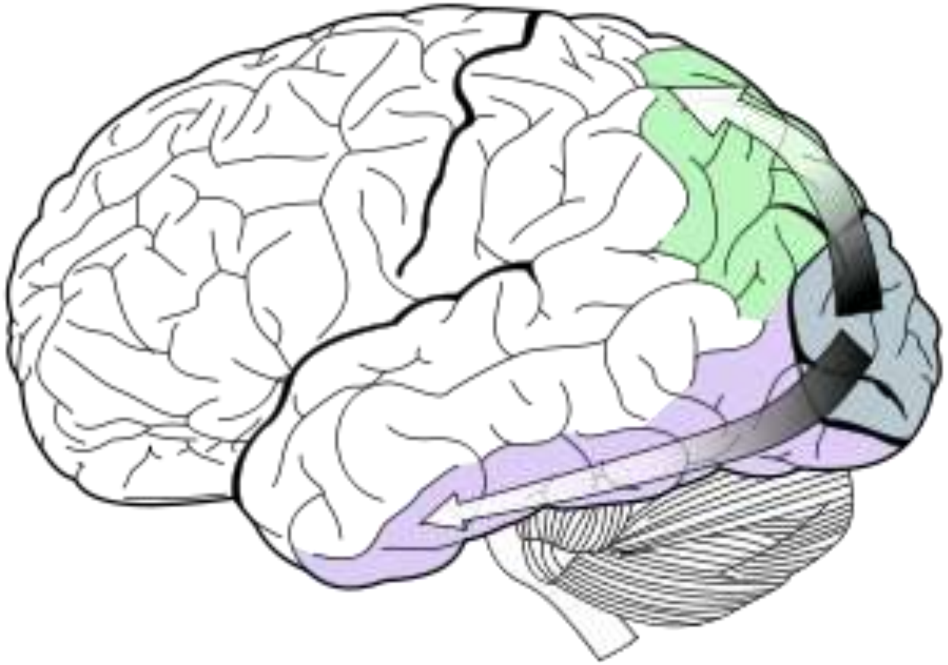
Human Visual System



Each hemisphere processes information from the opposite half of the visual field

Ventral vs. Dorsal Stream

- **Two-streams hypothesis** of higher-level vision:
  - Dorsal stream ("where")
    - Guiding behavior
    - Relatively fast
    - High temporal resolution
    - Input from full retina
  - Ventral stream ("what")
    - Recognition and visual memory
    - Relatively slow
    - High spatial resolution
    - Input mostly from fovea
    - Conscious perception



Radiometric Units

Units used when measuring electromagnetic radiation:

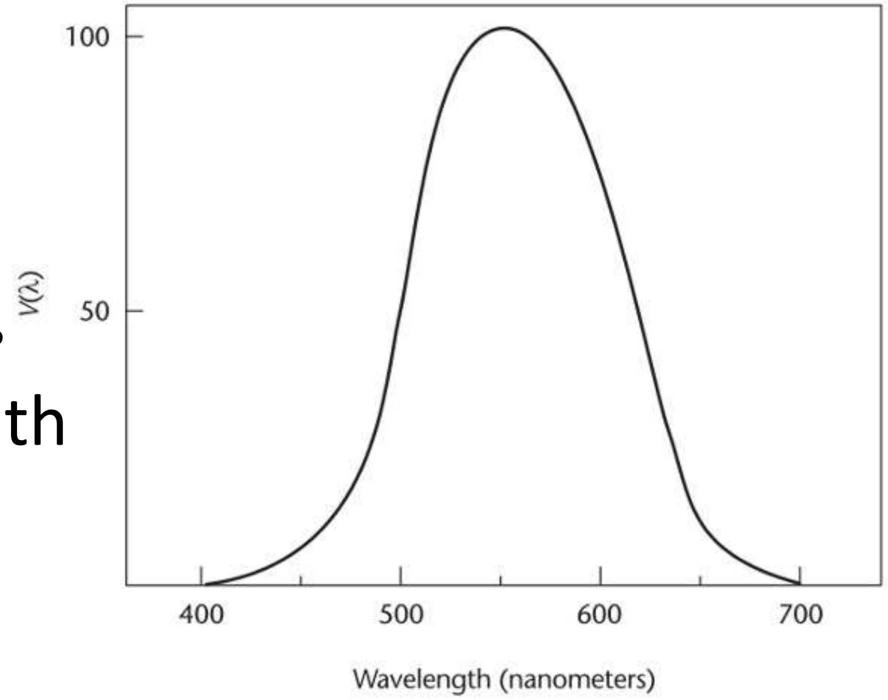
Name	Symbol	Unit
Radiant Energy	Q_e	J
Radiant Flux	Φ_e	$W = J/s$
Radiant Intensity	I_e	W/sr
Irradiance	E_e	W/m^2
Radiant Emittance	M_e	W/m^2
Radiance	L_e	$W/(m^2 sr)$

Luminance

- **Luminance** is the photometric counterpart to radiance. It quantifies the physical amount of visible light (not brightness)

$$L_v = \int V(\lambda) L_e(\lambda) d\lambda$$

  - Unit: candela / square meter
  - $V(\lambda)$ = luminous *efficacy*
    - Unit: lumen / Watt
    - Plot shows (normalized) luminous *efficiency*
  - Defined by *Commission Internationale de l'Éclairage* (CIE) based on experiments with standard human observers
    - Conducted for *photopic* vision, results would differ under mesopic / scotopic conditions



Photometric Units

Units used when measuring lights in terms of *perceived intensity* (weighted by luminous efficacy)

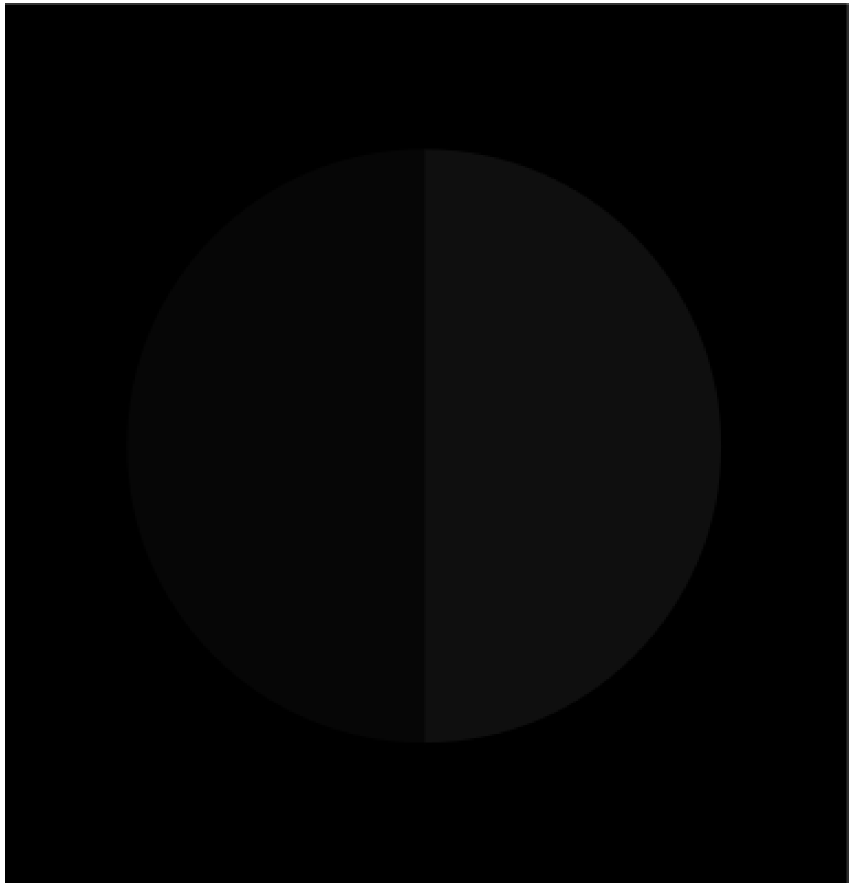
Name	Symbol	Unit
Luminous Energy	Q_v	$lm \cdot s$
Luminous Flux	Φ_v	lm
Luminous Intensity	I_v	$cd = lm/sr$
Illuminance	E_v	$lux = lm/m^2$
Luminous Emittance	M_v	$lux = lm/m^2$
Luminance	L_v	$lm/(m^2 sr)$

Brightness

- **Brightness** = *subjective* amount of light from a self-luminous object in a darkened room
  - Related to luminance roughly according to a **power law:** $B \approx L^n$
  - Power *n* depends on size of object, $n \approx 0.3 \ldots 0.5$
  - Similar power laws have been found to approximate perception of loudness, smell, taste, heaviness, force, and touch
    - Motivates use of logarithmic unit decibel to measure sound pressure

Monitor Gamma

- The relationship between pixel voltage on a CRT monitor and luminance also follows a power law: $L = V^{\gamma}$
  - Typically, $\gamma \approx 2.2$, leading to an approximately linear relationship between *V* and *B*
  - Modern LCDs are built to have similar γ
  - To ensure correct color reproduction, **monitor gamma has to be calibrated**

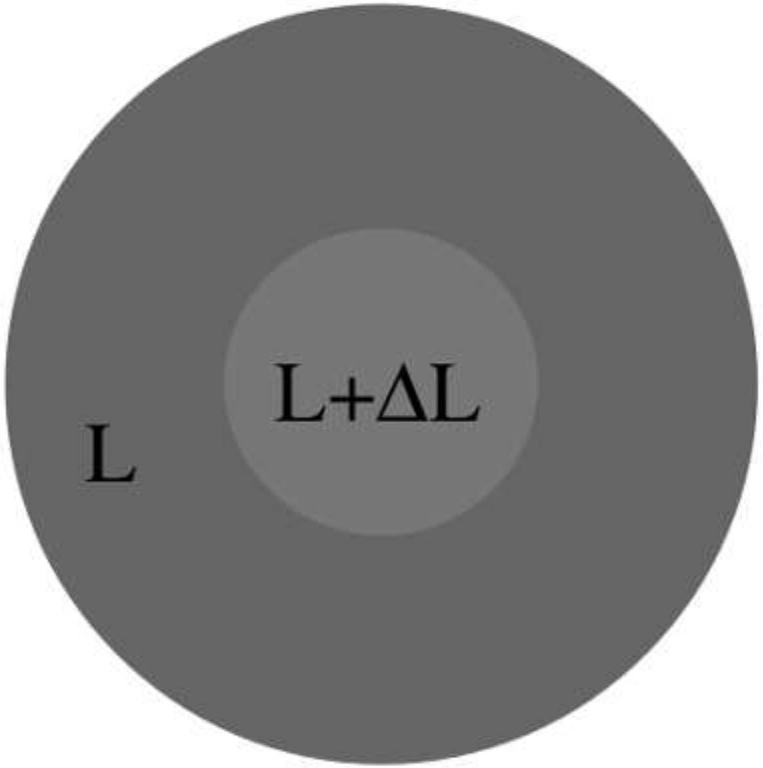


Luminance Illusions

The human visual system is **not** a reliable measurement device for luminance!

Weber-Fechner Law

- **Just Noticable Differences** (JNDs) are proportional to baseline brightness



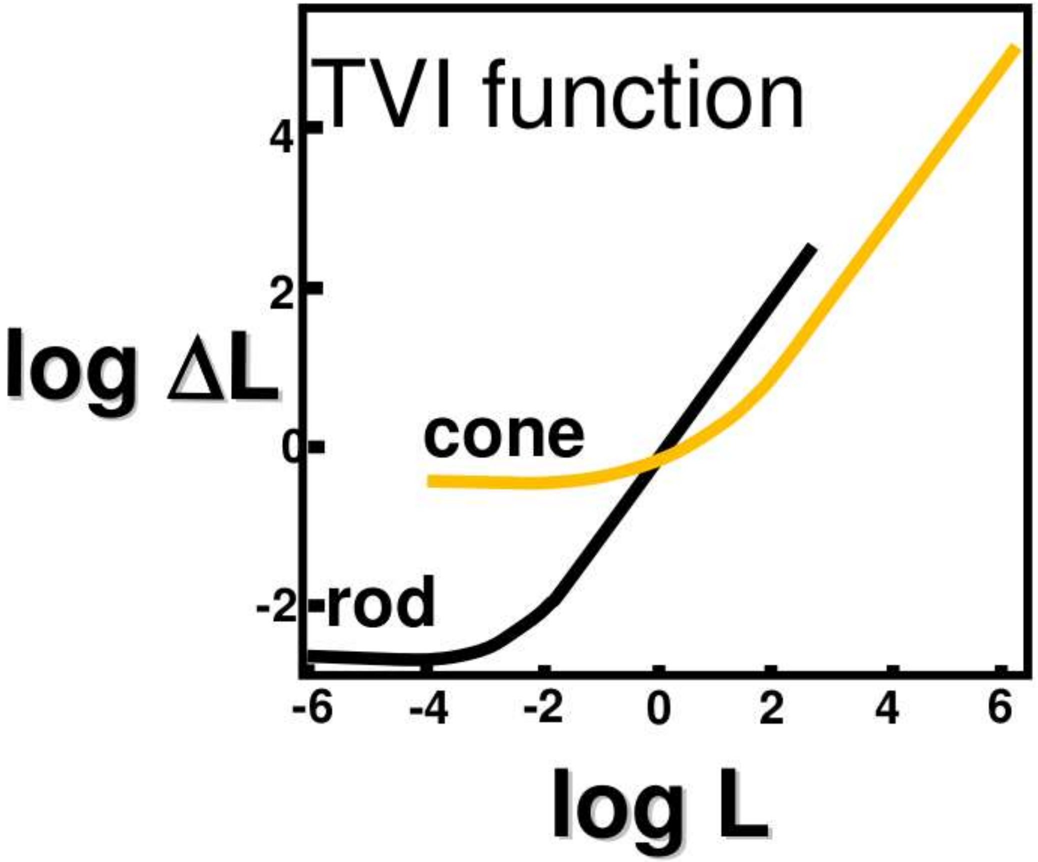
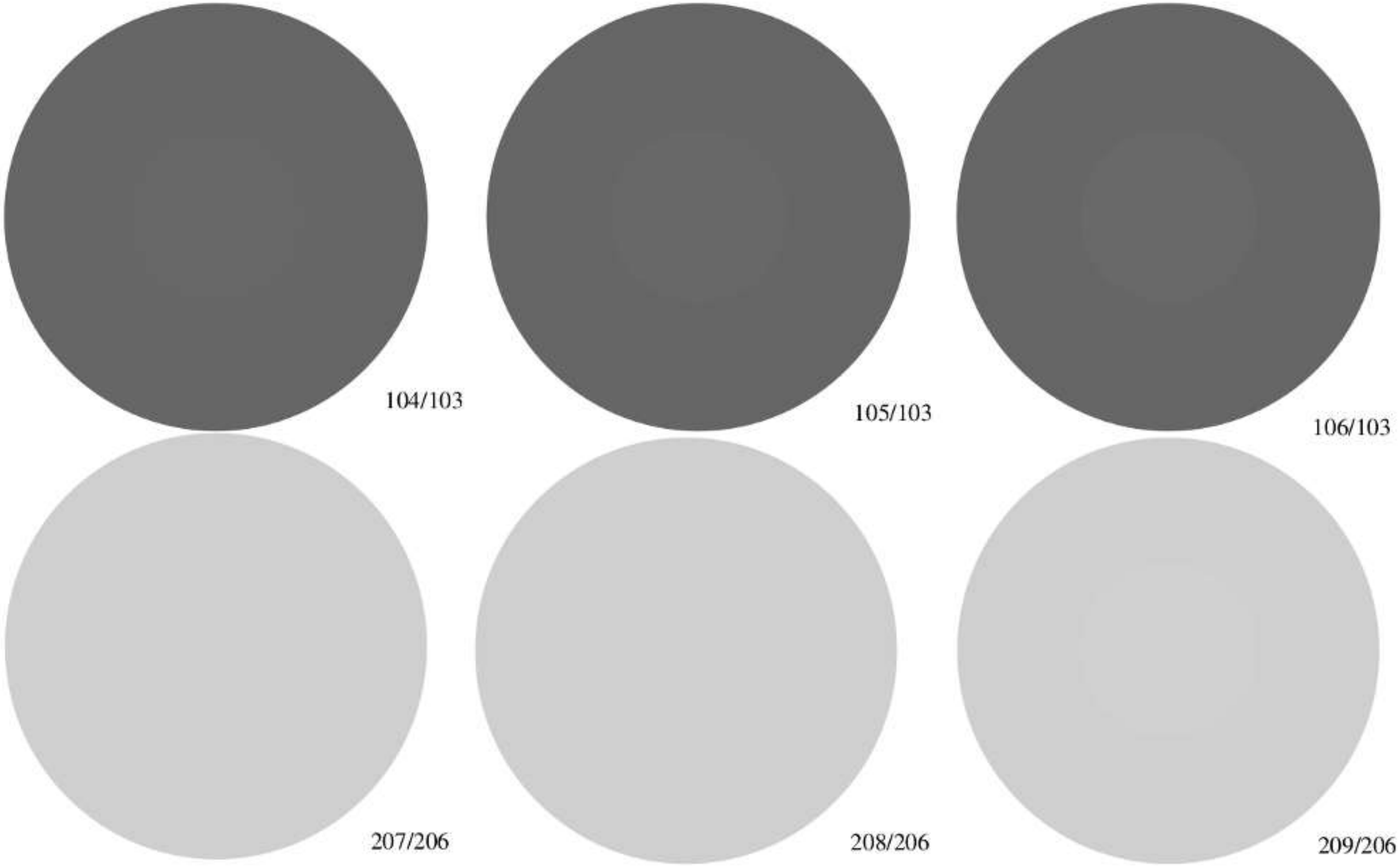


Illustration: Weber-Fechner Law

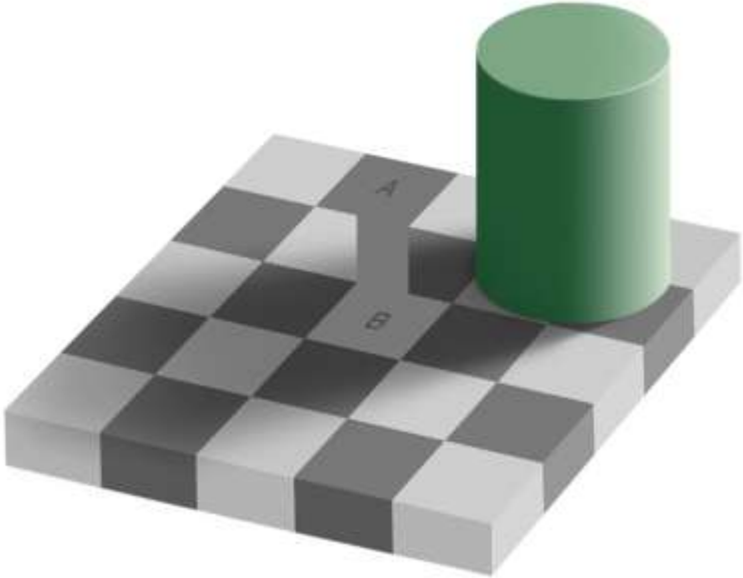


Lightness

- **Lightness** = subjective reflectance of a surface
  – Do we perceive an object as dark or bright?
  – Depends on many factors apart from luminance
    - In different environments, the same luminance can be perceived as “black” or as “white”
- **Lightness constancy**
  – Human visual system recognizes reflectance properties under a wide range of conditions
  – In everyday life, being able to recognize materials and objects is more important than to objectively measure luminance

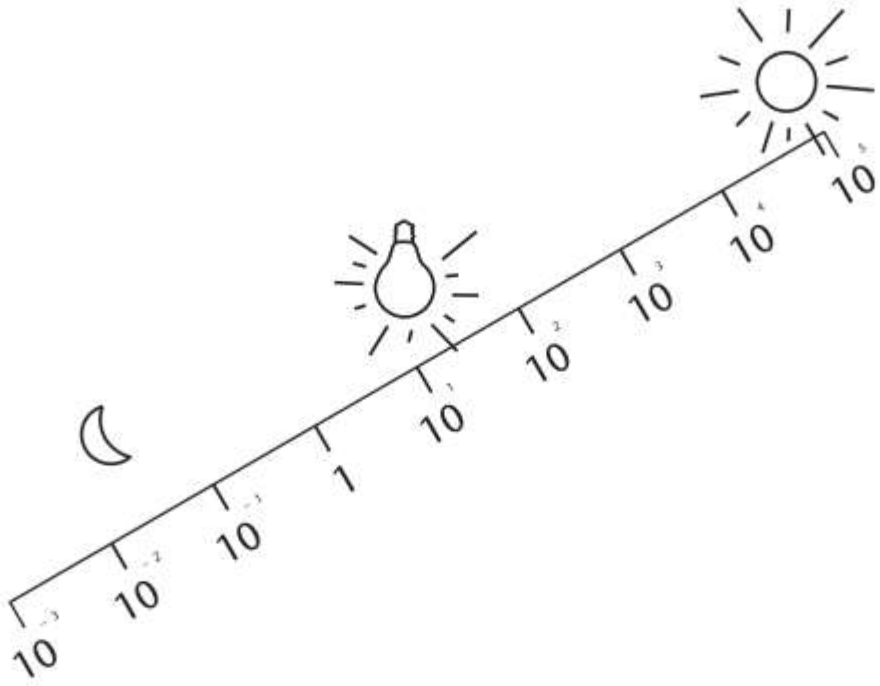
Lightness Illusions

- Perceived lightness is influenced by complex factors such as **surface orientation, shadows** and **positions of light sources**



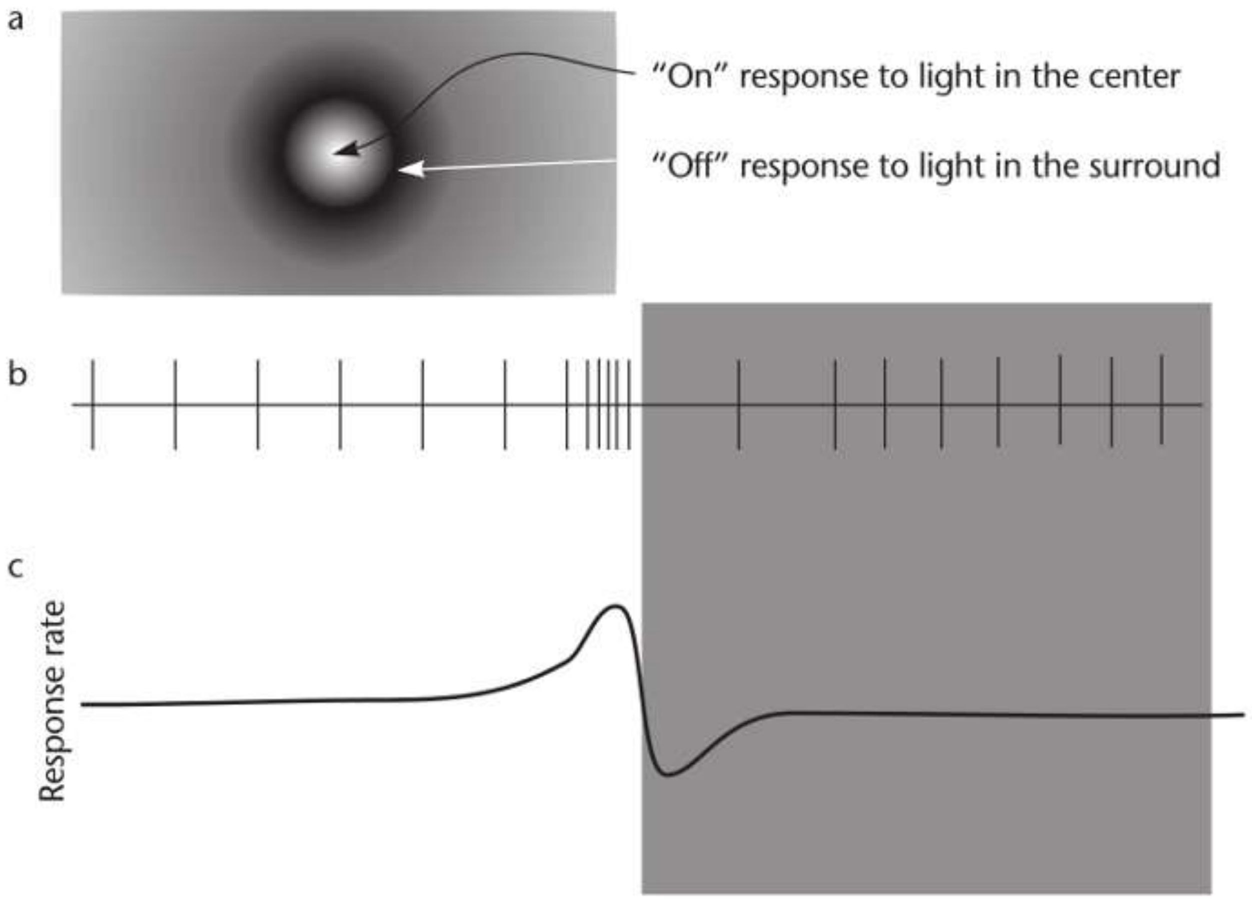
Adaptation

- **Adaptation** contributes to lightness constancy
  - The eye can see a contrast ratio of 1:1000, but can **adapt** over eight orders of magnitude
  - **Iris** opens and closes (factor 16-64)
  - Bleaching of **photopigment**
  - **Switchover** rods vs. cones
  - Full adaptation bright to dark:
Up to 30 minutes

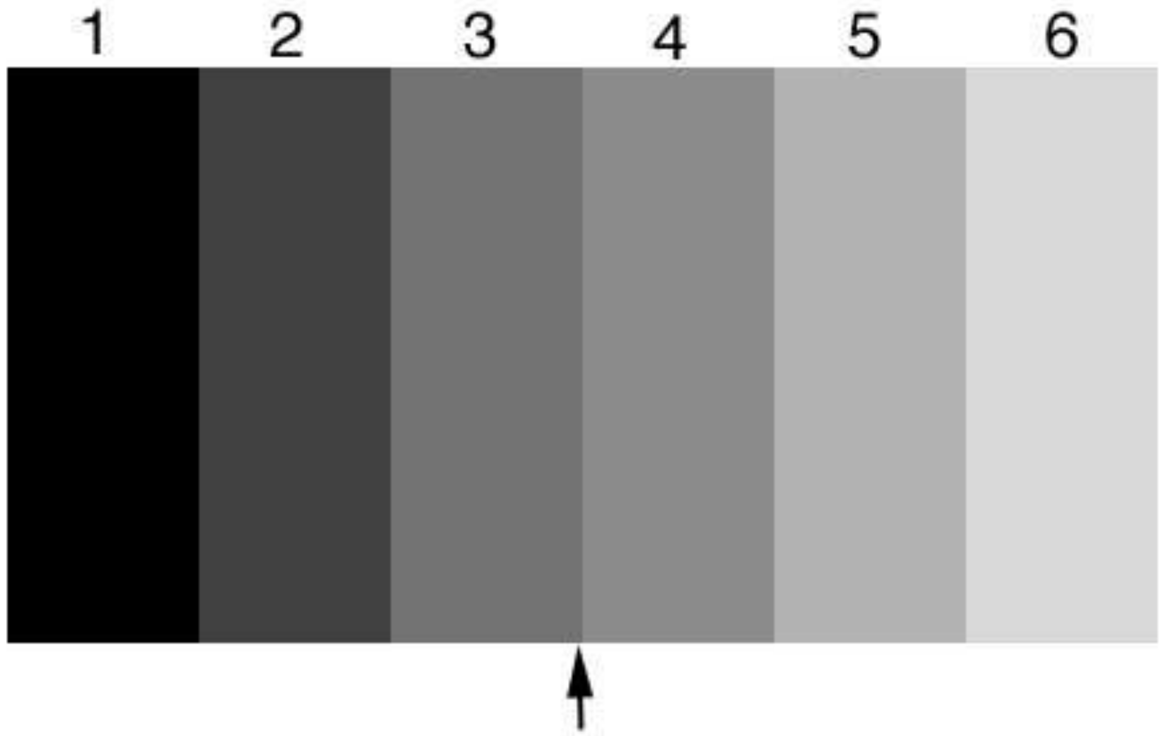


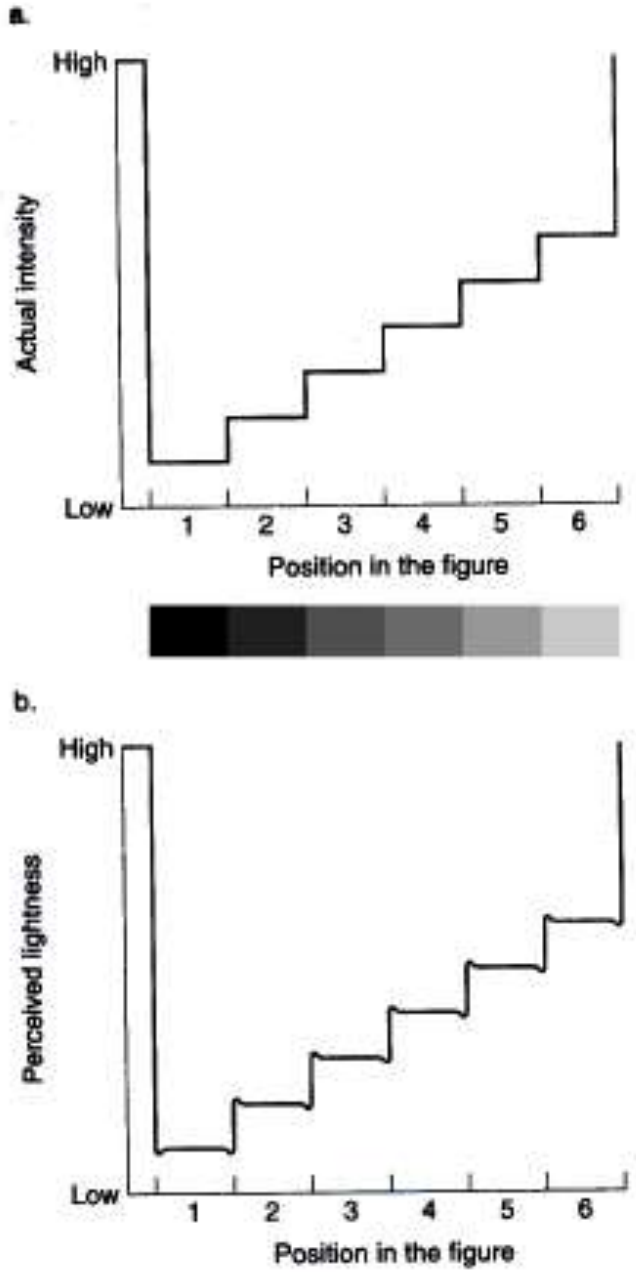
Contrast Vision

- Lightness constancy is partly achieved by **perceiving contrast** rather than luminance
  - Neural mechanism: **Lateral inhibition**
  - **Receptive field** with center/surround structure



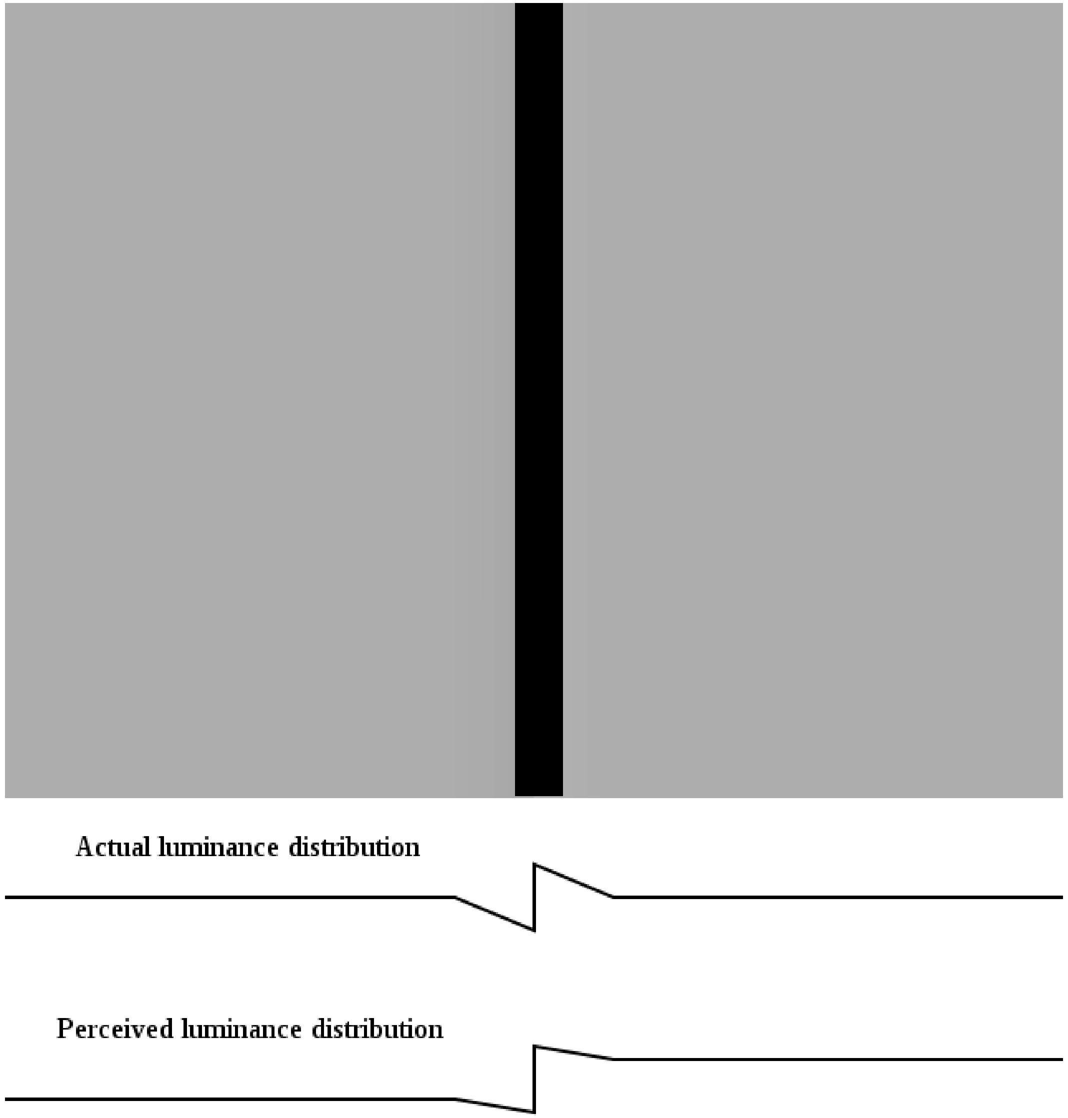
Mach Bands


1
2
3
4
5
6


a.
High
Actual intensity
Low
1 2 3 4 5 6
Position in the figure
b.
High
Perceived lightness
Low
1 2 3 4 5 6
Position in the figure

Cornsweet Illusion

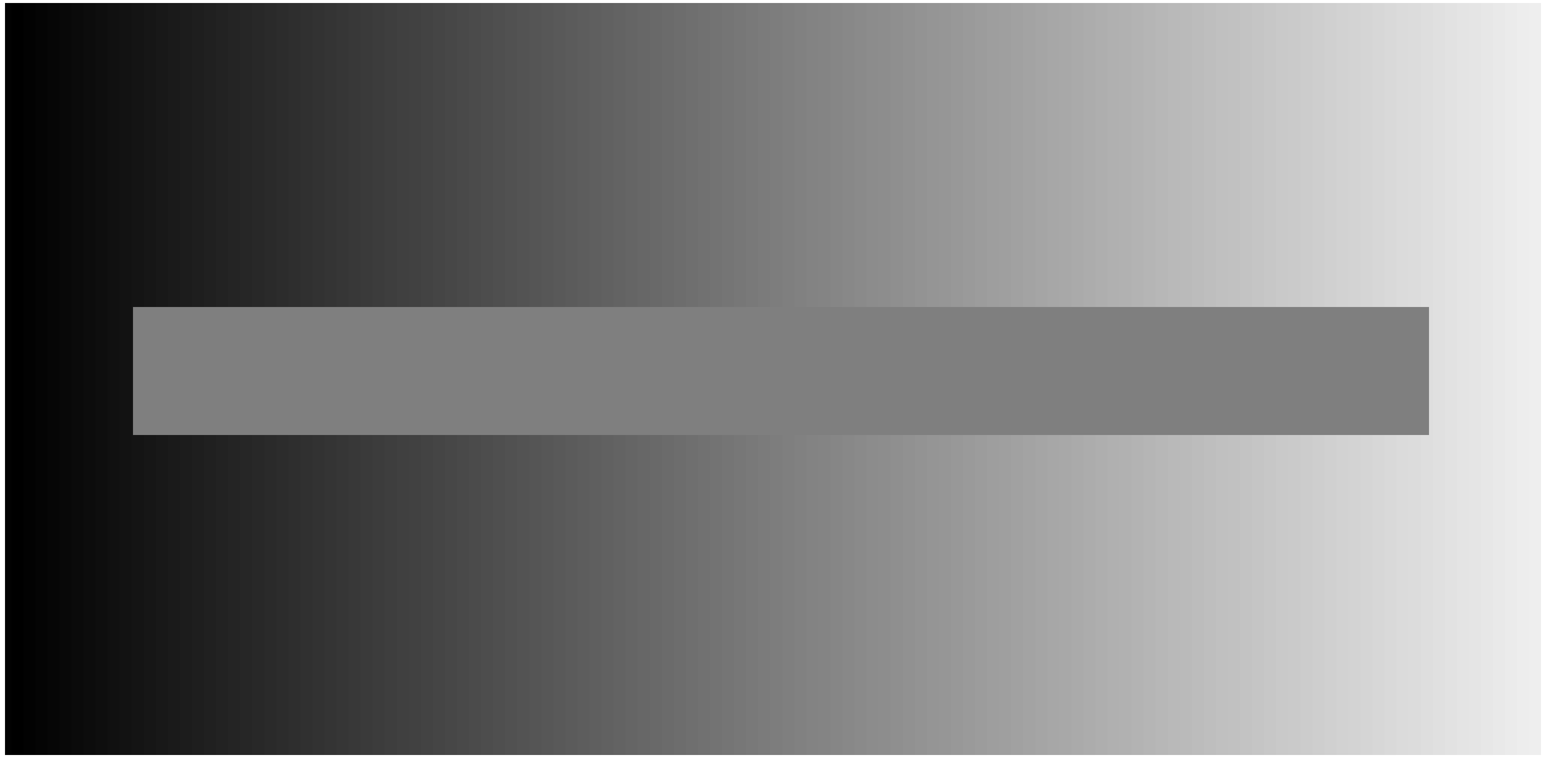
- Human visual system extrapolates edge information



Cornsweet Effect in Art

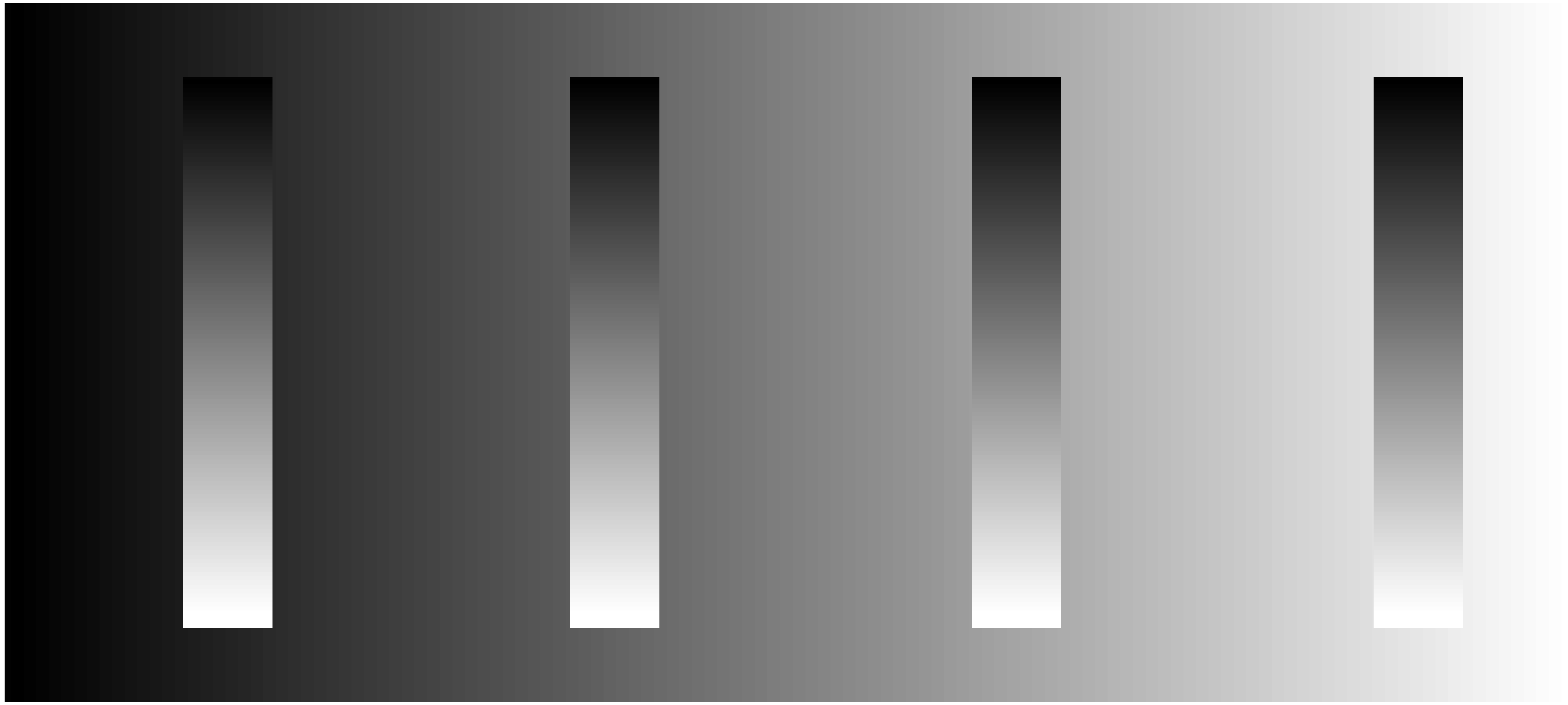
- Georges-Pierre Seurat (1859 – 1891):
  Bathers at Asnières

Simultaneous Contrast



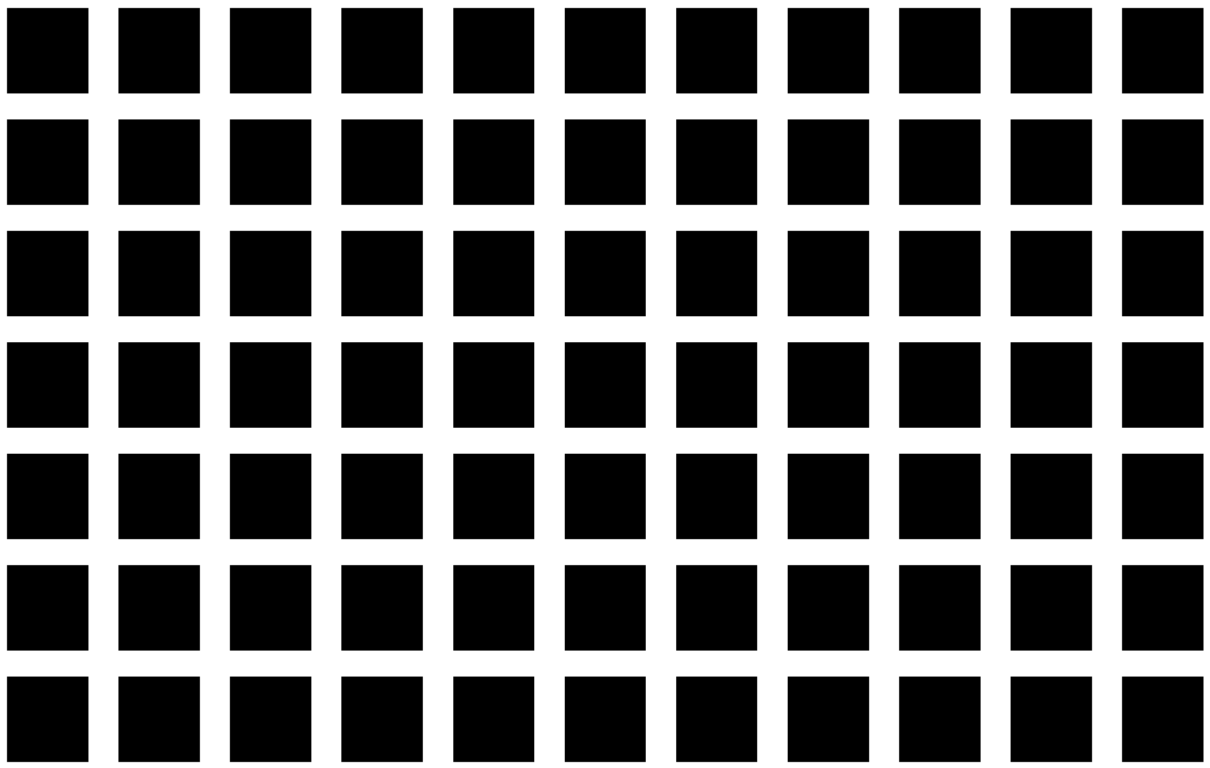
Contrast Crispening

- Sensitivity for small changes is best around the background intensity



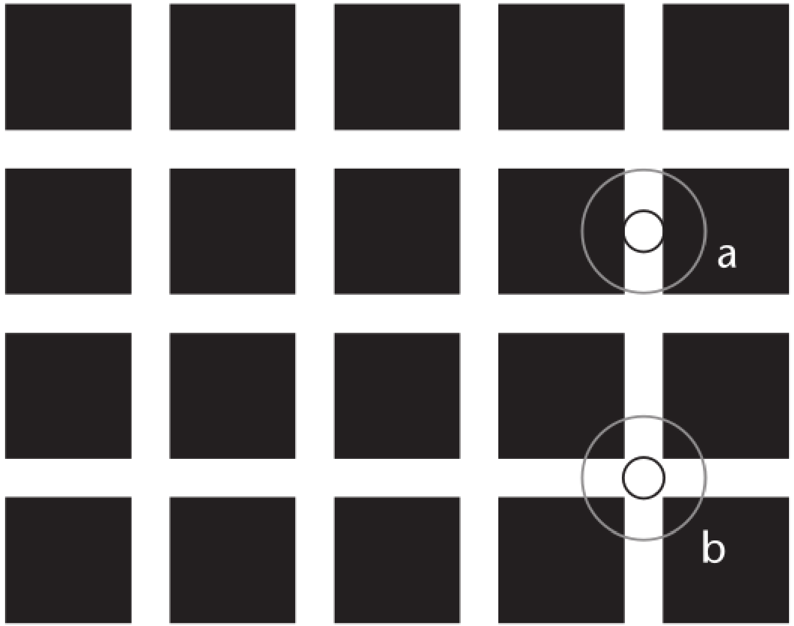
Contrast Illusions

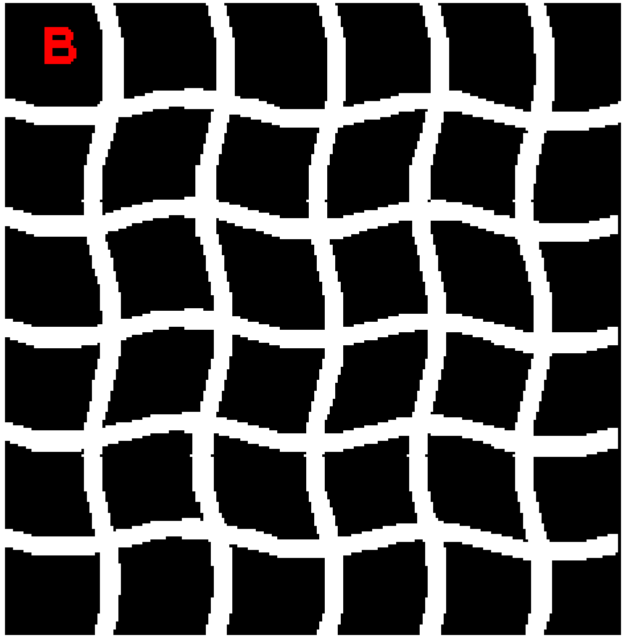
- **Hermann grid**
  - – Perception of smudges at the intersection between grid lines
  - – Vanish at the center of vision



Hermann Grid Explained?

- Traditional explanations in terms of center/surround structure incompatible with experimental evidence
- Effect might arise in visual cortex rather than the retina

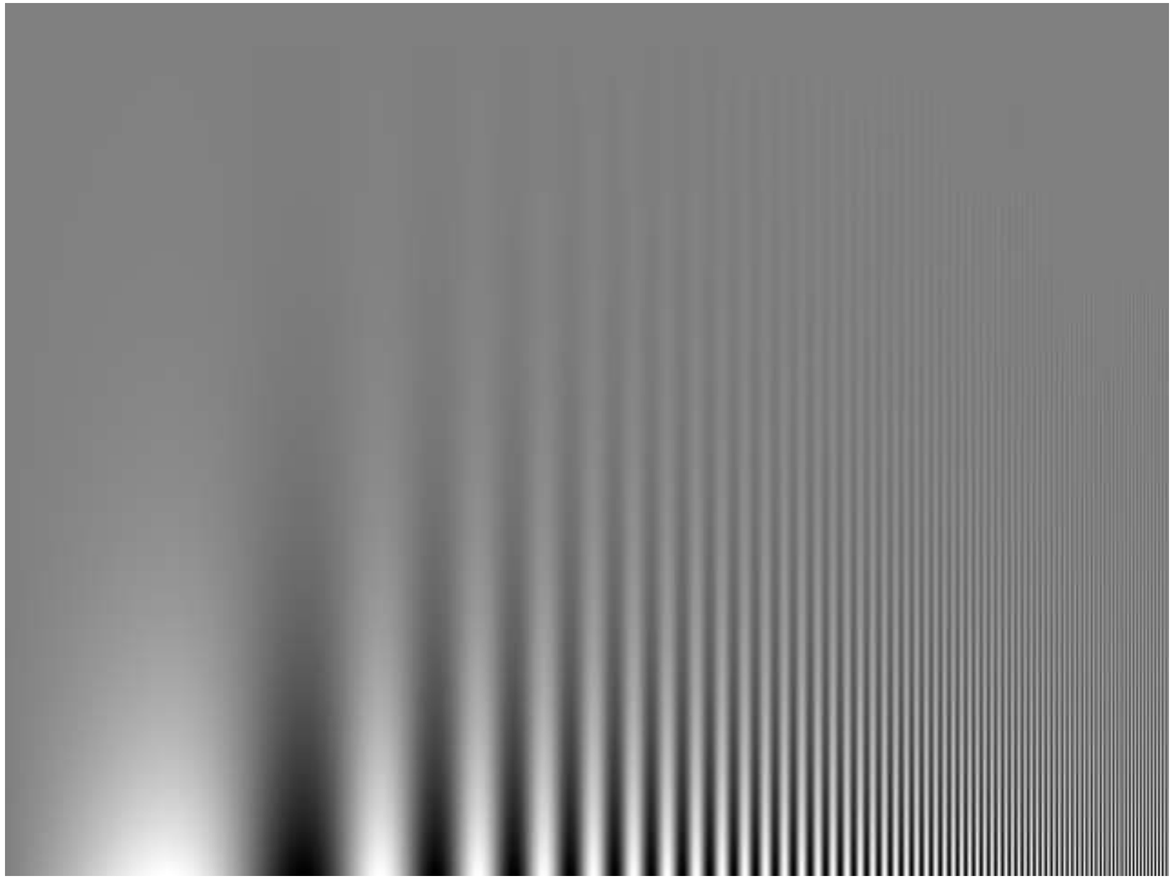




Source: https://web.mit.edu/bcs/schillerlab/research/A-Vision/A15-7.htm based on example from Janos Geier

Contrast Sensitivity

- The **Campbell-Robson Contrast Sensitivity Function** reflects our higher contrast sensitivity at medium spatial frequencies



Luminance vs. Color

"Colors are only symbols. Reality is to be found in luminance alone..." (Pablo Picasso)

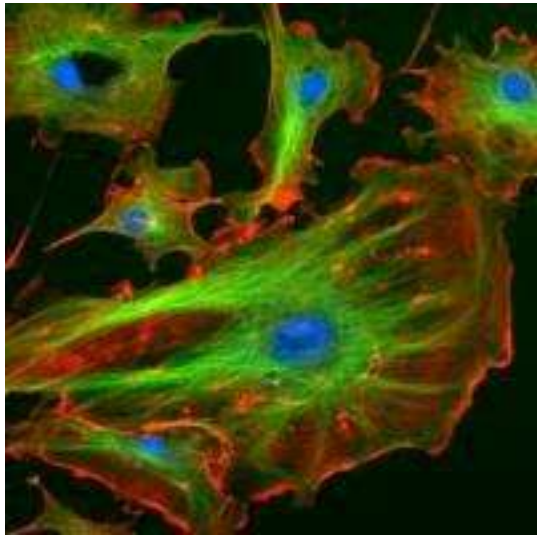
The Tragedy (1903)

Typically, luminance is the key to object recognition and colors are merely perceived as object attributes.

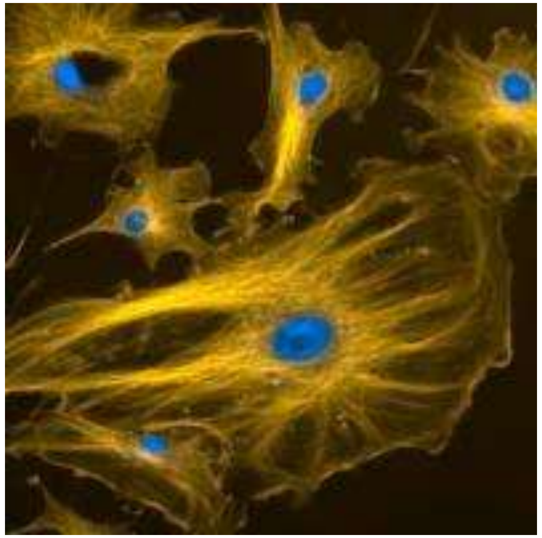
Recoding Luminance to Color

- Object detection becomes hard (or even impossible) when **mapping the black-white axis onto a color axis** (e.g.,yellow-blue)

Color Impairment



- About 8% of the male and about 1% of the female population suffer from some kind of **impaired color vision**
  - Most common: Red/green weakness
  - Should keep this in mind when designing visualizations



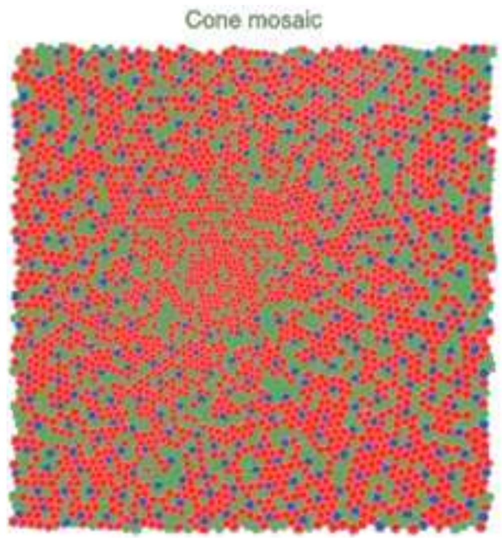
- Is color not that important after all?

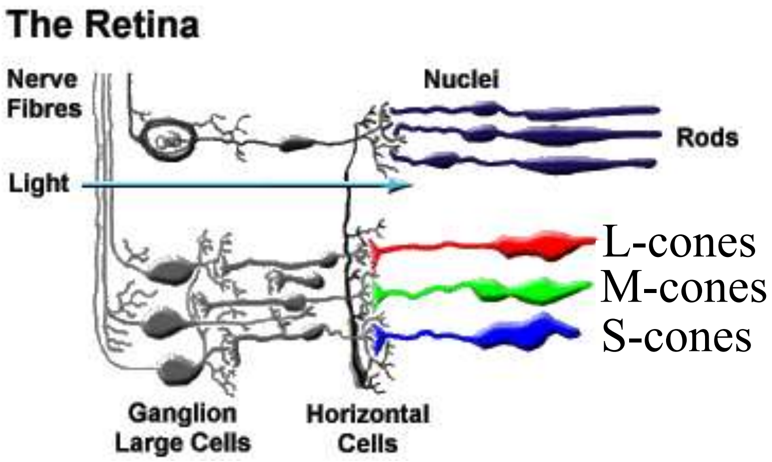
Importance of Color

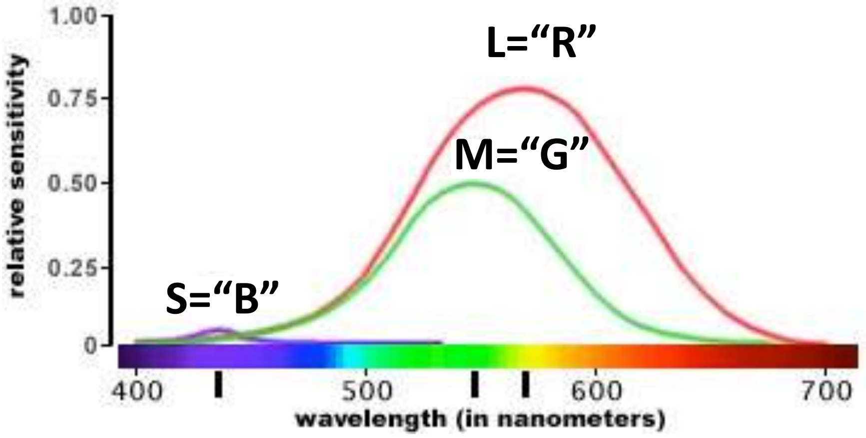
- Color can be helpful to quickly tell apart different **types of objects,** especially if they are similarly shaped

Trichromacy

- Color vision is possible due to the presence of **three different types of cones**
  - Respond more strongly to different parts of the spectrum (short/medium/long)
  - Human color perception is **fundamentally three-dimensional**
    - Eye projects continuous light intensity $I(\lambda)$ onto three numbers, e.g., $S = \int s(\lambda)I(\lambda)d\lambda$

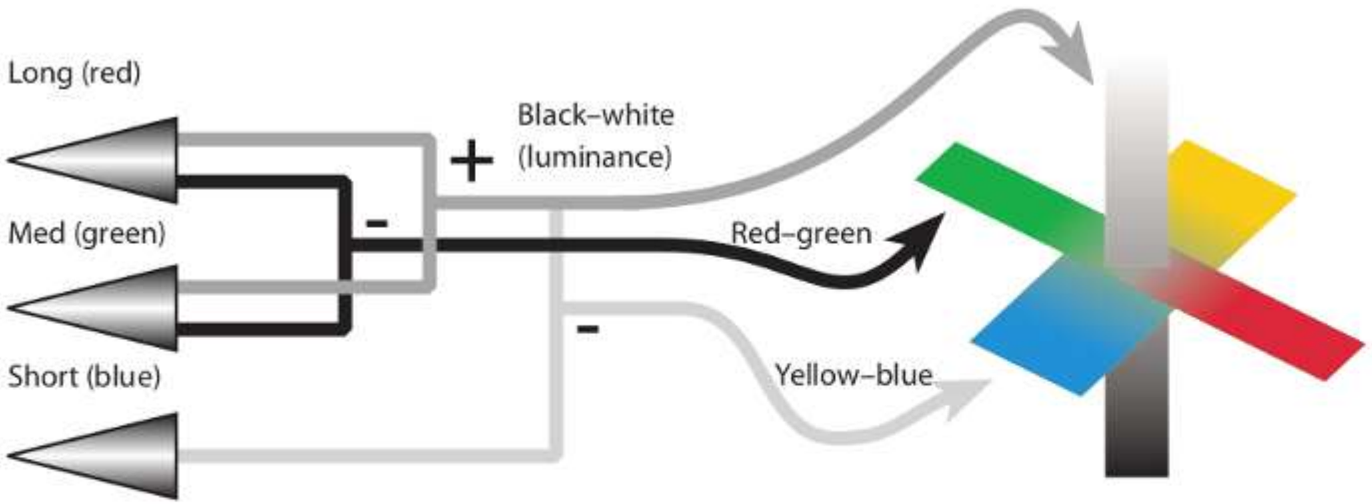






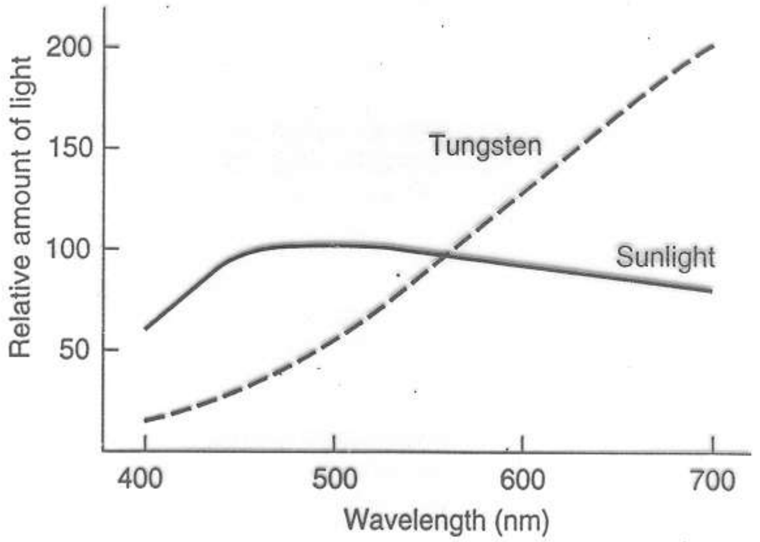
Opponent Colors

- **Opponent Color Theory**
  - First proposed by Ewald Hering (around 1900)
  - Six elementary colors that form **three pairs**:
    - Black-white, red-green, yellow-blue
  - Supported by
    - Naming (greenish blue vs. reddish green)
    - Cross-cultural naming (languages with few color words)
    - Neuro-anatomy



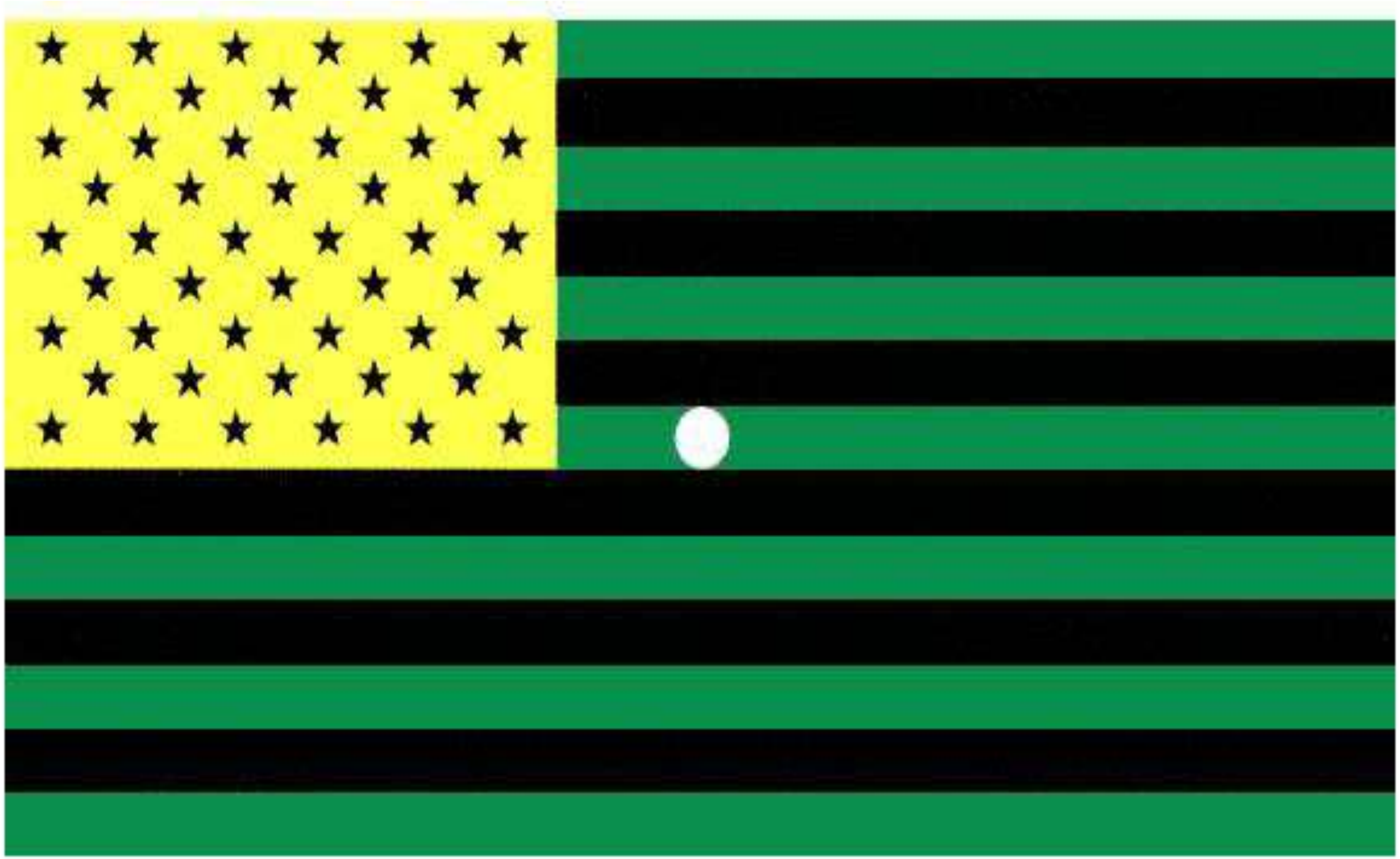
Color Constancy

- **Color constancy** is even more complex than lightness constancy
  - Colors are perceived to be relatively stable under a wide range of incident spectra
  - **Color adaptation** and **color contrast** contribute
  - **“Reference white”** in scene plays a role

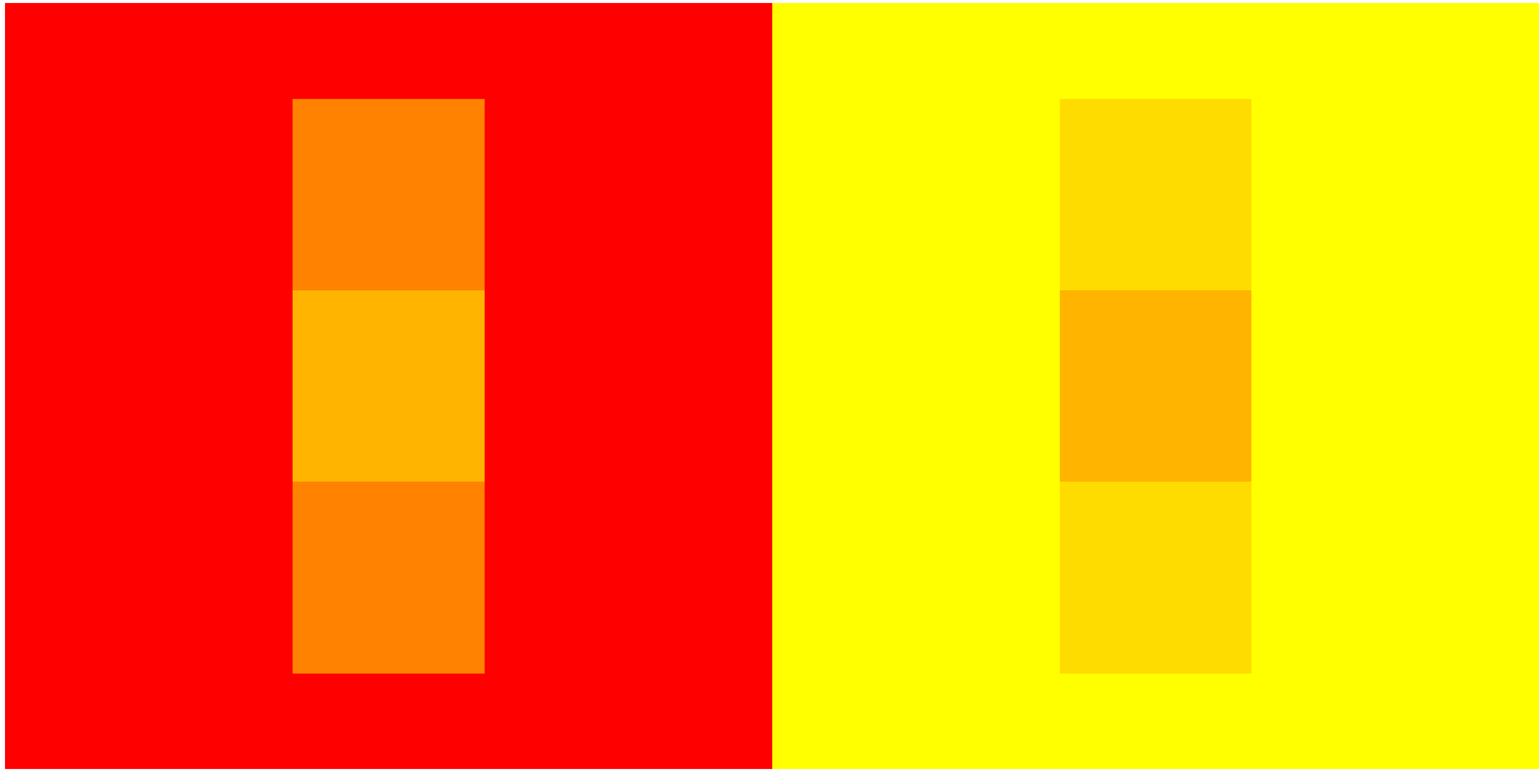


White balance in photography

Color Adaptation



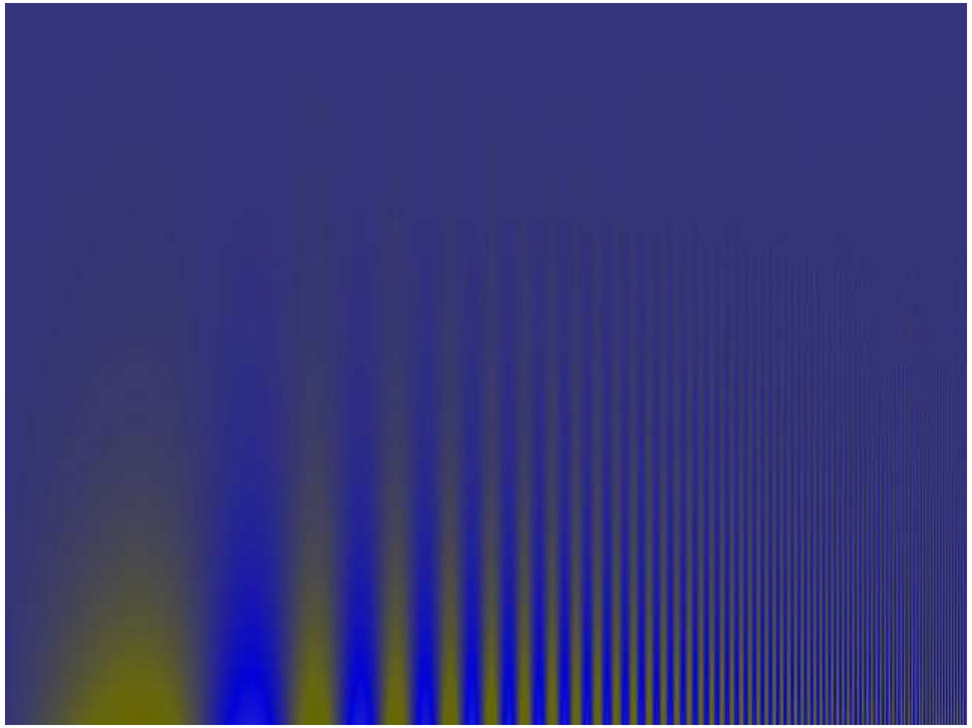
Color Contrast



Source: www.psy.ritsumei.ac.jp/~akitaoka/shikisai2005.html

Color Contrast Sensitivity

- Our sensitivity for **color contrast** is lower than for luminance contrast
  - Exploited in **image compression:** More bits allocated to luminance than to color



How Would You Call This Slides's Background Color?

How Would You Call This Slides's Background Color?

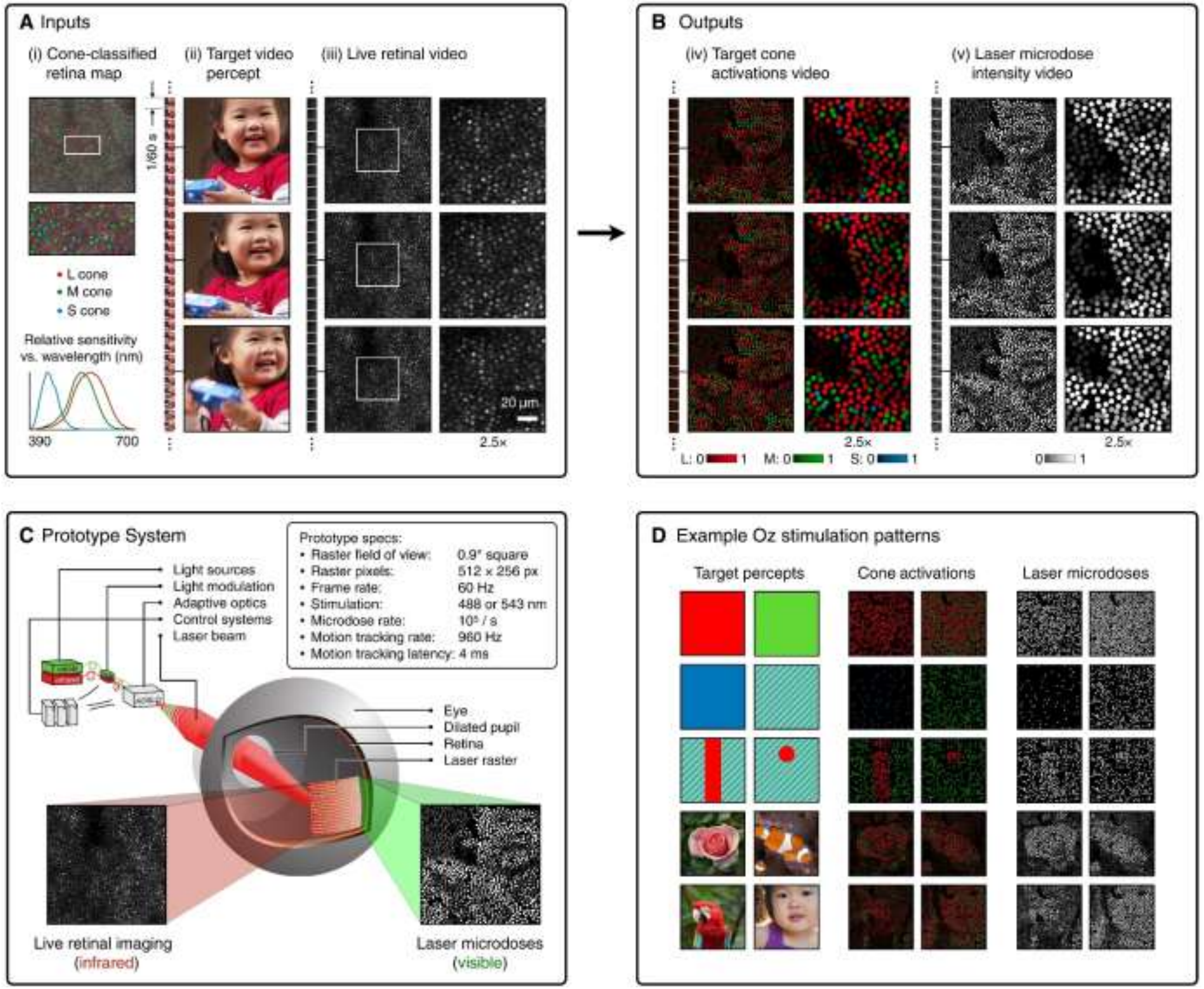
Brown

- Dark shades of yellow / yellowish orange are perceived as **brown**
  - Opposed to dark shades of red / green / blue
  - Brown is not part of the spectrum, but still perceived as a distinct color
  - Perception of brown requires presence of other colors (e.g., white) for reference

- **Important for visualization:**
  - When encoding class membership using color, dark and light shades of yellow and orange might not be recognized as belonging together

Fun Fact: Inventing New Colors

- [Fong et al. 2025] display color via selective cone activation
  – 10^5 targeted microdoses per second of a single wavelength (“green”) laser
  – Exclusive activation of type-M cones led to new color perception “olo”, which was described by test subjects as hypersaturated teal

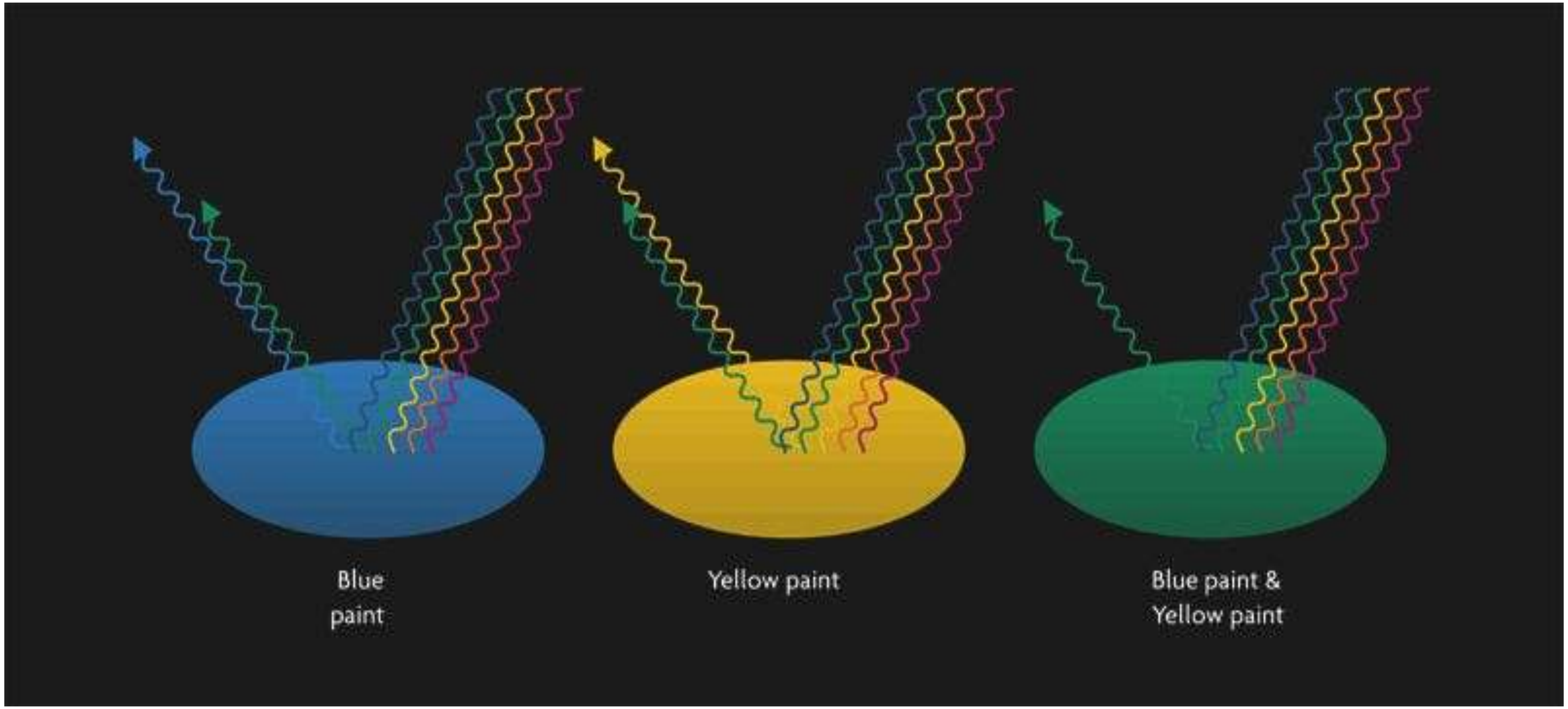


Summary: Color Perception

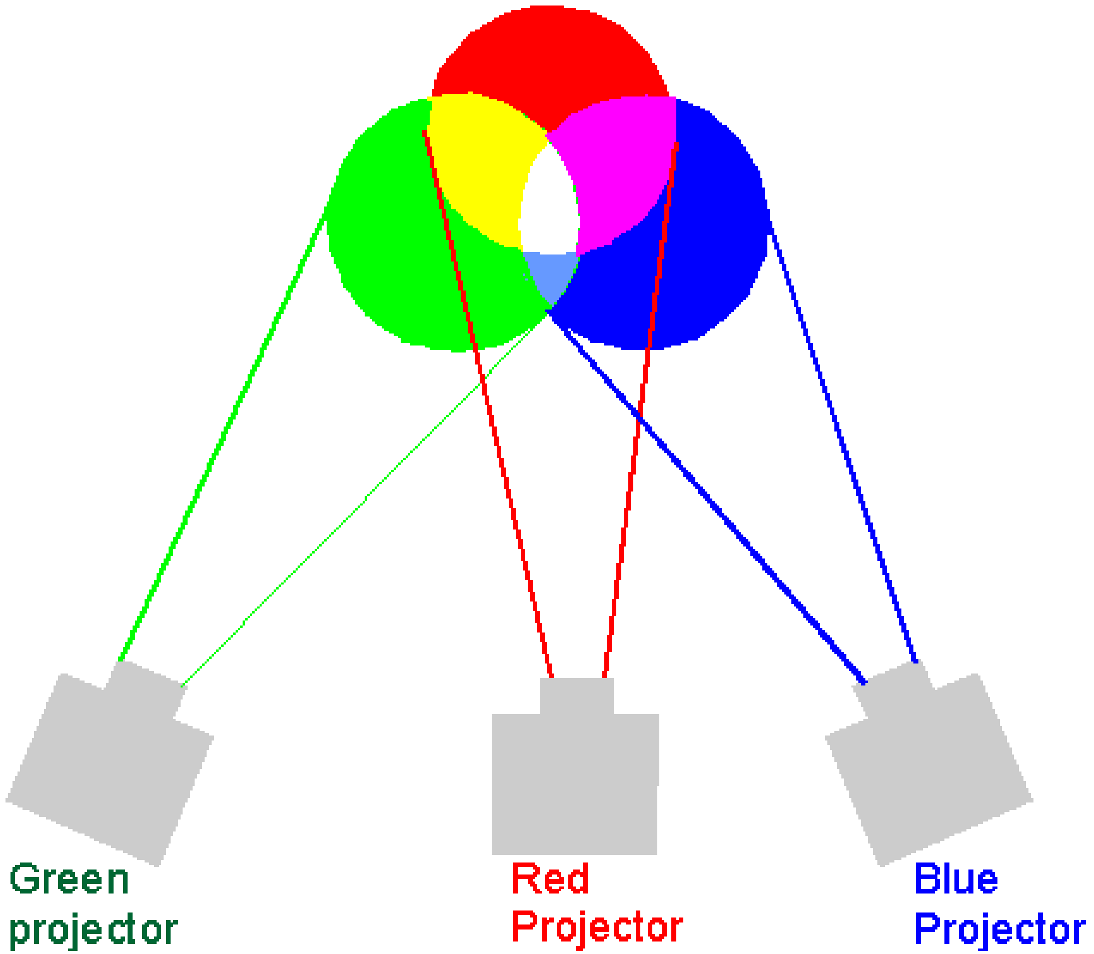
- **Color perception** is…
  - a highly complex and subjective process
  - trichromatic, recoded to black/white, red/green, yellow/blue
- **Luminance** is…
  - most important for recognizing shapes etc.
  - perceived logarithmically (brightness)
- **Lightness** and **color constancy** are achieved…
  - by adaptation and contrast perception
  - using high-level information (shape, illumination etc.)
- **Visualization should not assume that human visual system performs objective measurements**

Section 4.2:
Color Spaces

Subtractive Color Mixing

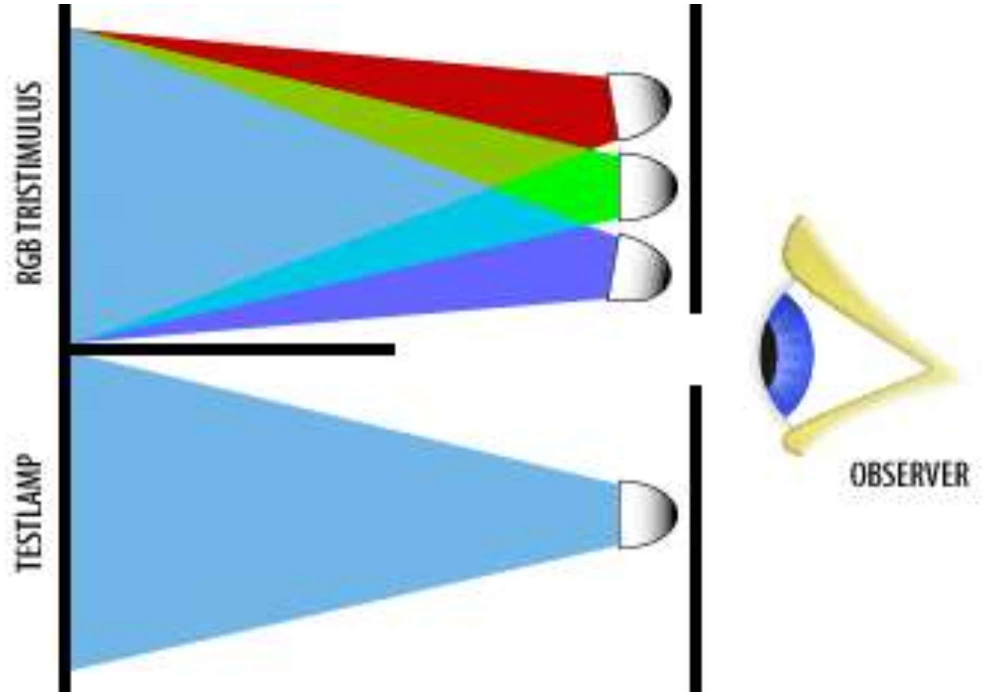


Additive Color Mixing



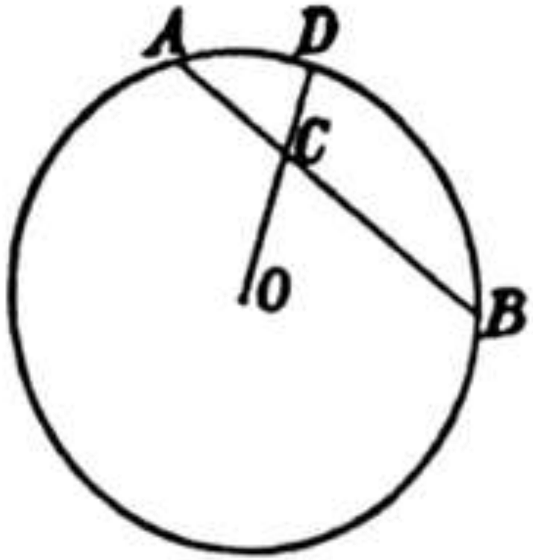
Primary Colors

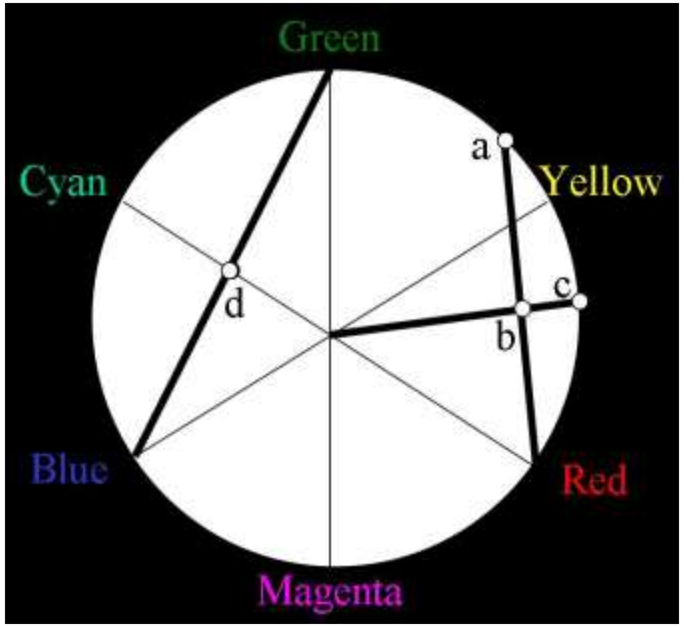
- It's possible to match any color by mixing three **primary colors**
  - Might require "negative light", i.e., adding one of the primaries to the reference
  - **Metamers** = physically different spectral distributions perceived as the same color
  - Typical primary colors (digital cameras, computer screens):
    - Red, Green, Blue



Grassman's Law

- Hermann Günther Graßmann (1809–1877):
  - "If two simple but non-complementary spectral colors be mixed with each other, they give rise to the color sensation which may be represented by a color in the spectrum lying between both and mixed with a certain quantity of white."



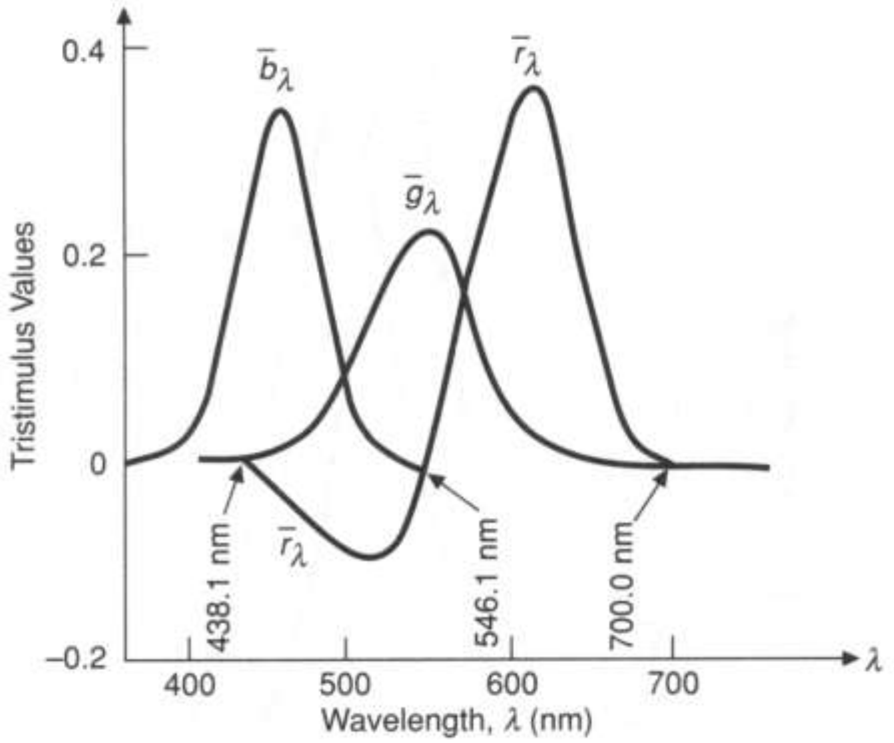


Changing Primaries

- Due to projection of the continuous spectrum onto three types of cones, color can be described as a **3D vector space**
  - Representing the same color using a **different set of primaries** amounts to **changing the basis**
  - Row *i* of M_{CB} given by coefficients needed to match new primary *i* using the old primaries

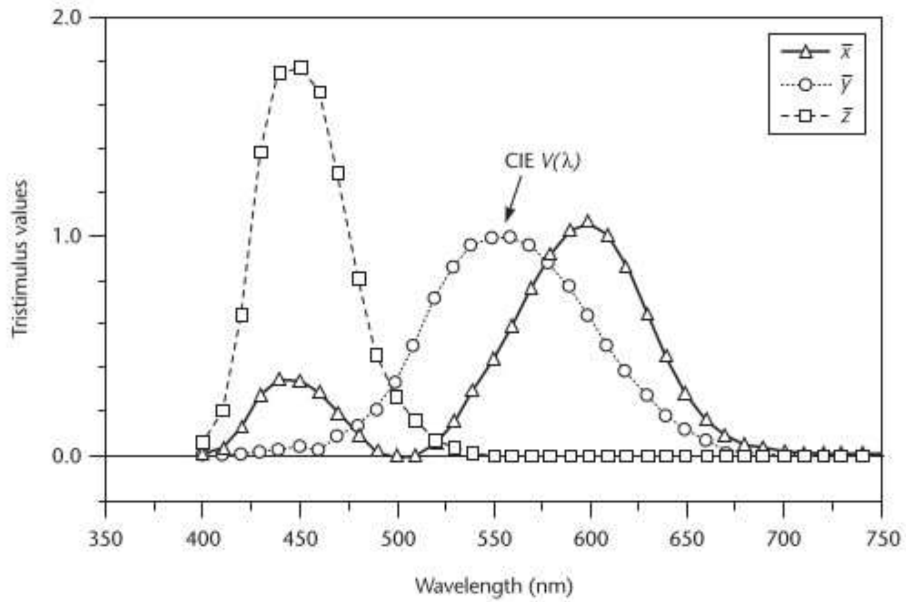
CIE 1931 Standard Observer

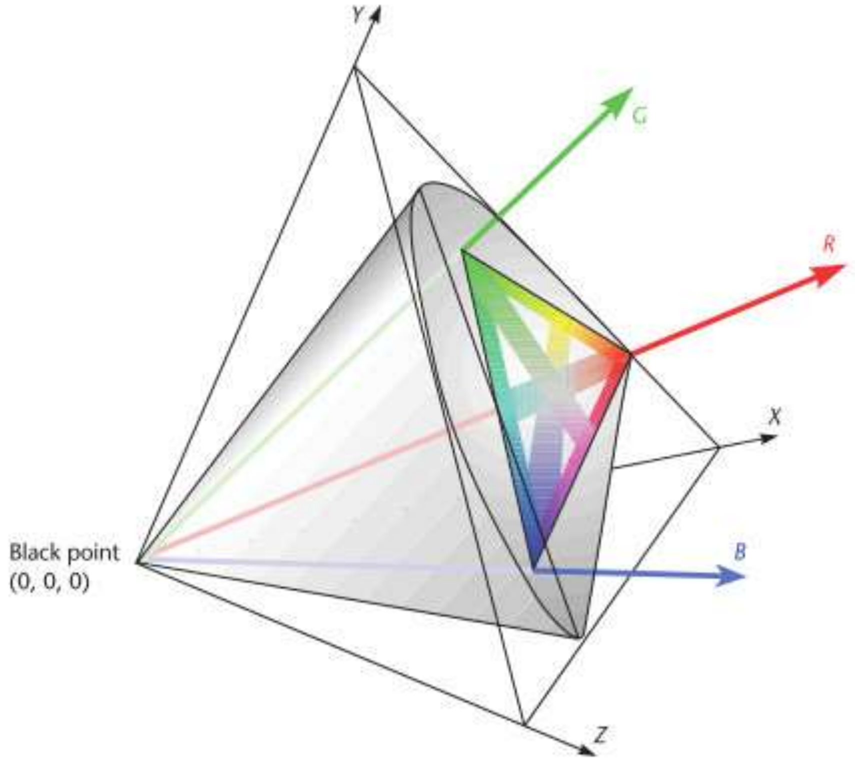
- In 1931, the CIE provided the **first mathematical definition** of a color space
  - Based on matching pure spectral colors with primaries using a **“standard observer”**
    - Group of 15-20 trained individuals
  - **Primaries:** R = 700 nm, G = 546 nm, B = 438 nm
  - Wavelengths around 500 nm require **negative contribution** of red
    - No triple of physically feasible primary colors lead to non-negative weights



CIE XYZ Color Space

- For the XYZ color space established by the 1931 standard, the CIE defined basis vectors which can reproduce any visible wavelength **without negative contributions**
  - "Imaginary" **XYZ primaries** do not correspond to any physical colors





Properties of XYZ Color Space

- **CIE XYZ** remains the reference for defining other color spaces
  - Y = luminance $V(\lambda)$
  - X/Z can be thought of as having zero luminance

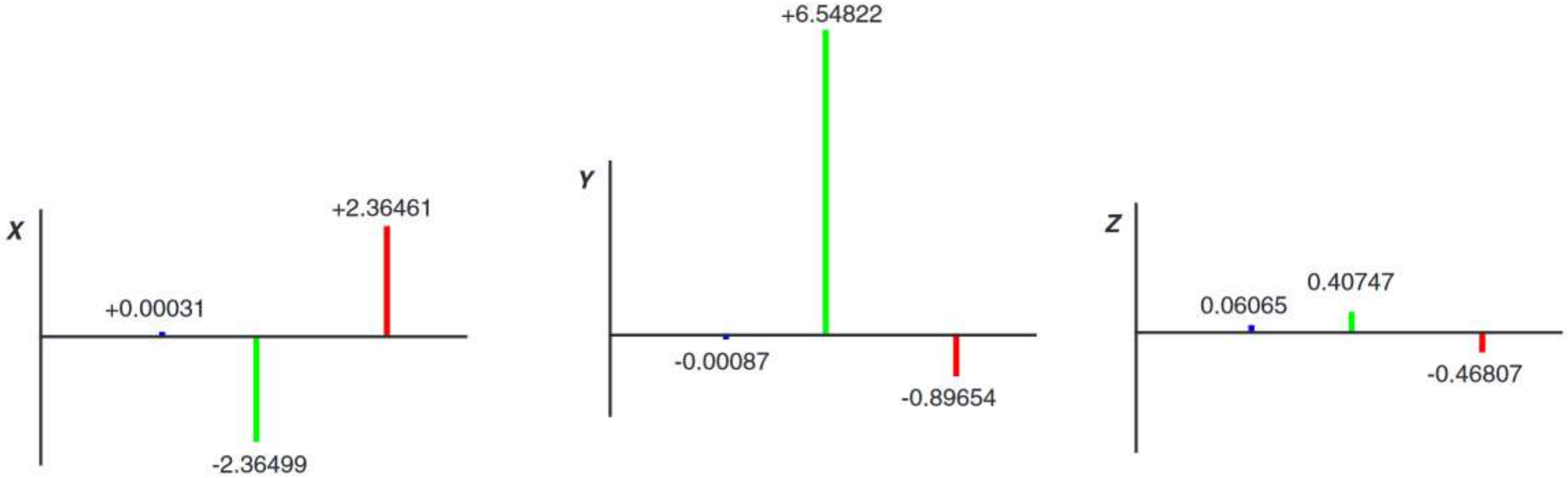
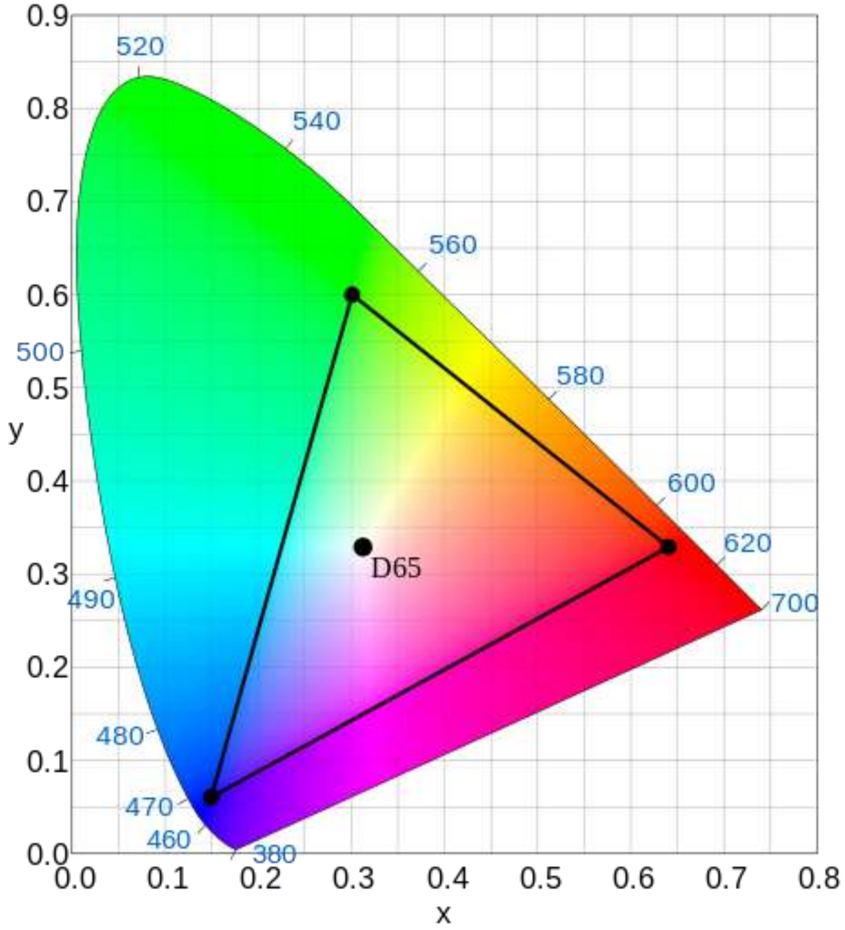


Image source: http://docs-hoffmann.de/ciexyz29082000.pdf

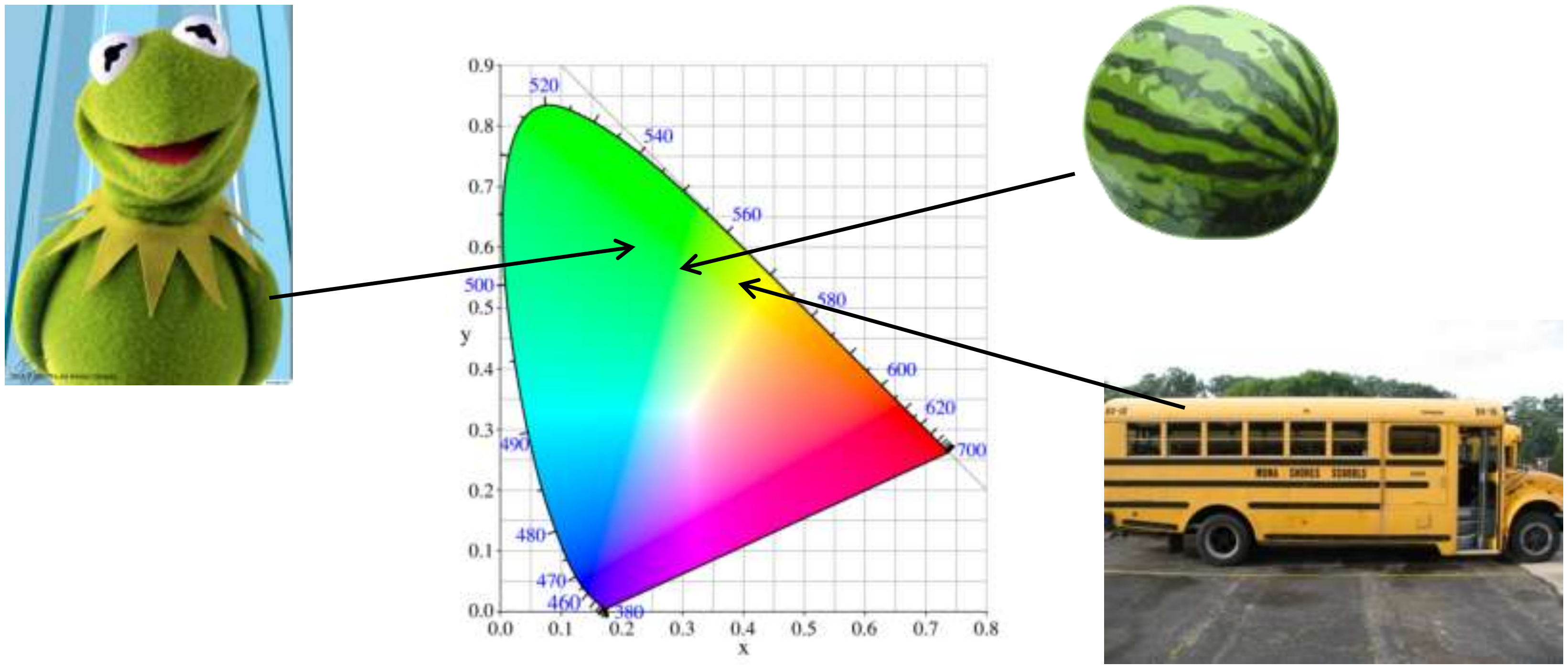
Chromaticity Coordinates

- **Chroma** is defined by normalizing by the overall amount of light (X+Y+Z)
  - $x = \frac{X}{X+Y+Z} \quad y = \frac{Y}{X+Y+Z}$
  - Unaffected by changes in luminance
  - Leads to **CIE chromaticity diagrams**
    - Spectral locus
    - Purple boundary
    - sRGB gamut
    - Complementary color: Reflect at whitepoint



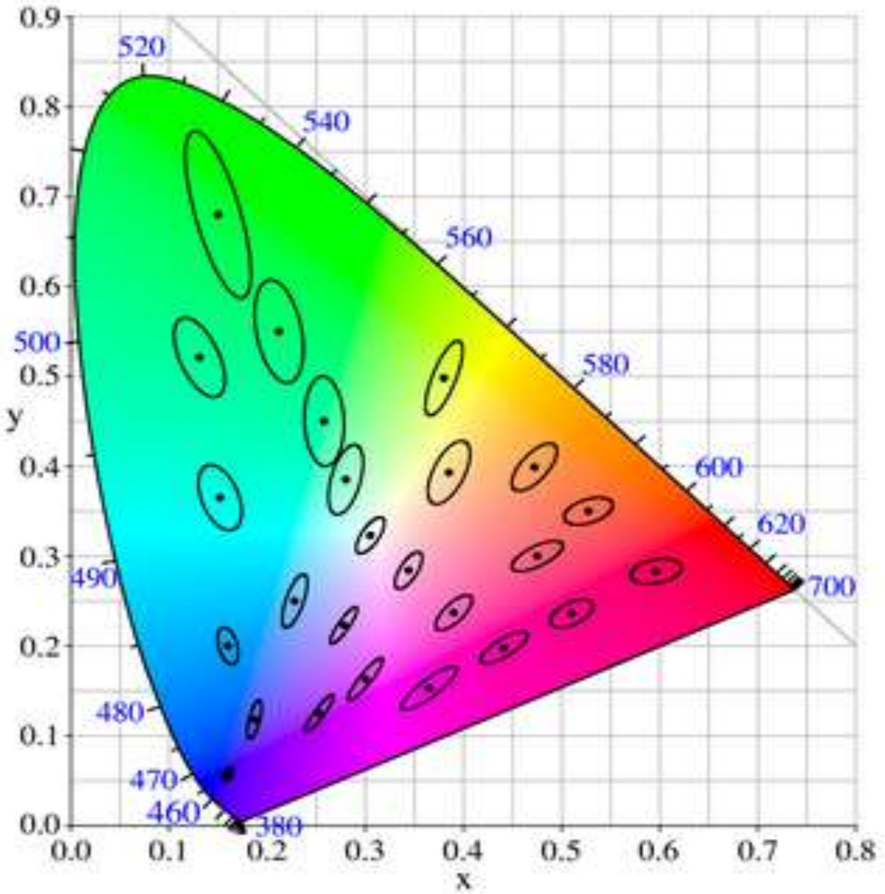
Perceptual Uniformity?

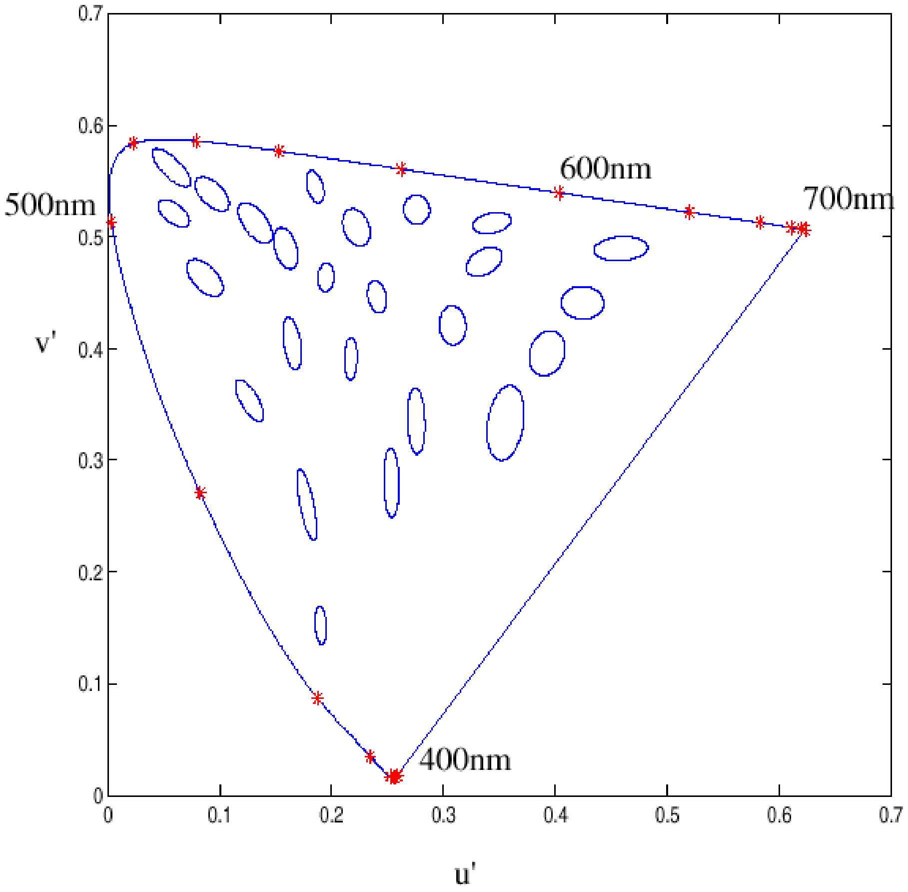
- Distances in chroma do not reflect perceptual differences
  - Problematic for quantization



Uniform Color Spaces

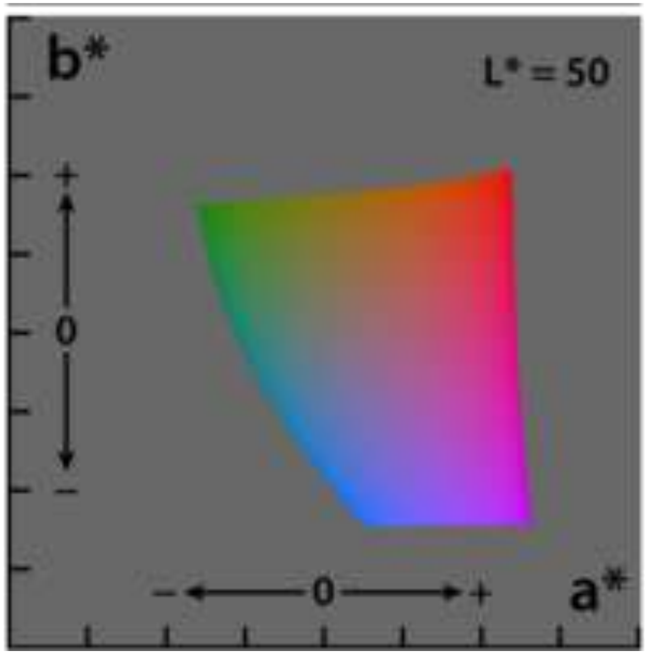
- **McAdam ellipses** illustrate just noticeable differences in color
- **Uniform color spaces** attempt to warp them to circles





CIELAB

- In 1978, CIE proposed two perceptually (more) uniform color spaces, CIELAB and CIELUV
  - Require reference white X_n, Y_n, Z_n
  - CIELAB: Axes roughly aligned with lightness (L*), red/green (a*), yellow/blue (b*) channels
    - $L^* = 116 f\left(\frac{Y}{Y_n}\right) - 16$
    - $a^* = 500\left[f\left(\frac{X}{X_n}\right) - f\left(\frac{Y}{Y_n}\right)\right]$
    - $b^* = 200\left[f\left(\frac{Y}{Y_n}\right) - f\left(\frac{Z}{Z_n}\right)\right]$
    - $f(t) = \begin{cases} \sqrt[3]{t} & \text{if } t > \left(\frac{6}{29}\right)^3 \\ \frac{1}{3}\left(\frac{29}{6}\right)^2 t + \frac{4}{29} & \text{else} \end{cases}$



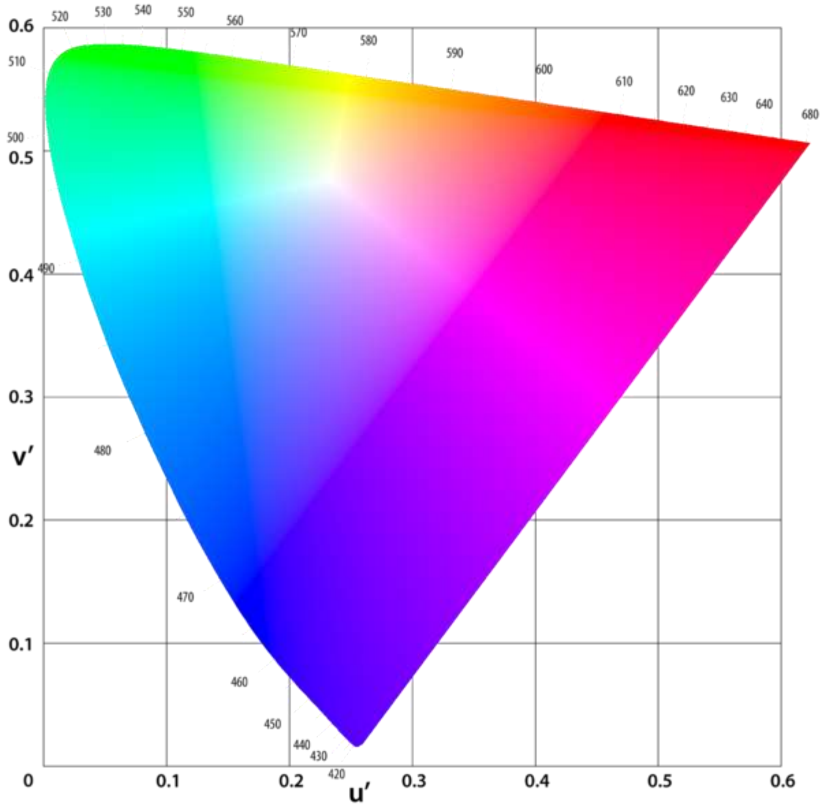
CIELUV

- In u^* and v^* of CIELUV color space, reference chroma is subtracted rather than divided out; CIE recommends it for (additive) color display
  - $u' = \dfrac{4X}{X+15Y+3Z} \quad v' = \dfrac{9Y}{X+15Y+3Z}$
  - To account for the fact that it becomes harder to perceive differences in color at lower lightness:

    L^* as before

    $u^* = 13L^*(u' - u'_n)$

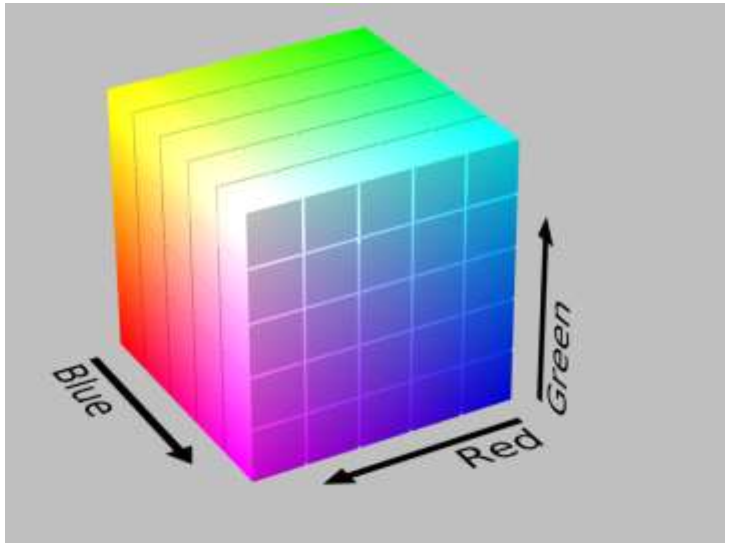
    $v^* = 13L^*(v' - v'_n)$
  - Refined color difference equations in 1994 and 2000, outside our scope

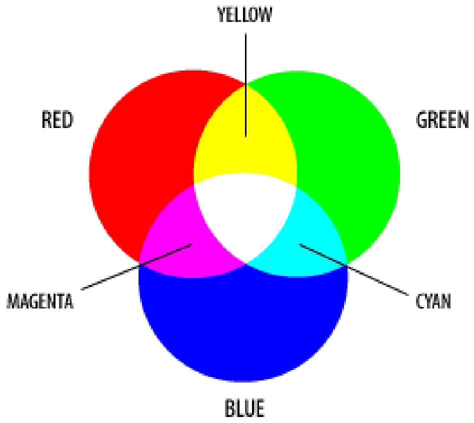


Uniform Chromaticity Scale

RGB Color Space

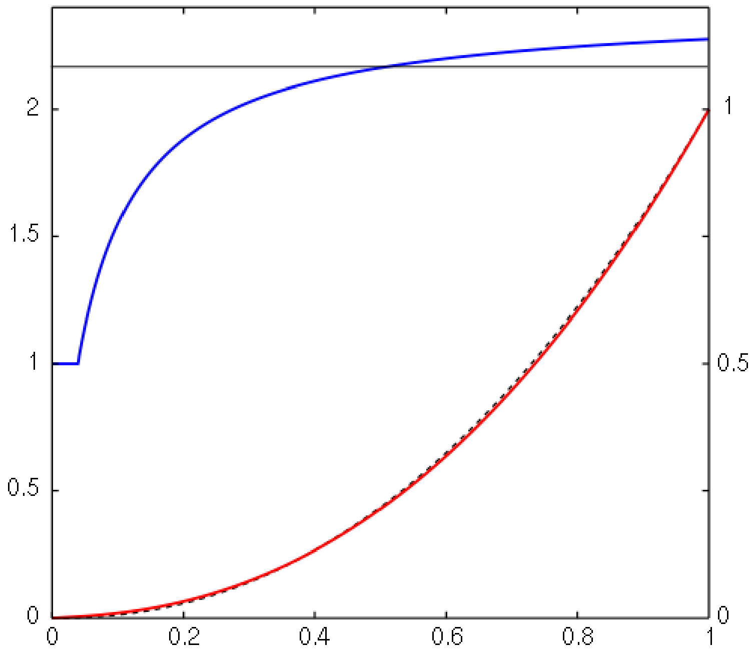
- **RGB:** Natural color space for devices that additively mix red, green, blue primary colors
  - Coefficients denote intensities of three primaries
    - Often either scaled to [0,1] or to [0,255]
  - Axes not aligned with opponent colors
  - Not perceptually uniform
  - Device-specific





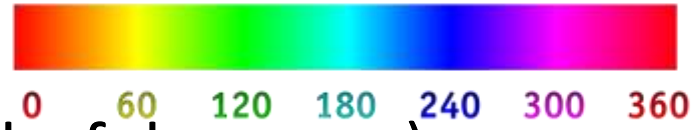
sRGB Color Space

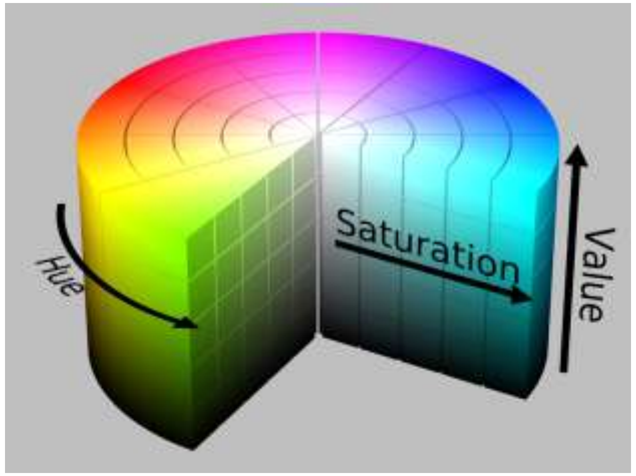
- **sRGB:** Device-independent color space
  - Created by HP and Microsoft in 1996
  - Optimized for monitors, printers, internet
  - Fixes primary colors and a non-linear mapping from coordinates to intensities
    - Efficient use of quantized coefficients
    - Red line: Intensity vs. values
      - Dashed black line (below): gamma=2.2
    - Blue line: Effective gamma
  - Many modern devices map sRGB to device-specific RGB

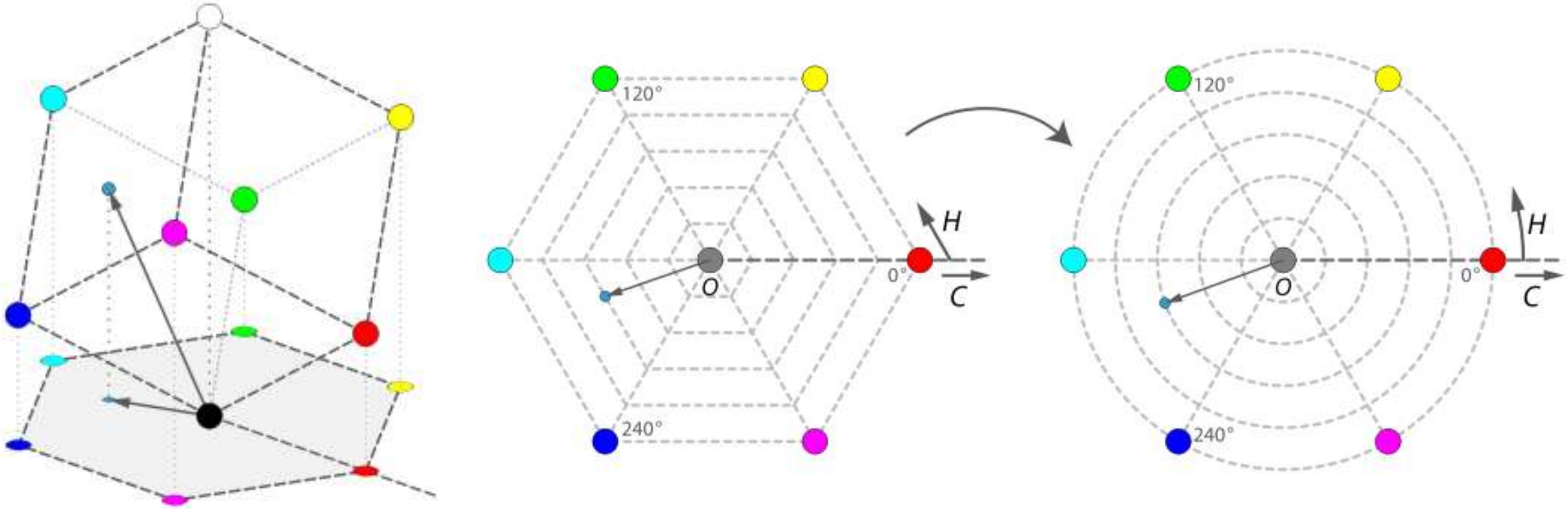


HSV Color Space

- **HSV:** More intuitive for color specification
  - Hue
  - Saturation (colorful vs. gray)
  - Value (dark vs. bright)

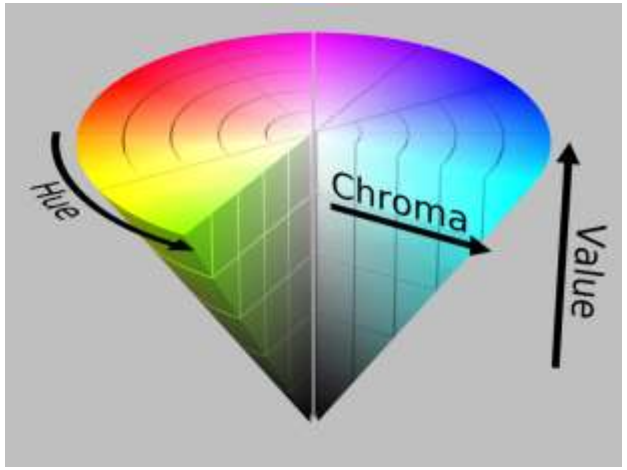


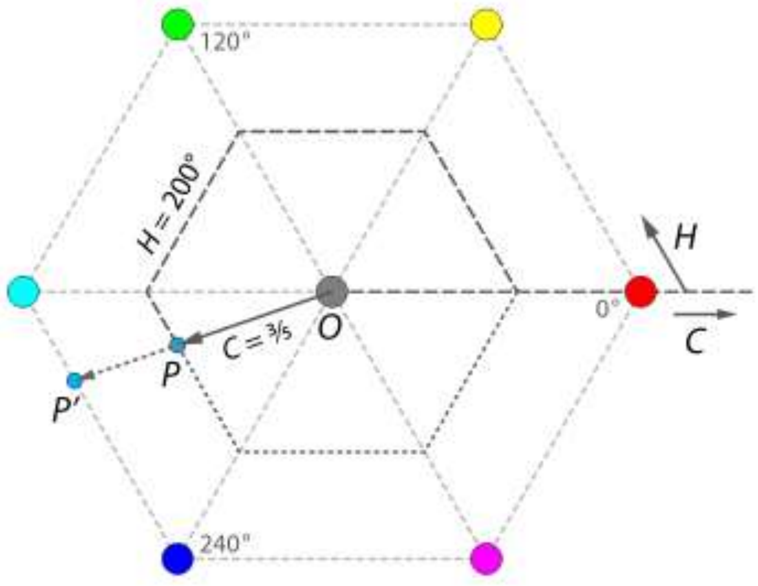




RGB to HSV

- How to convert RGB ∈ [0,1] to HSV:
  - Let M=max(R,G,B); m=min(R,G,B)
  - V=M
    - puts primary and secondary colors into plane with white
  - $S = 1 - \frac{m}{M} = \frac{C}{V}$ with C=M-m
  - $H = 60° \times \begin{cases} 0 & \text{if } C = 0 \\ \frac{G-B}{C} \bmod 6 & \text{if } \mathrm{M} = \mathrm{R} \\ \frac{B-R}{C} + 2 & \text{if } \mathrm{M} = \mathrm{G} \\ \frac{R-G}{C} + 4 & \text{if } \mathrm{M} = \mathrm{B} \end{cases}$





Perceptual Incorrectness of HSV

- Holding S in HSV constant only provides a **rough approximation** to perceptually exact equi-saturation

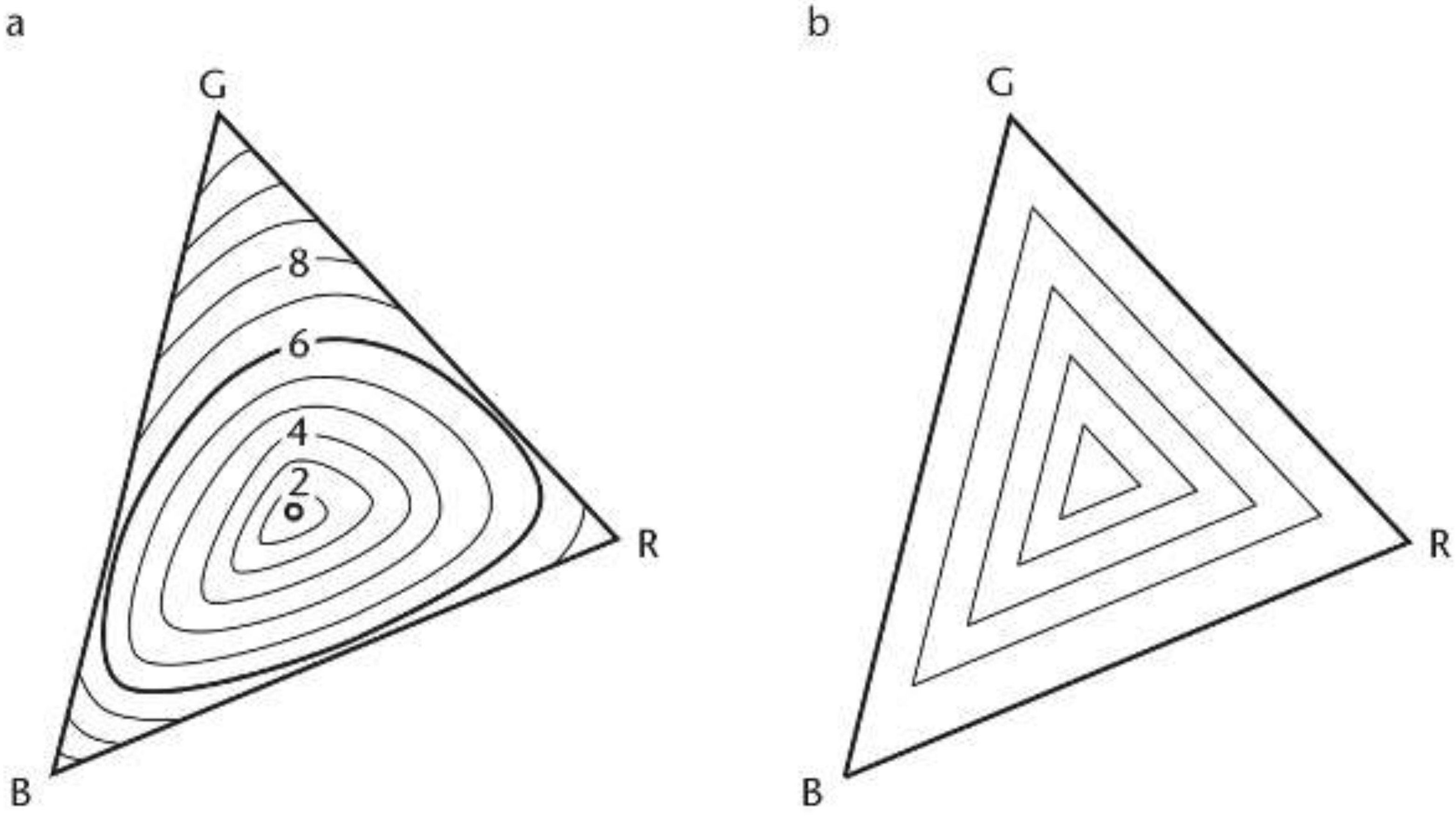
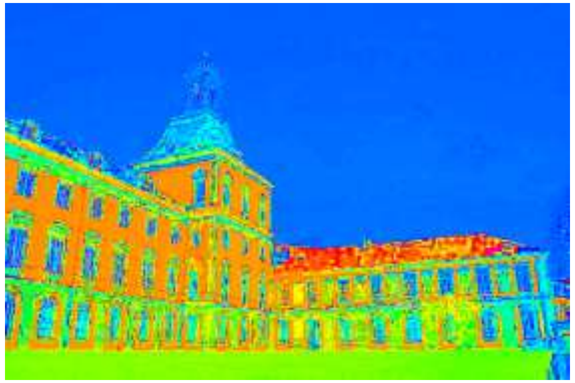


Illustration: RGB vs. HSV

Red

Green

Blue



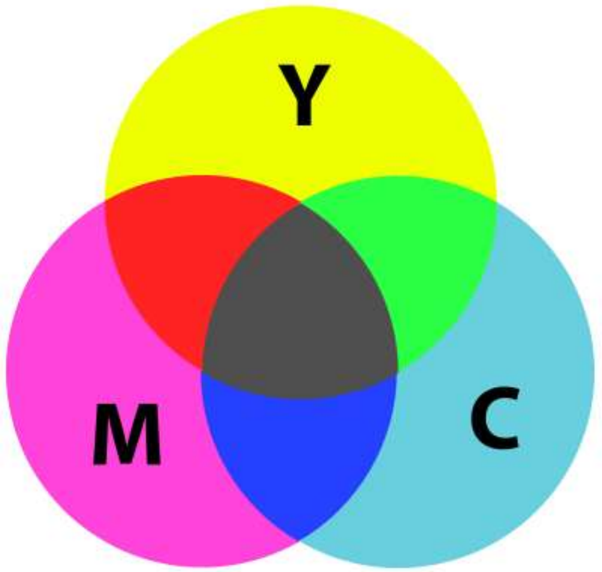
Hue

Saturation

Value

CMYK

- Primaries for subtractive color mixing (e.g., printing) are cyan, magenta, yellow
  - Red/green/blue obtained as secondary colors
  - Interactions between pigments make results of subtractive mixing much harder to predict than for additive mixing on computer screens
  - Black (K) added, since C+M+Y often does not produce a perfect black



Summary: Quantifying Color

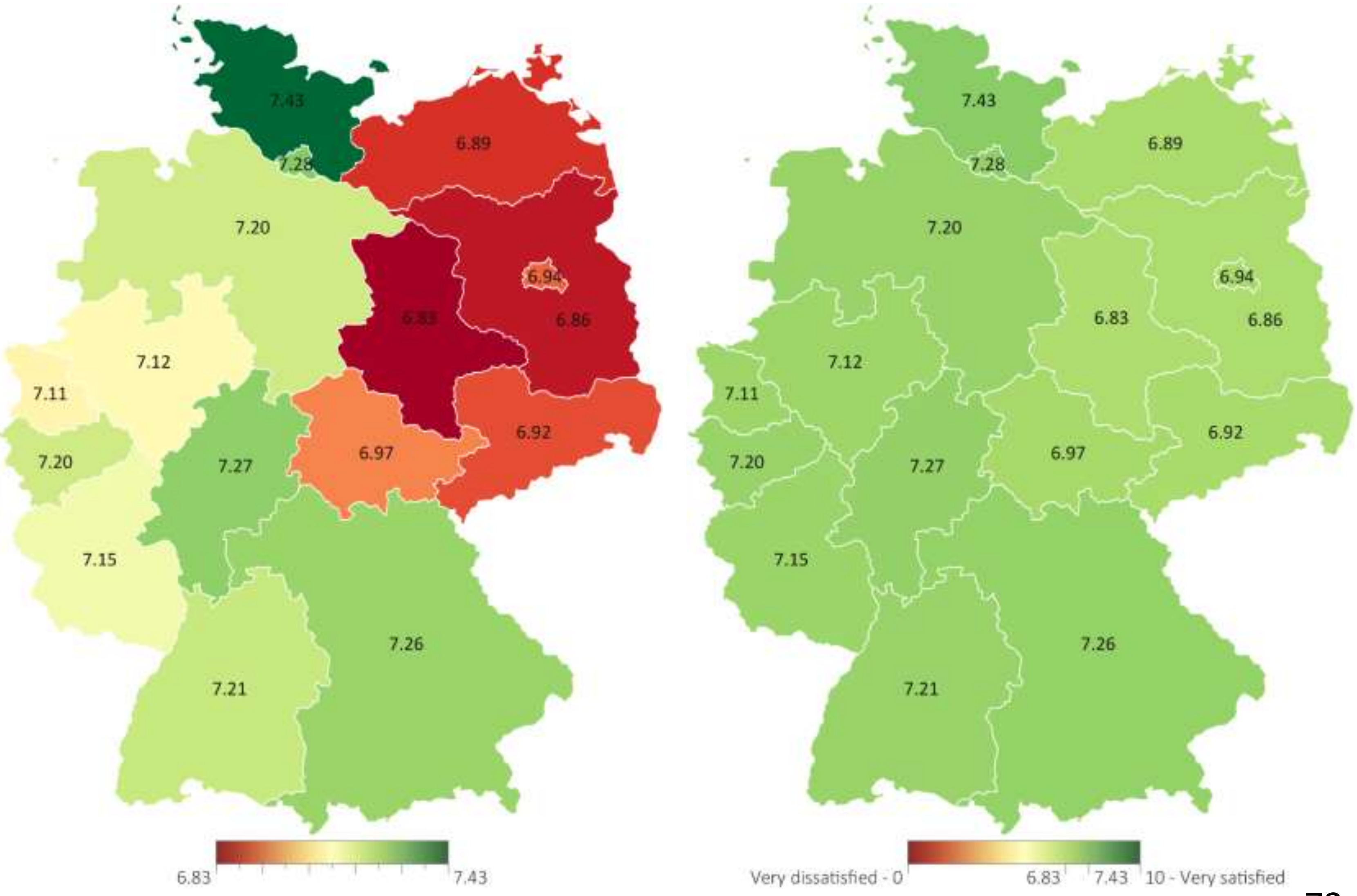
- Any color can be matched using three **primary colors**
  - Can treat color as a vector space
- Different **color spaces** for different use cases:
  - CIE XYZ, sRGB
    - Standardized reference spaces
  - CIELAB, CIELUV
    - Approximate perceptual uniformity
  - HSV
    - Navigating / selecting colors
  - RGB, CMYK
    - Device-specific color spaces

Section 4.3:
Color for Visualization

Impact of Color Maps on Interpretation

The choice of color can guide (or bias) data interpretation

- *Example*: Survey of life satisfaction in Germany



Figures from *Tominski/Schumann: "Interactive Visual Data Analysis". AK Peters 2020*

Illustration: Effects of Different Color Maps

All these examples show exactly the same data:

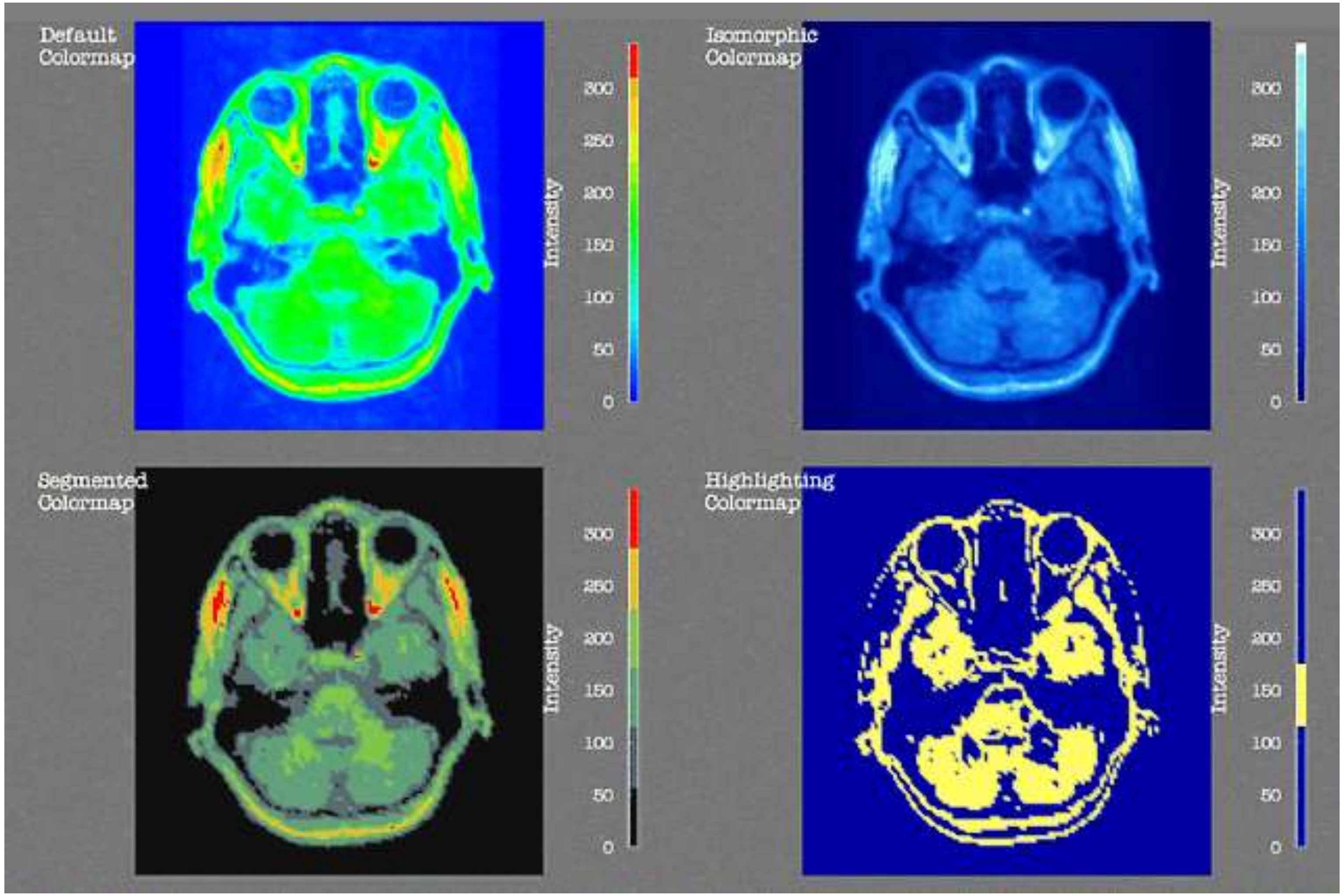


Image Source: Rogowitz et al.

Pick Your Colors Wisely!

In this case, unnecessary surgery was performed based on a poorly adjusted color map:

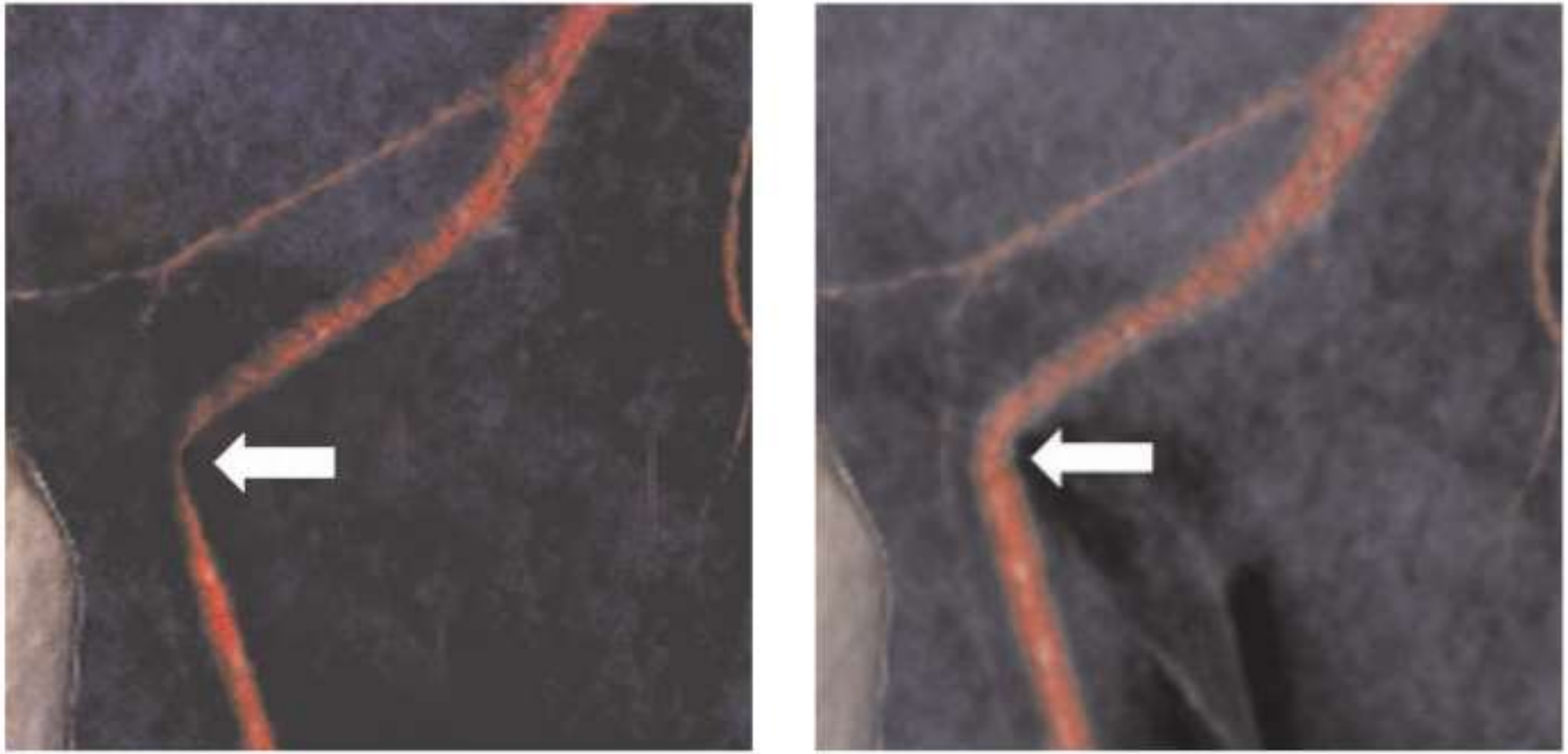


Image Source: Lundström et al.

Scales of Measurement

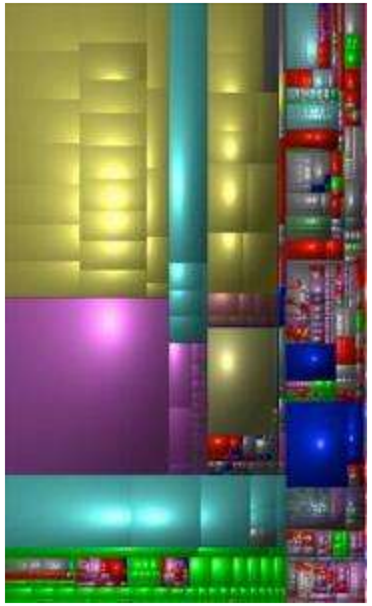
- Four **scales of measurement** according to Stanley Smith Stevens

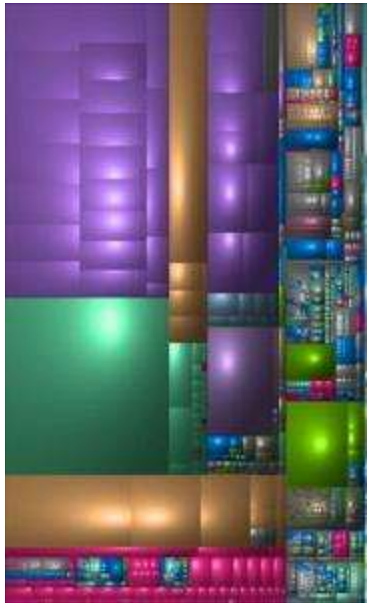
Operation	Nominal	Ordinal	Interval	Ratio
$=$ / $\neq$	Yes	Yes	Yes	Yes
$>$ / $<$	No	Yes	Yes	Yes
$+$ / $-$	No	No	Yes	Yes
$\times$ / $\div$	No	No	No	Yes

Scales of Measurement and Color

- **Principle:** Visualization *V(D)* of data *D* should reflect intrinsic symmetries of *D*
  - *Example:* Color coding nominal values should be symmetric under permutation

Example by Gordon Kindlmann

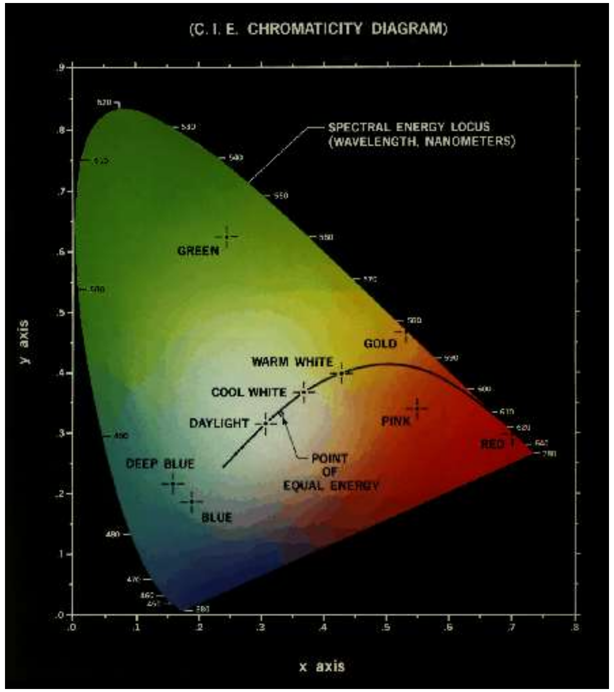


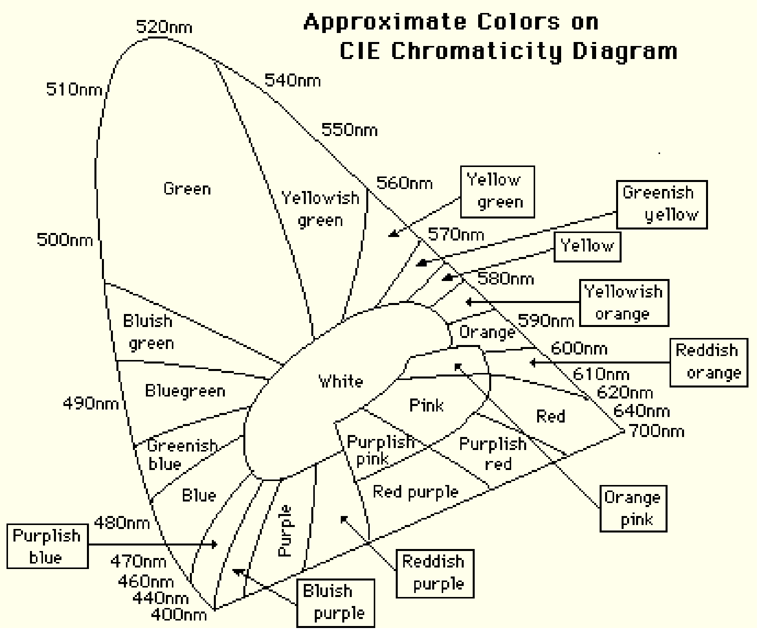


Color Mapping Nominal Data

Around 9 basic colors are reliably distinguished:

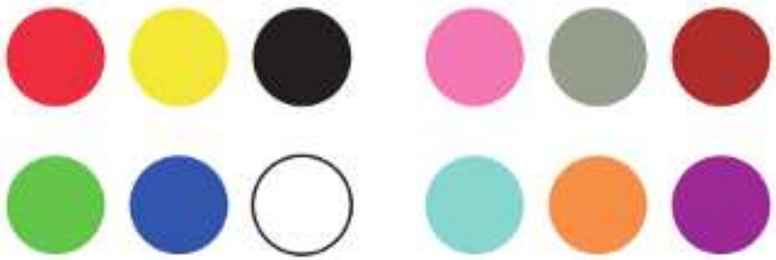
- Magenta (430 nm)
- Blue (476 nm)
- Blue-green (504 nm)
- Green (515 nm)
- Yellow green (556 nm)
- Yellow (582 nm)
- Orange (596 nm)
- Reddish orange (610 nm)
- Red (642 nm)

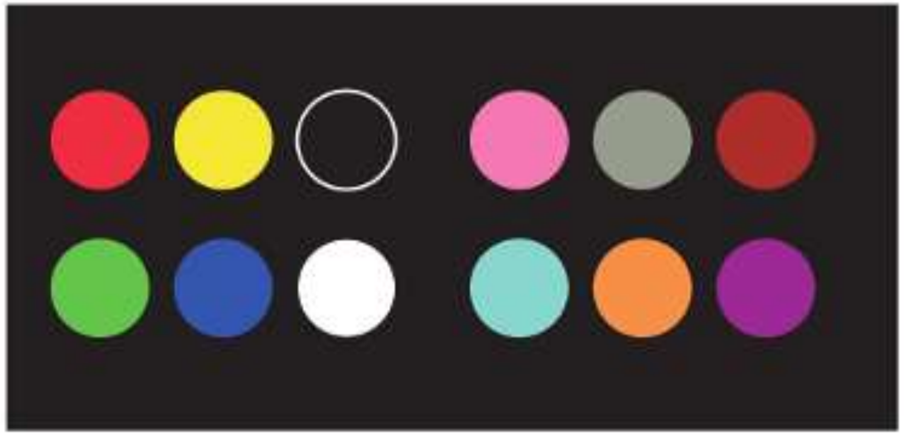




Use Opponent Colors First

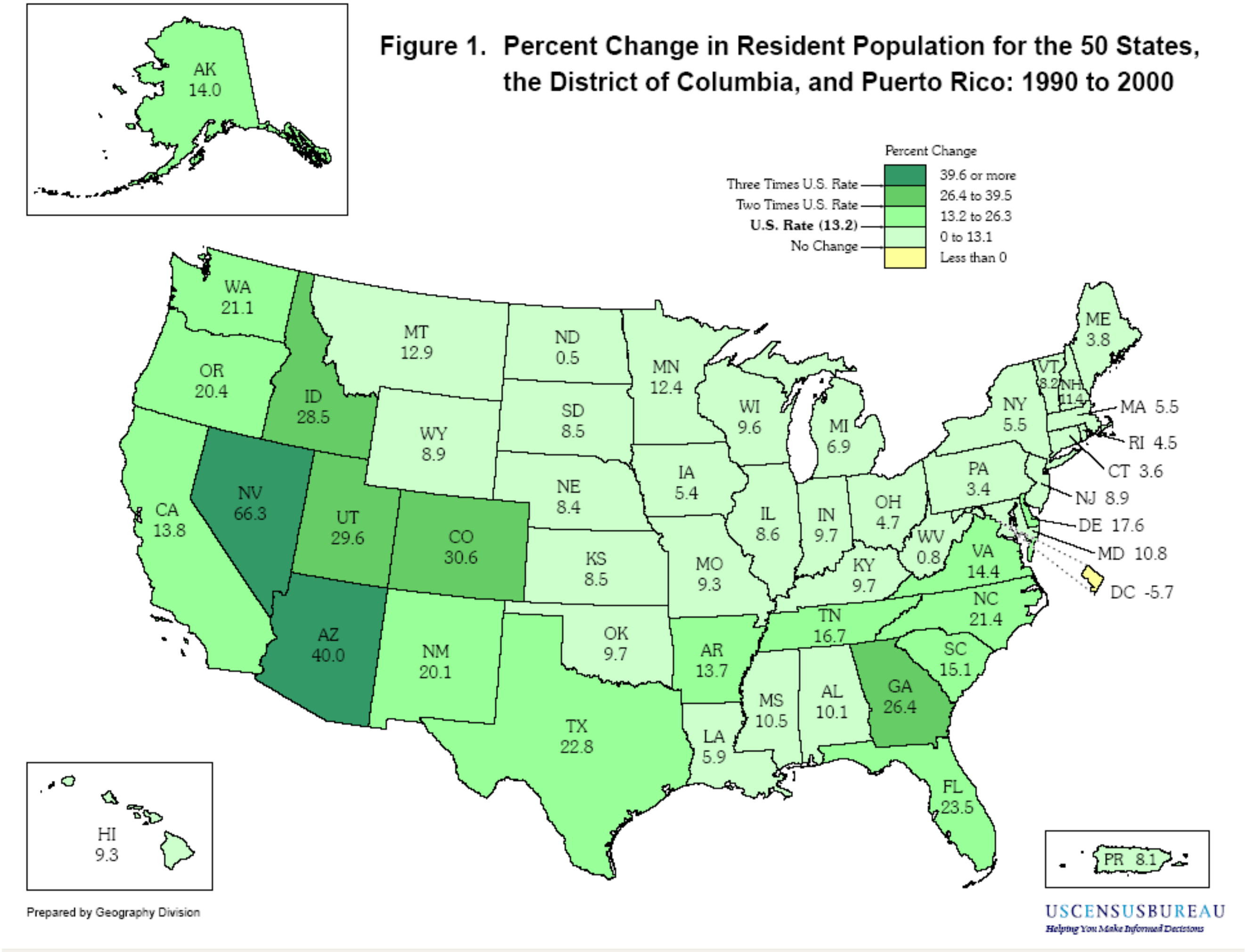
- For labeling nominal data, the six **opponent colors** are a natural choice
  - In deuteranomaly, ability to distinguish between red and green is preserved if color contrast and luminance are sufficient
  - Pink / cyan / gray / orange / brown / purple are recommended by Ware in case more than six colors are needed





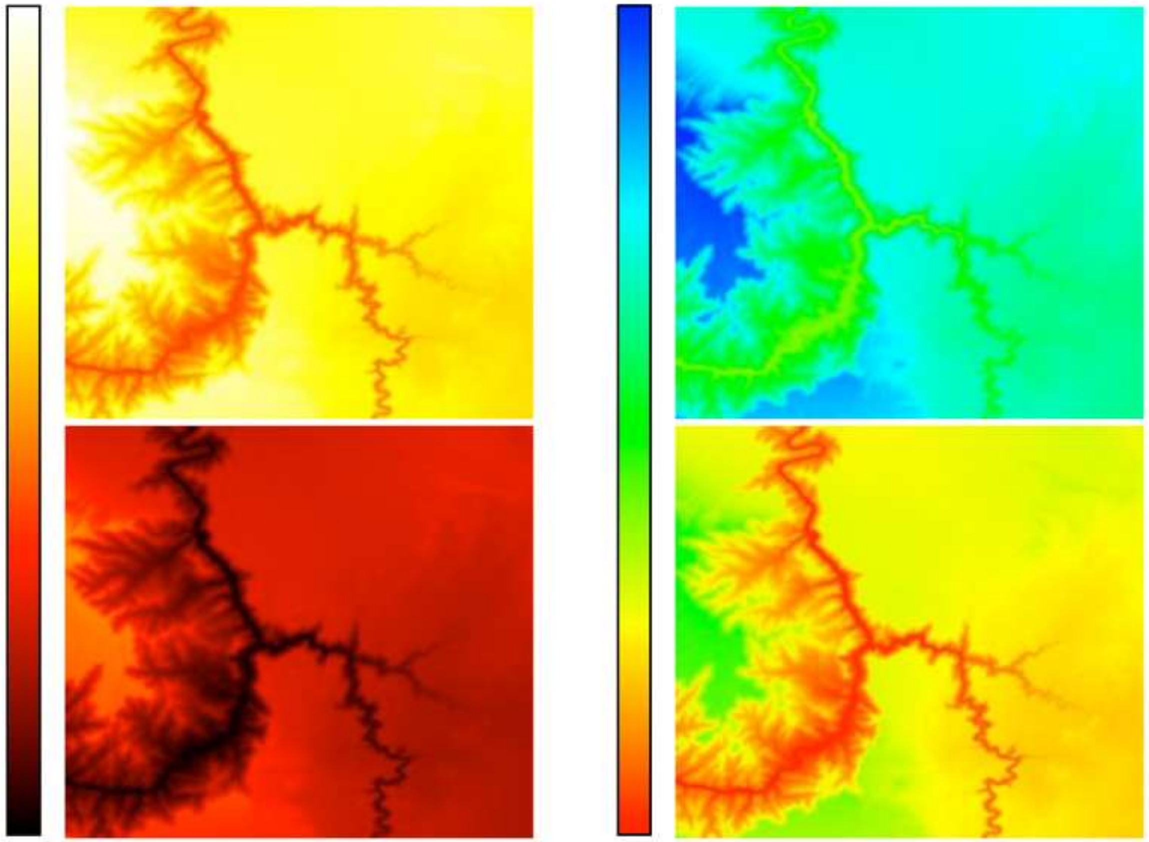
Color Mapping Ordinal Data

- Color Map needs to clearly suggest an ordering



Color Mapping Interval Data

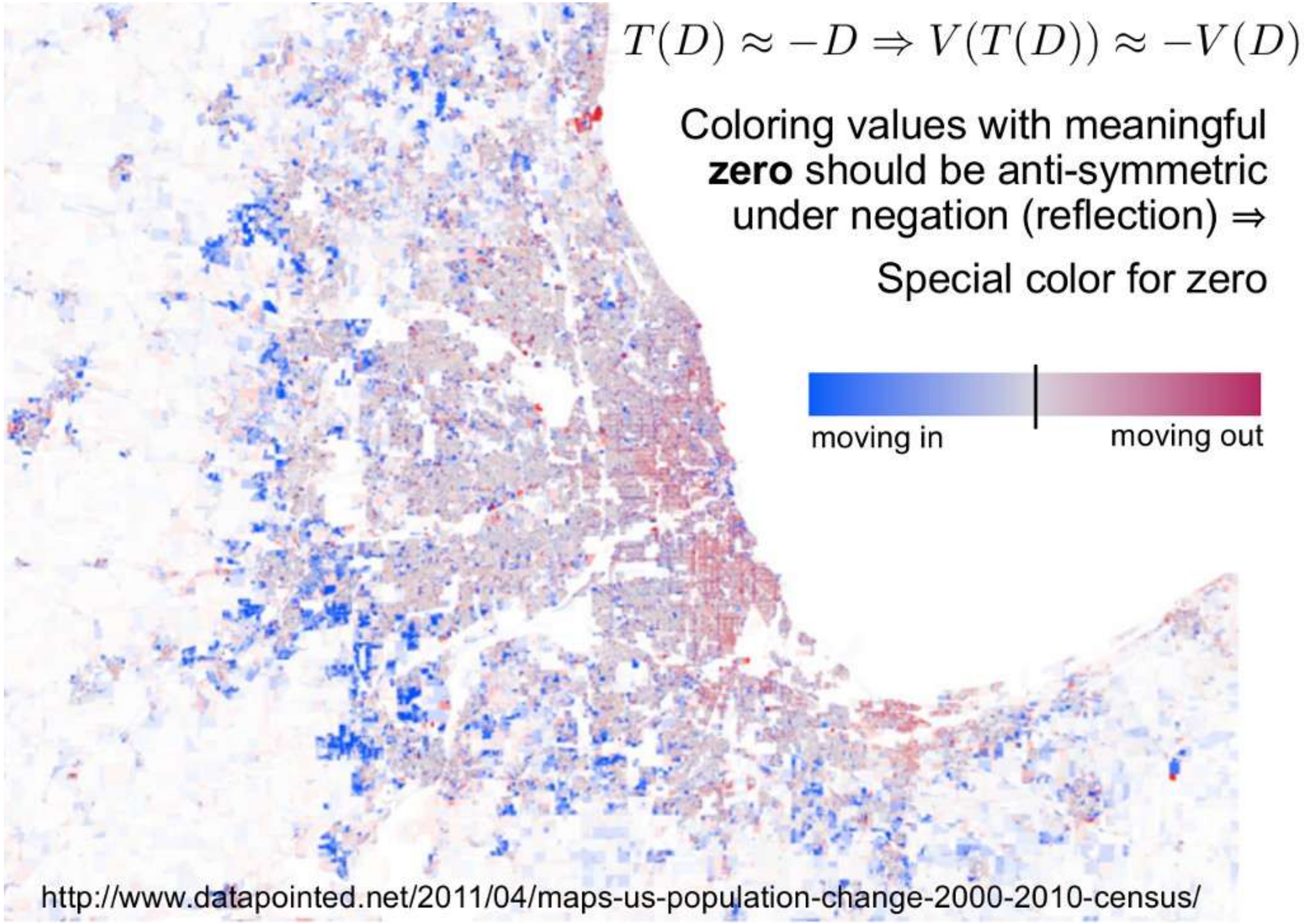
- Coloring of values on an interval scale should still show similar contrast after adding a constant



$V(D + c) \approx V(D)$ $V(D + c) \neq V(D)$

Example by Gordon Kindlmann

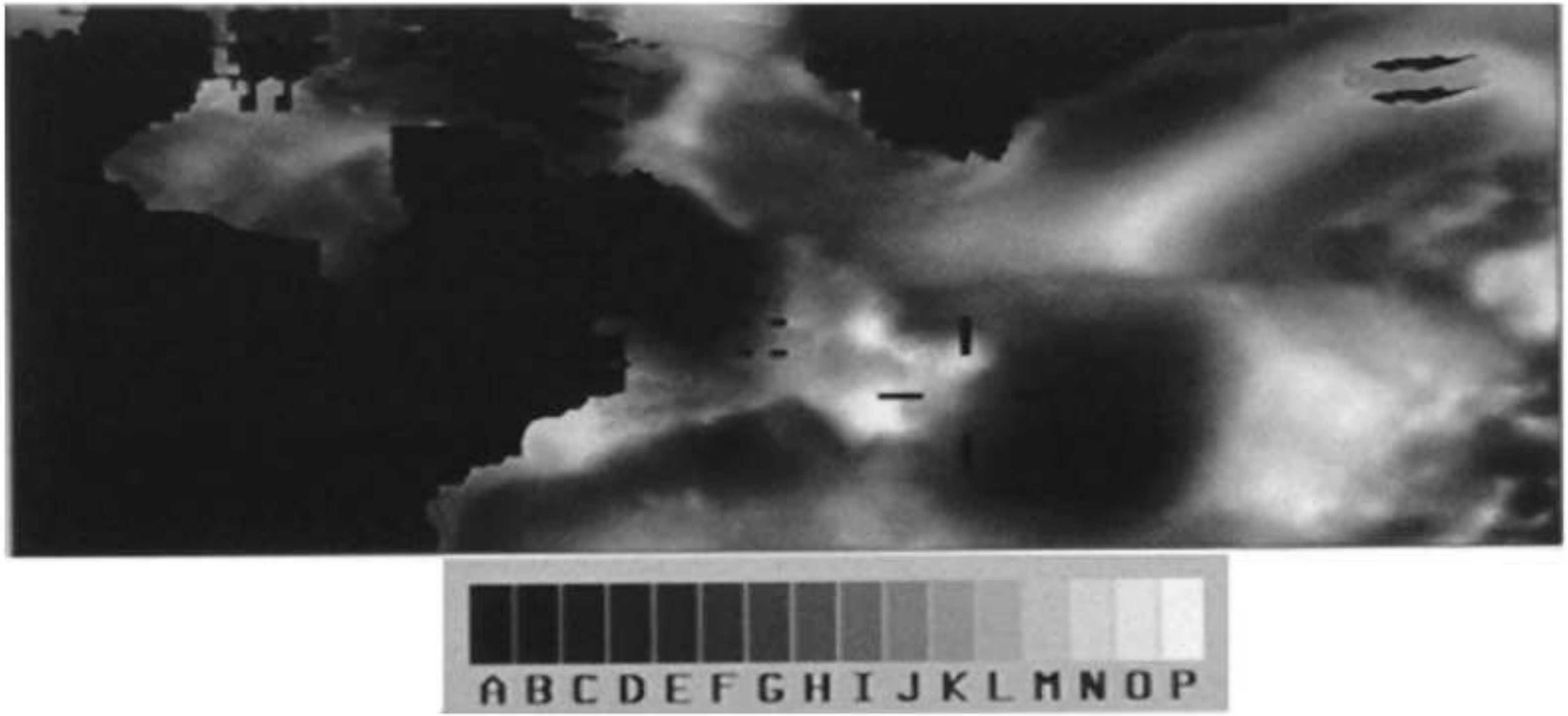
Color Mapping Ratio Data



Example by Gordon Kindlmann

Caveat: Grayscale for Color Coding

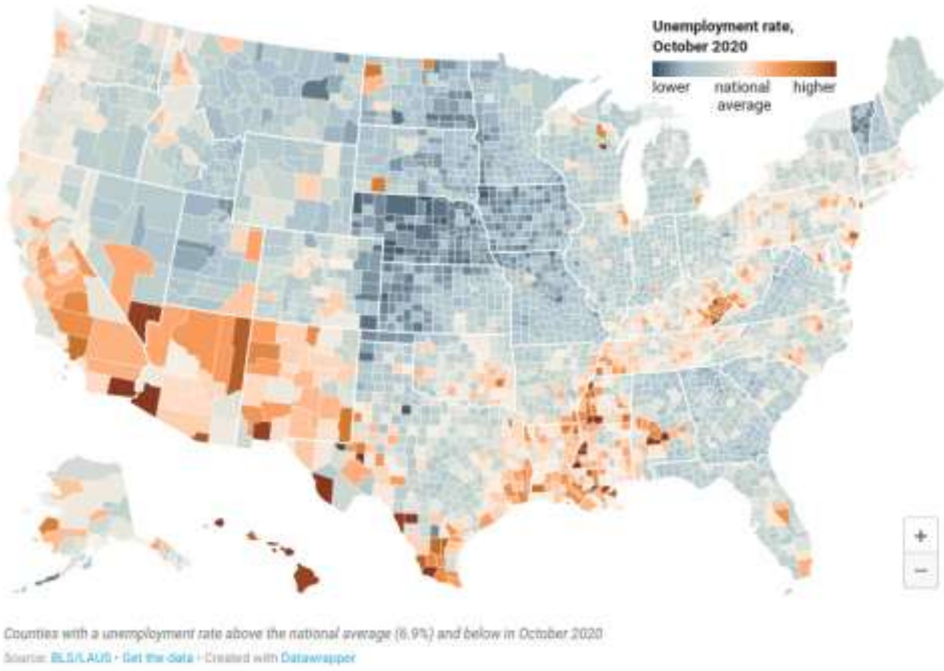
- Due to **contrast effects,** large readoff errors (20% of range) occur when using grayscale to encode continuous quantities

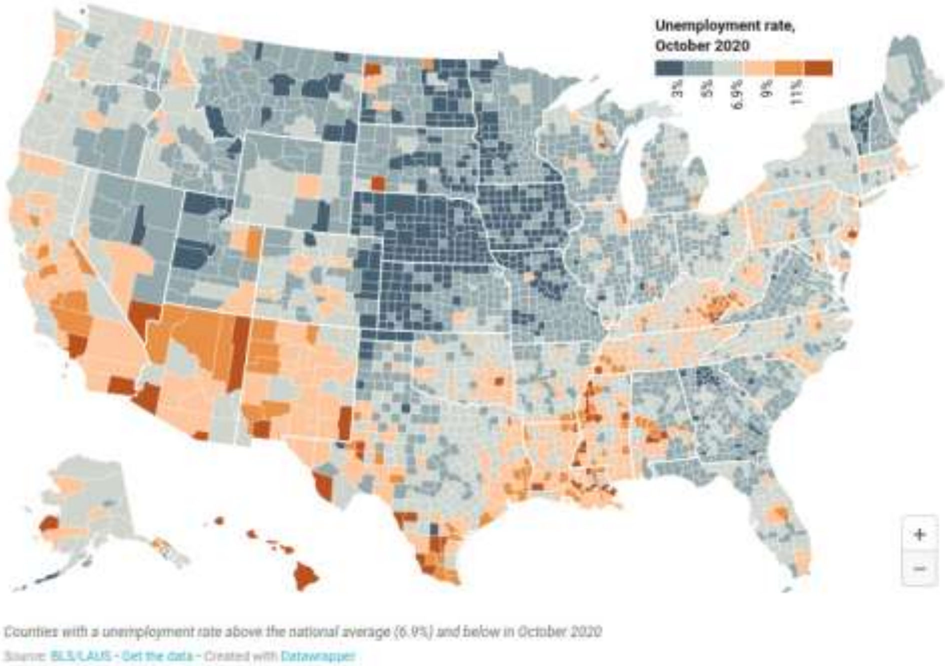


Discretizing Color Maps

Discretizing (“binning”) continuous quantities for color mapping

+ Makes it easier to read off values
+ Clearly communicates statistical breaks (e.g., above/below average)
- Prevents comparisons within regions of similar values
- Does not distinguish between smooth and abrupt transitions



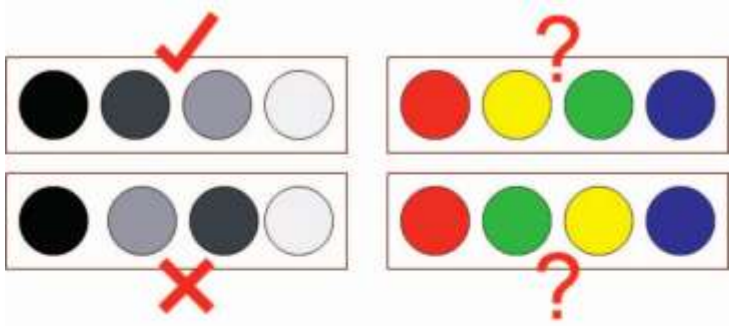


Images from https://blog.datawrapper.de/classed-vs-unclassed-color-scales/

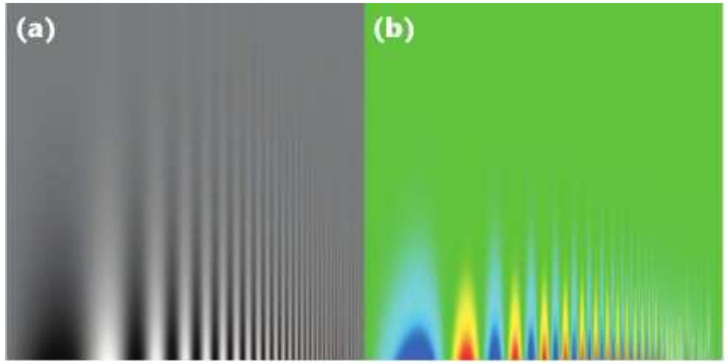
Caveats: Rainbow Color Map

Default color map in many visualization systems, eye-catching, improves read-off accuracy, **but…**

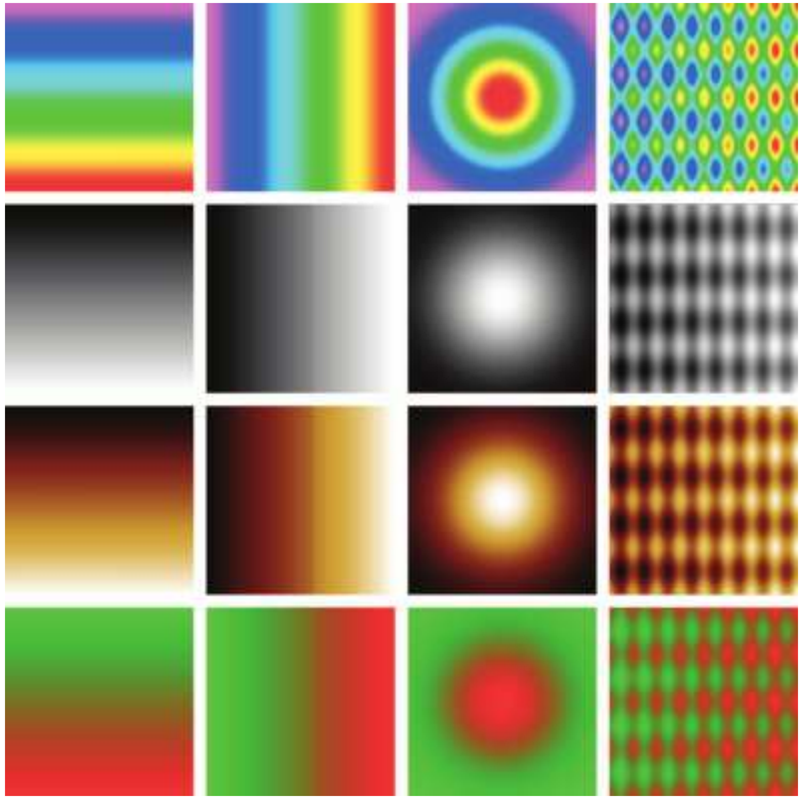
1. No intuitive ordering:



2. Can reduce sensivity:



3. Artifactual gradients:

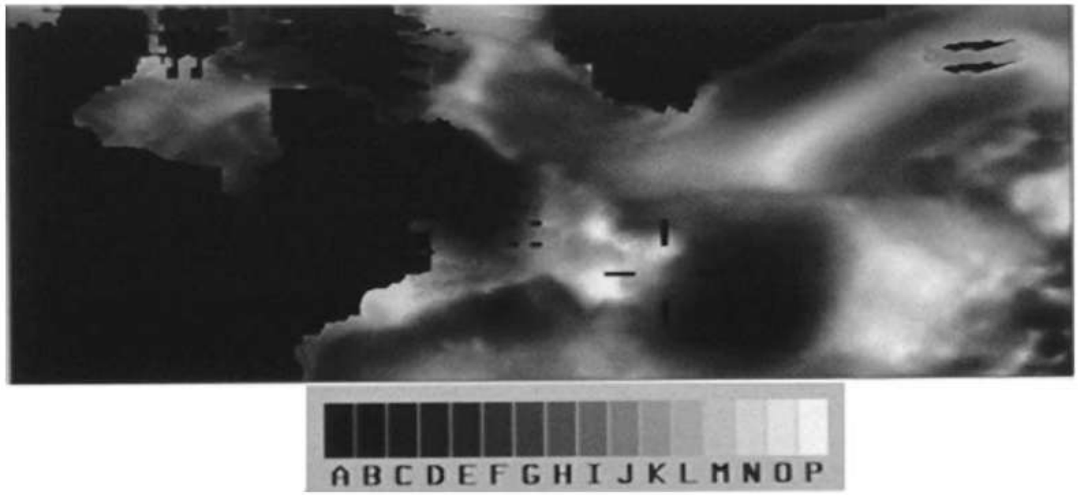


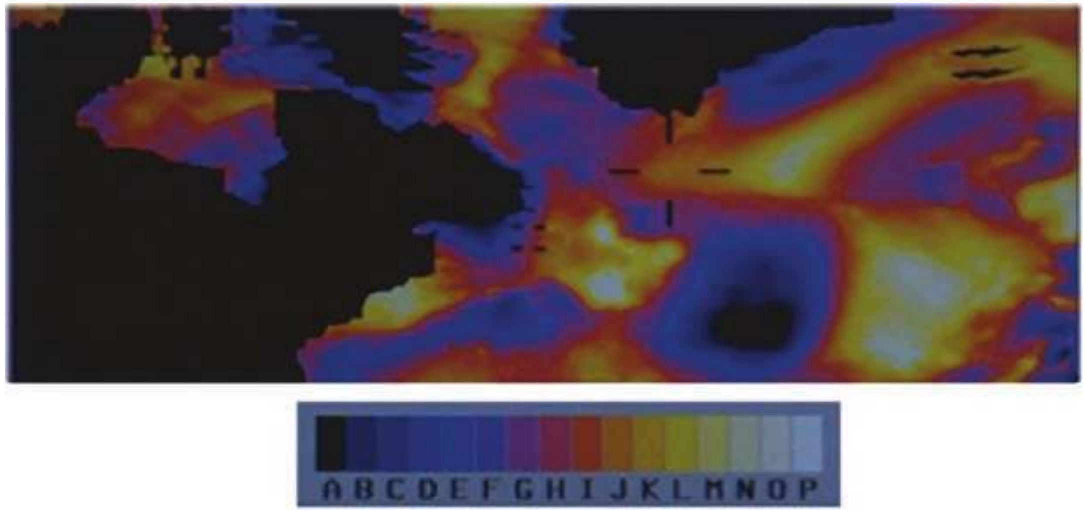
4. Relies on red-green vision

Reference: Borland/Taylor, Rainbow Color Map (Still) Considered Harmful, CG&A 2007

Guidelines for Color Map Design

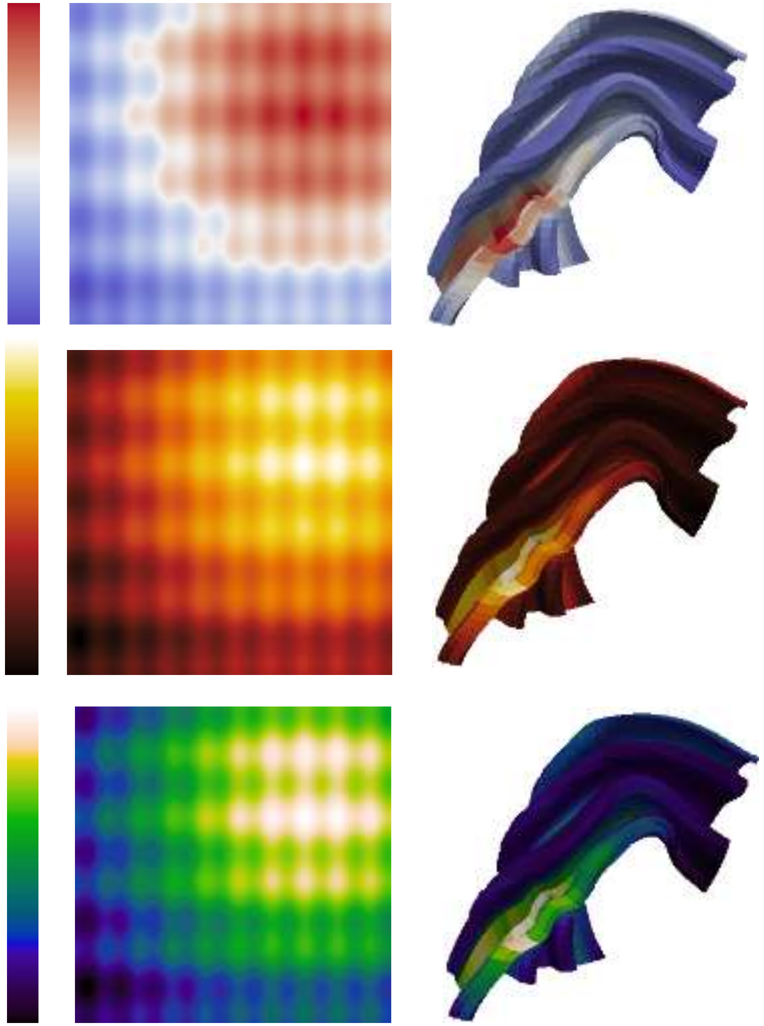
- **Good general strategy:** Combine changes in hue with monotonically increasing luminance
  - Avoid combining red and green
  - Enforce perceptual uniformity
  - Supports reading off values more precisely
  - Still maintains an intuitive impression of high and low values





Choosing Color Maps

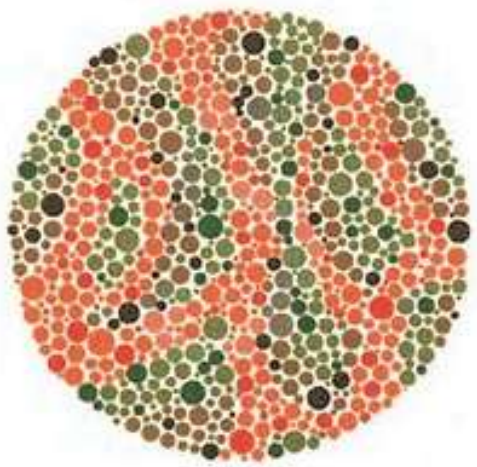
- Choice depends on
  - Whether to highlight “neutral” (zero) value
  - Use in 2D or on surface (avoid dark colors!)
  - Aesthetics / personal preference
- Visualization packages (paraview, matplotlib) increasingly offer good choices



Examples from Kenneth Moreland, http://www.kennethmoreland.com/color-advice/

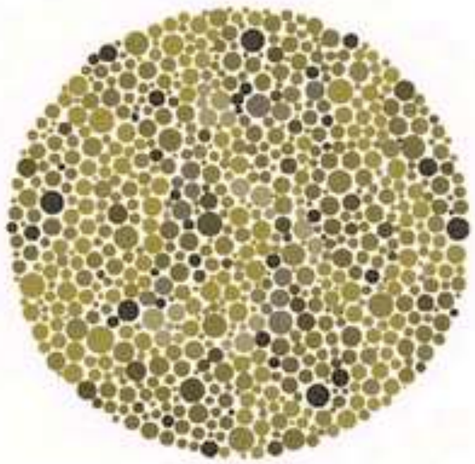
Keep Color Impairment in Mind

- **Color impairment** (especially red-green weakness) affects a substantial part of the population!
  - Tools for simulating color blindness and improving color accessibility available online, e.g.
    http://www.color-blindness.com/coblis-color-blindness-simulator/
    http://www.daltonize.org/



Intitial Image

Corrected Image

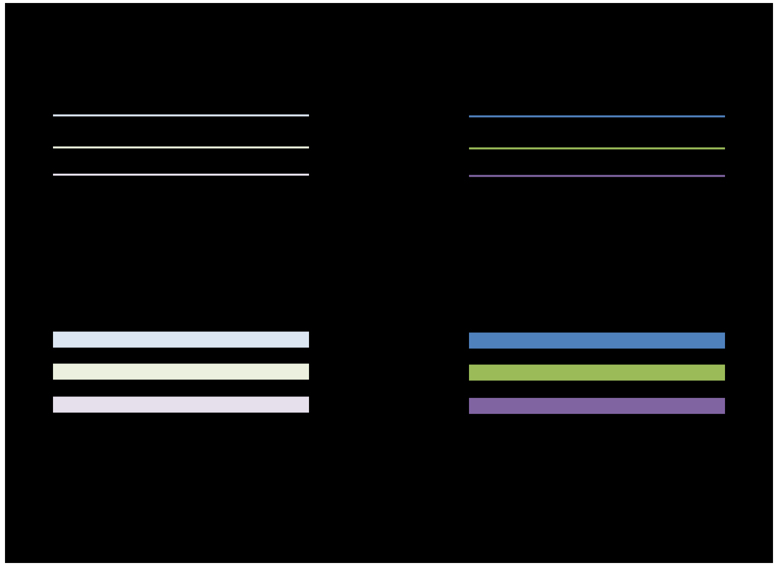


Initial Image
Simulated Protanope

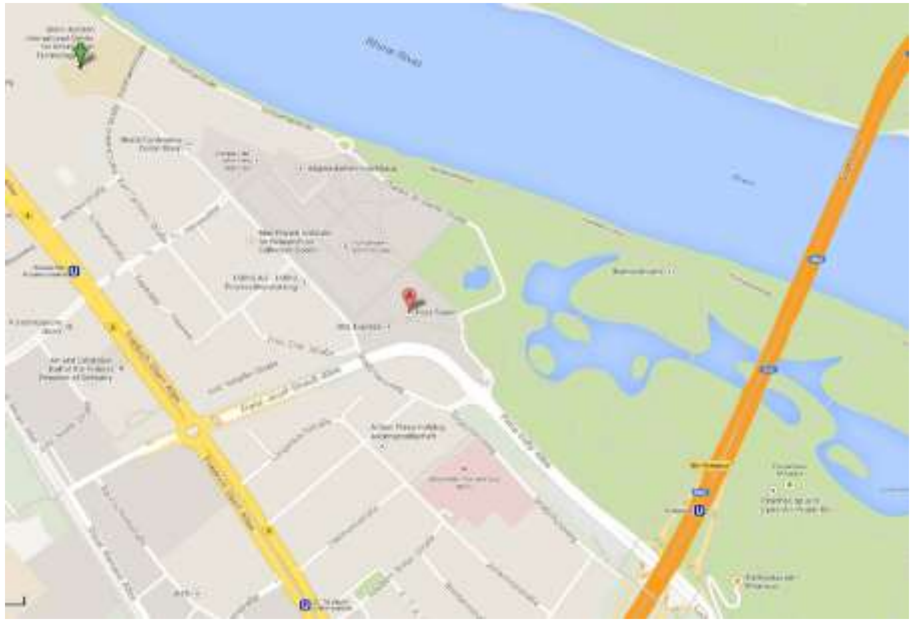
Corrected Image
Simulated Protanope

Saturation

- Colors on **small shapes** or **fine details** need to be strong in order to be distinguishable
- Soft (pastel) colors should be used for **backgrounds** or for **segmenting larger areas**
- Do not use saturation levels to encode >3-4 values



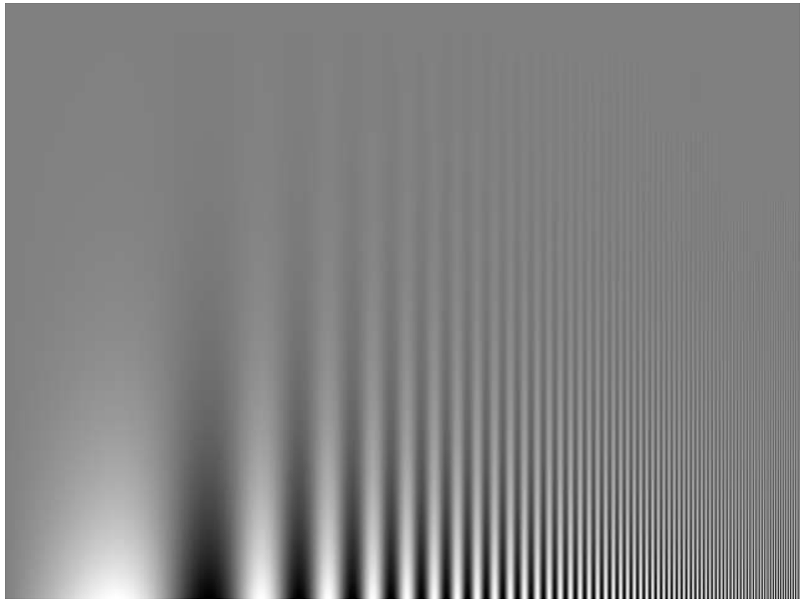
What will a dark background change?



Fine Detail

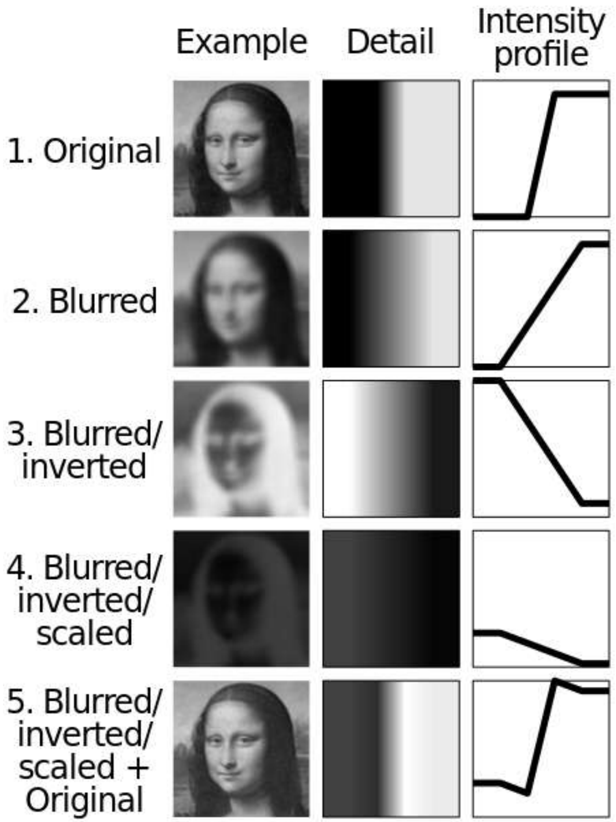
- For visualizations that include fine detail (such as text), use enough **luminance contrast**
  - At least 3:1, better 10:1
  - If that's not an option, add a strong border (or a Cornsweet edge)

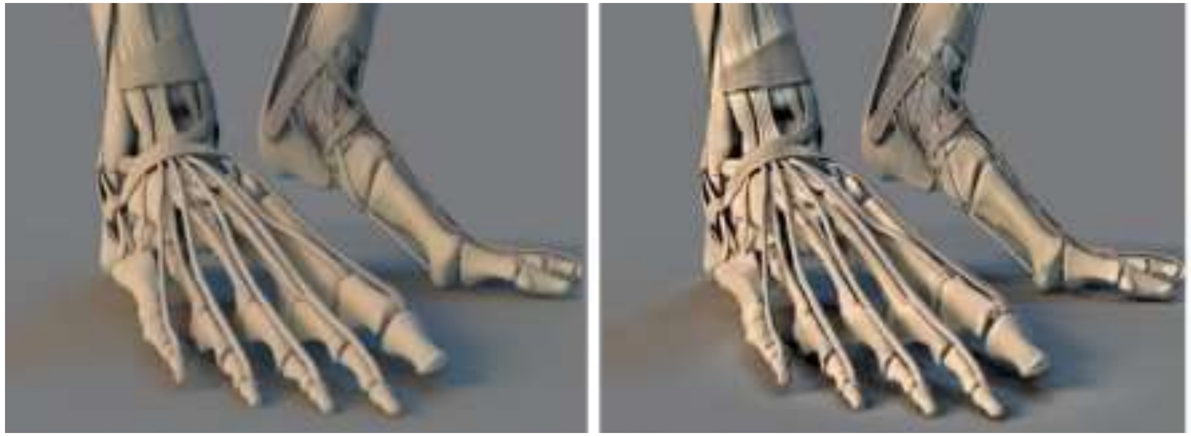
Small details such as those found in text are especially hard to see when shown on an isoluminant background.



Consider Using Cornsweet Effect

- **Cornsweet edges** can increase apparent contrast in an image (e.g., unsharp masking)

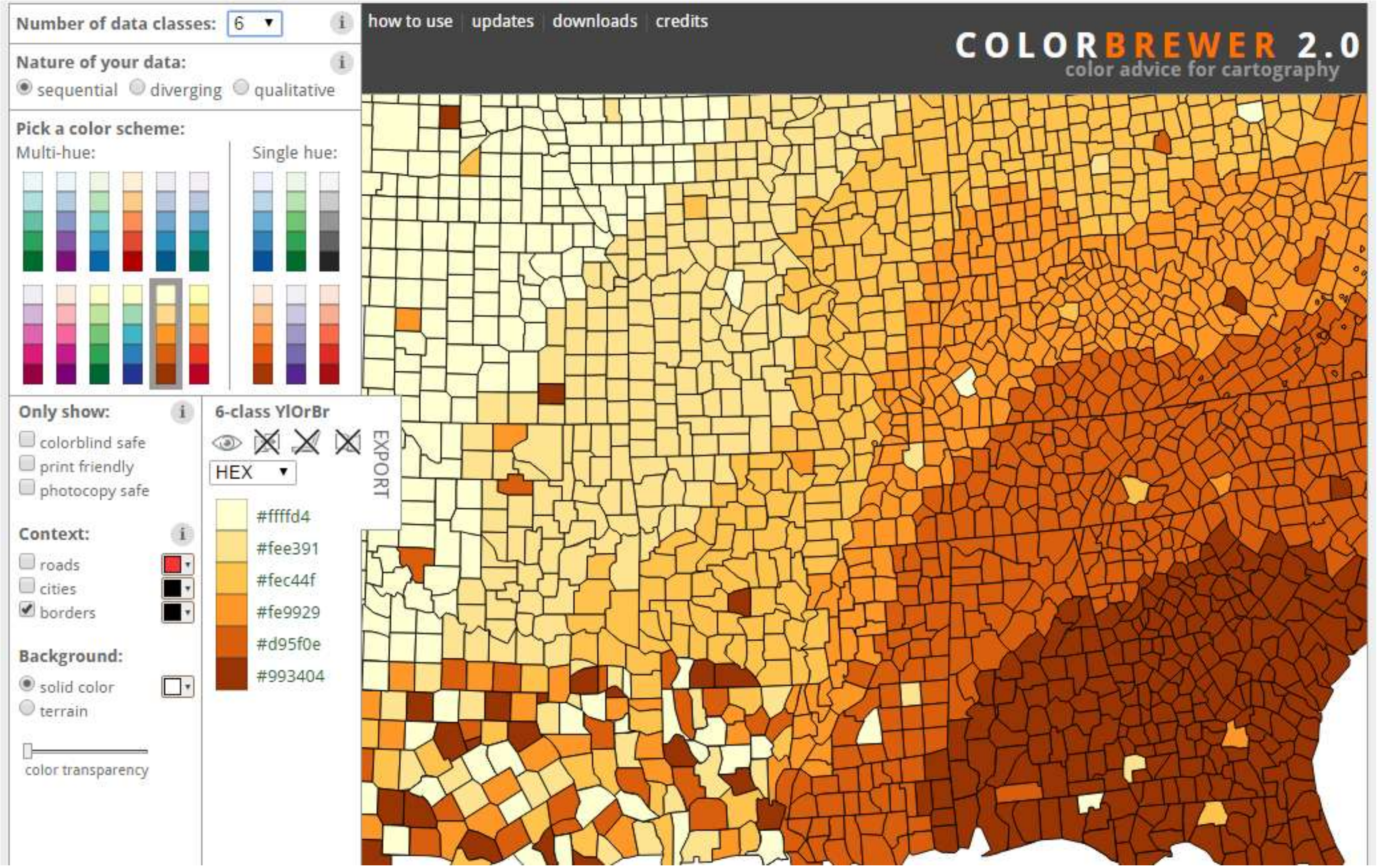




[Ritschel et al., SIGGRAPH 2008]

Final Note: A Useful Resource

- Visit colorbrewer2.org



Summary: Color in Visualization

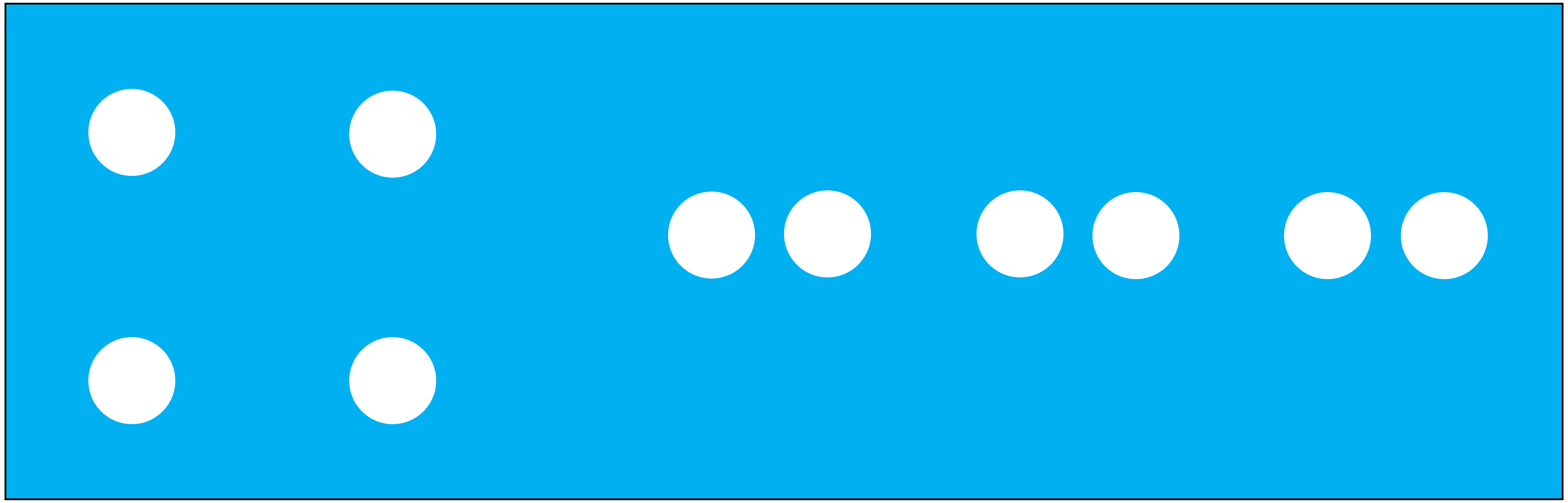
Practical take-home messages:

- Good **color maps** should account for
  - the **scale of measurement** of the data
    - nominal, ordinal, interval, ratio
  - avoiding read-off errors caused by **contrast effects**
  - avoiding **artifactual gradients**
  - **color impairment**
- Use professionally designed **perceptually uniform color maps**
- Use **contrast in luminance** where it is called for

Section 4.4:
Gestalt, Perspective, and Depth

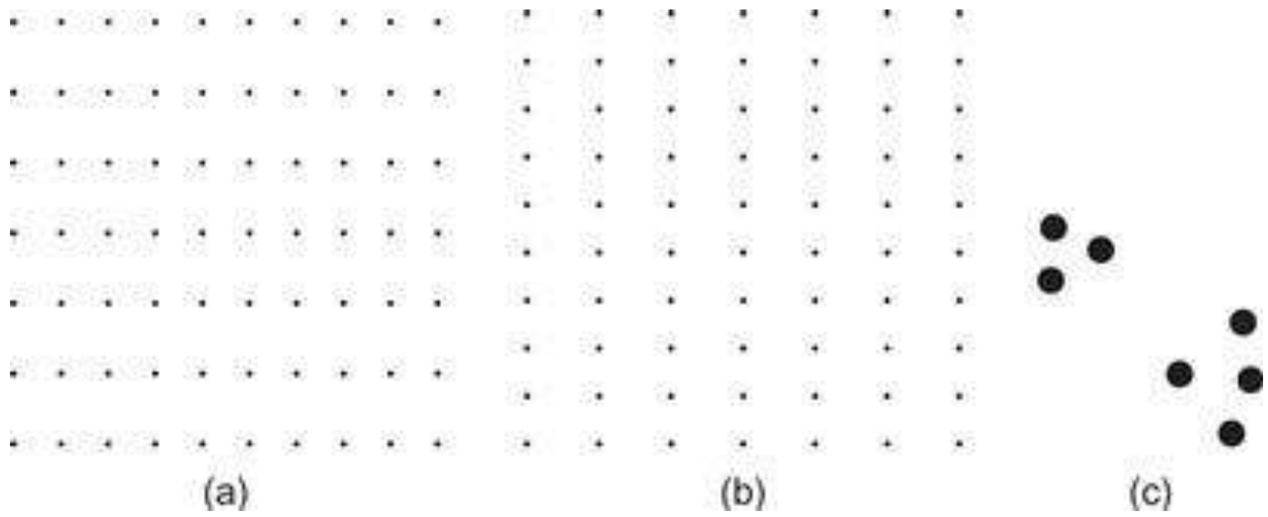
Gestalt Principles

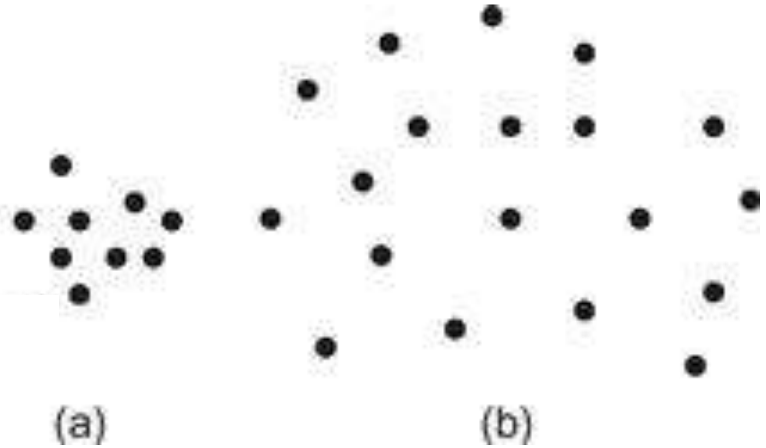
- **Gestalt:** form / configuration / pattern
  - School of perceptual psychology, founded in 1912
- *Idea:* perception of forms or patterns that are not “objective attributes” of the stimuli that create them.
  - Why do patterns emerge?
  - Under what circumstances?



Gestalt Law of Proximity

- **Law of Proximity**
  - Elements that are close together will be perceived as a group
  - Spatial concentration / similarity in density also plays an important role

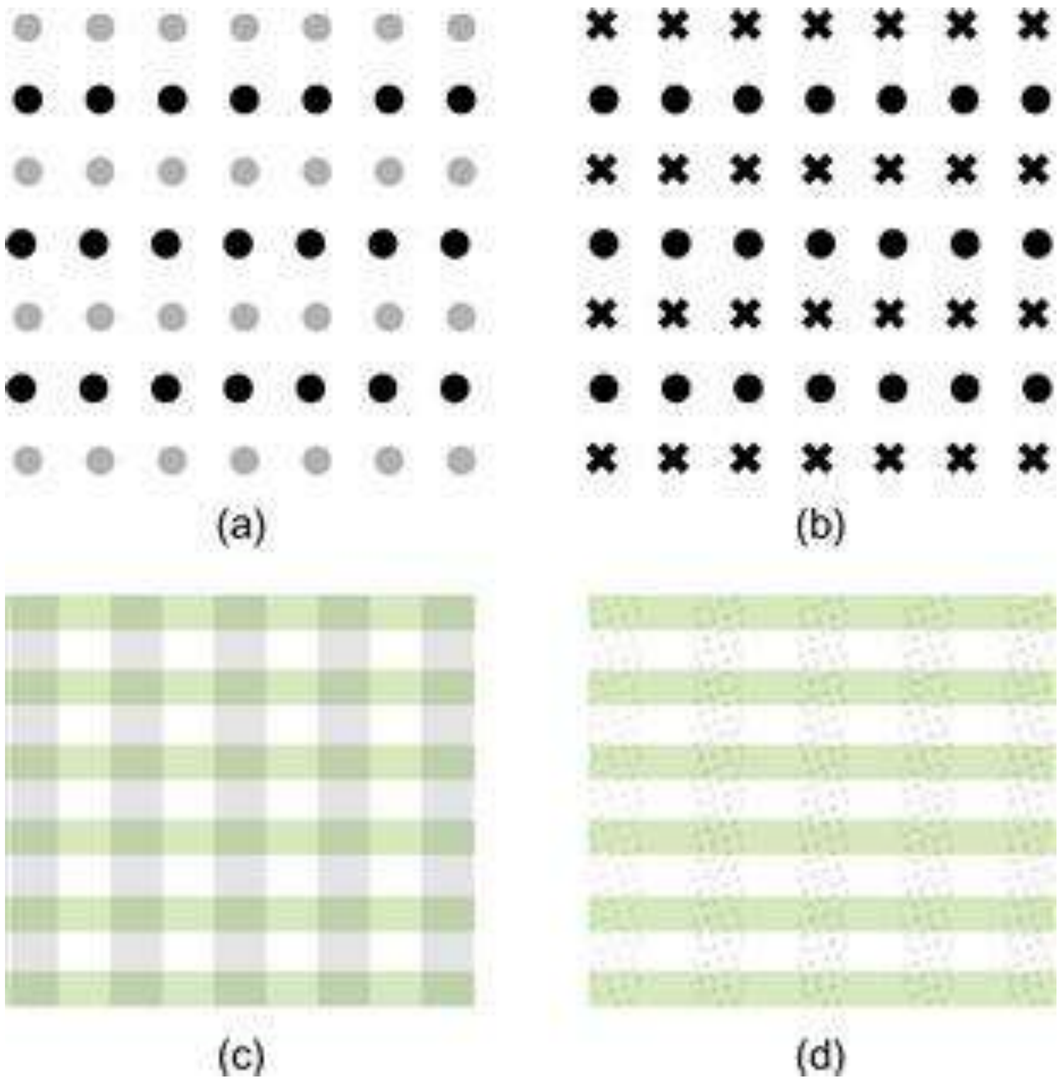




Illustrations from Colin Ware: Information Visualization, 3rd edition

Gestalt Law of Similarity

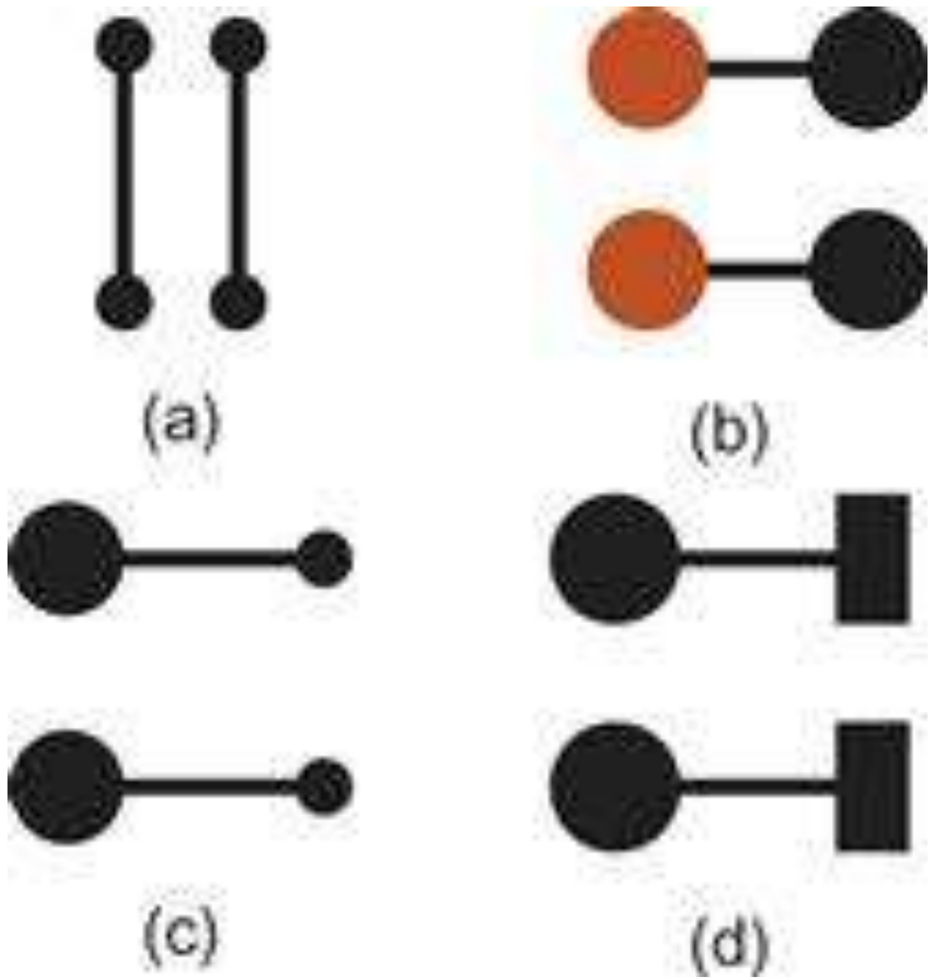
- **Law of Similarity**
  - Elements that are similar in shape, color, etc. are perceived as belonging together
  - The more modern concept of channels (Chapter 5) will help us reason about the multi-faceted nature of "similarity"



Illustrations from Colin Ware: Information Visualization, 3rd edition

Principle of Connectedness

- **Law of Connectedness**
  - Not among the original Gestalt laws, but very powerful
  - Connecting objects can override other cues such as proximity, shape, color etc.
  - Fundamental to node-link diagrams (e.g., graphs)



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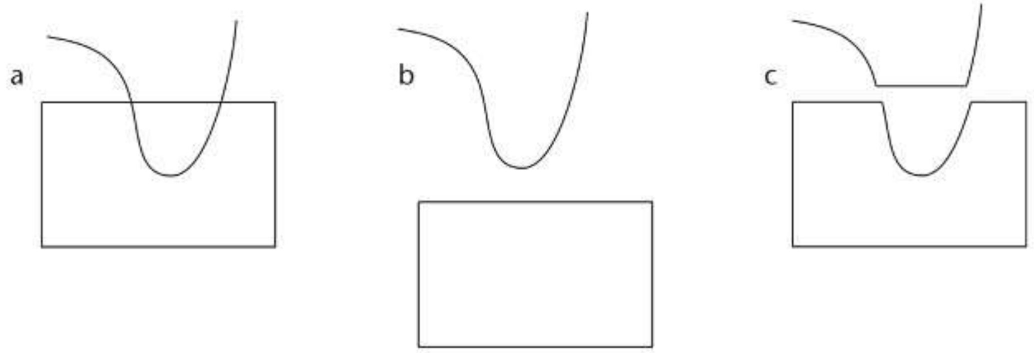
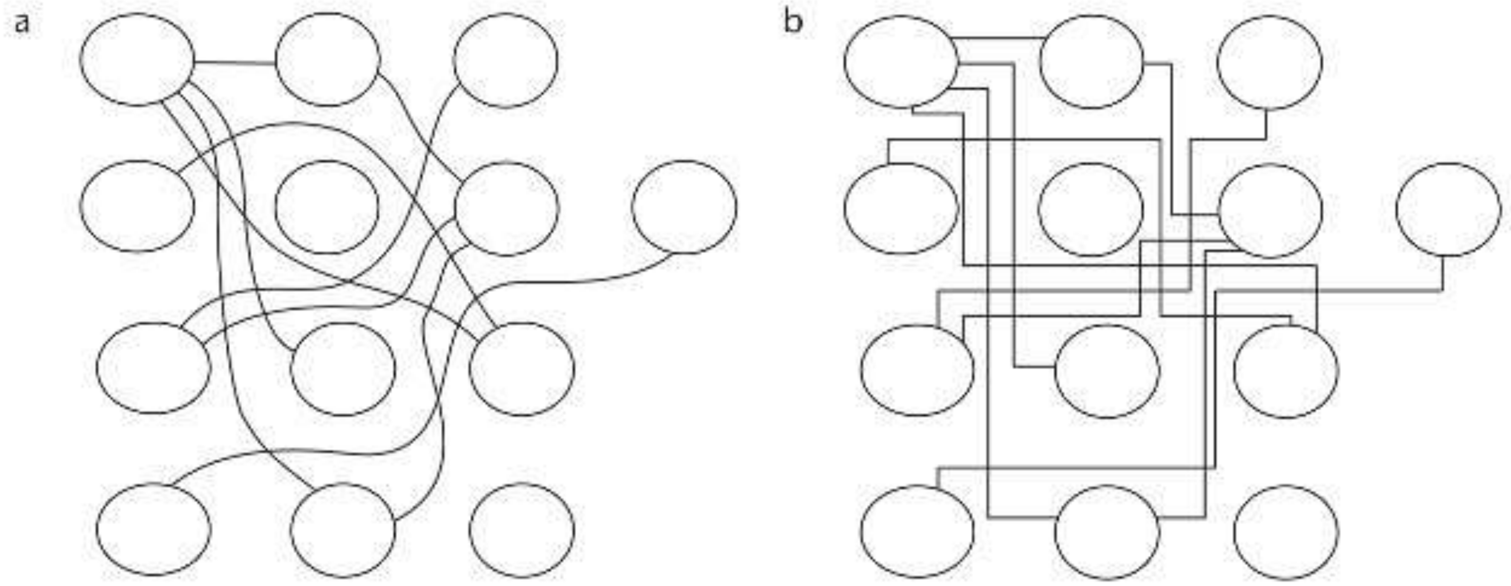
Gestalt Law of Continuity

- **Law of Continuity**
  - Explanations in terms of continuous objects are favored
  - Smooth connections in diagrams are easier to visually trace





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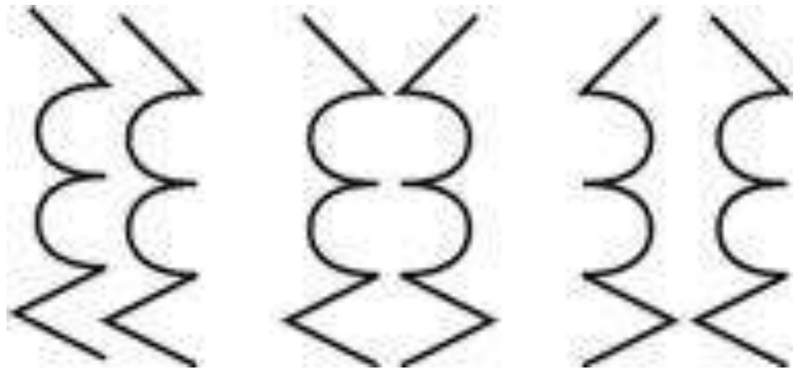
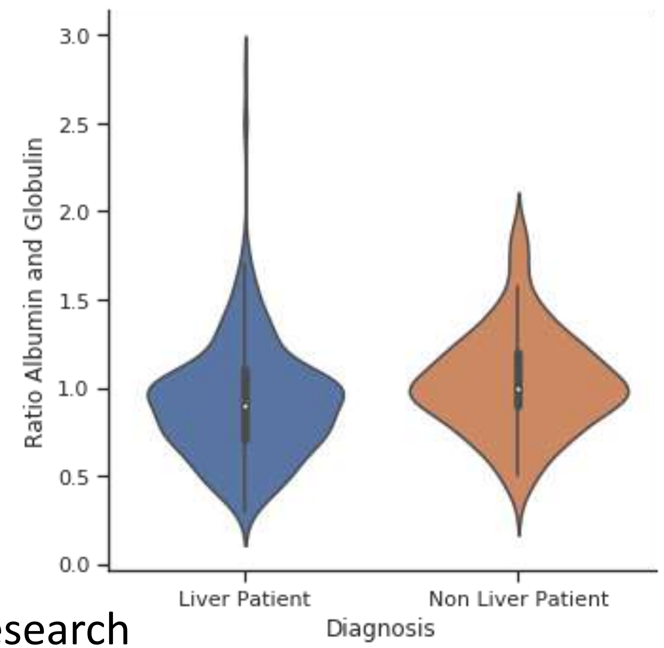
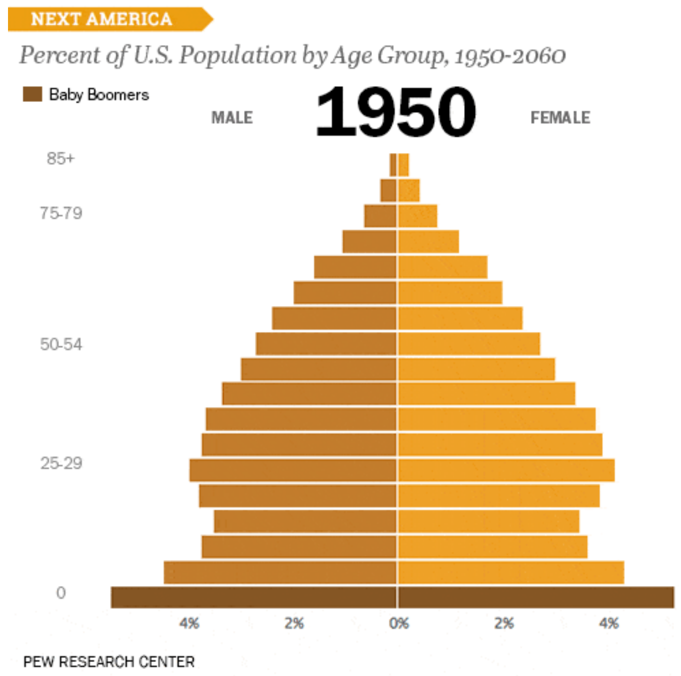
Gestalt Law of Symmetry

- **Law of Symmetry**
  - Compared to parallel layouts, symmetric arrangements are more readily perceived as forming a whole
  - Horizontal and vertical axes are perceptually preferred
  - Examples in visualization: Violin plots, demographic plots



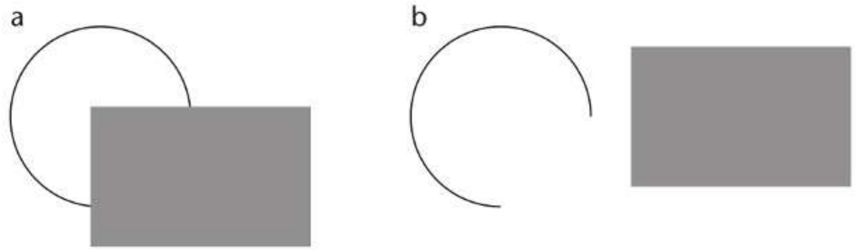


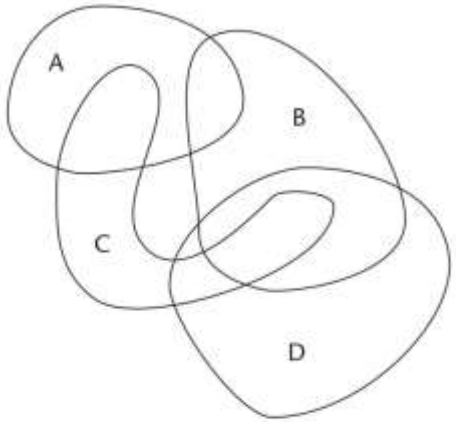


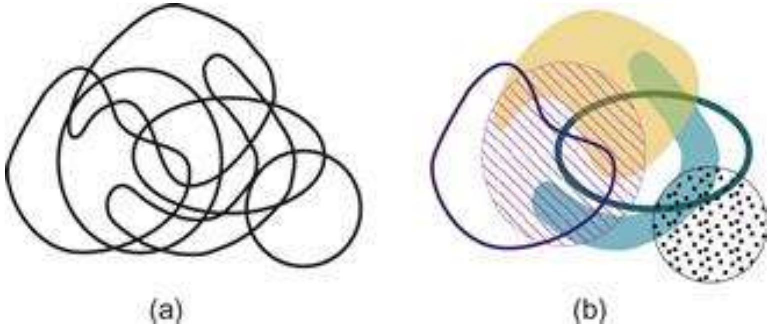
Illustrations from Colin Ware / own work / Pew research

Gestalt Law of Closure

- **Law of Closure and Common Region**
  – Explanations in terms of closed objects are favored
  – Closed contours define common regions, another strong cue for belonging together
    - Venn Diagrams
    - Windows-based user interfaces
  – Consider color / texture as additional cues for complex shapes and intersections



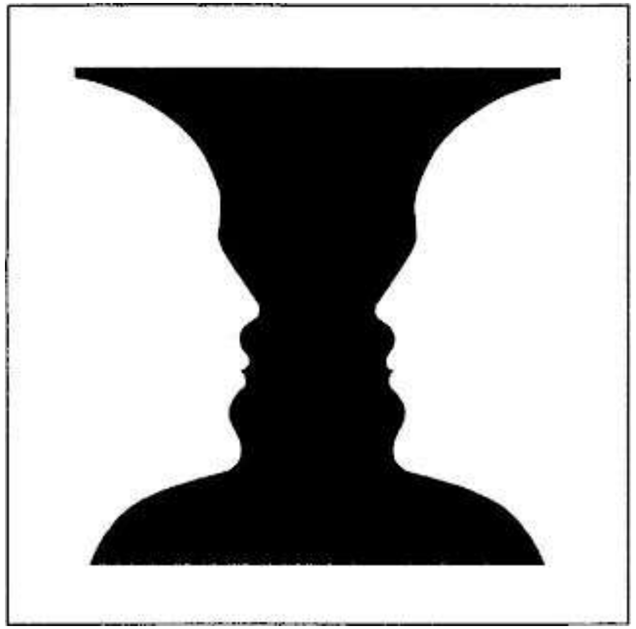


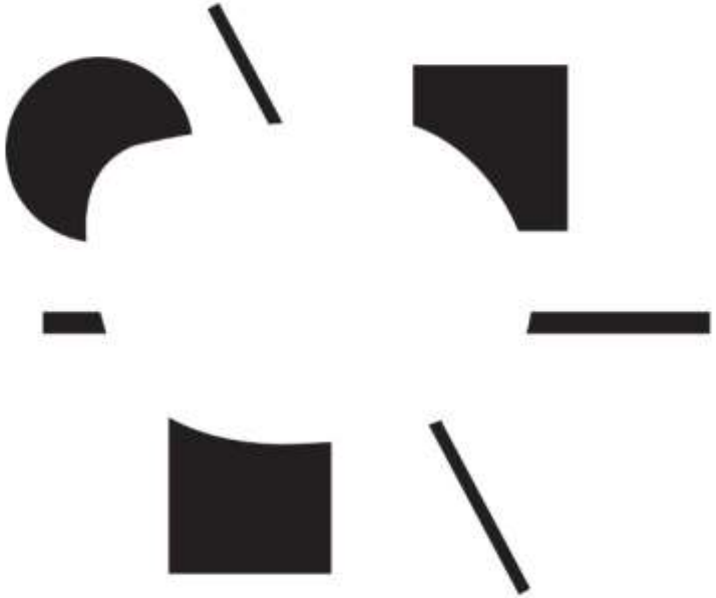


Illustrations from Colin Ware: Information Visualization, 3rd edition

Figure-Ground Effect

- **Figure/Ground Perception**
  - As part of object identification, we attempt to explain stimuli in terms of foreground figure + background
  - Can lead to perception of subjective contours

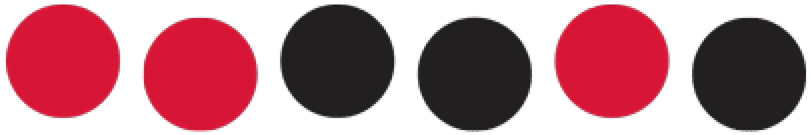


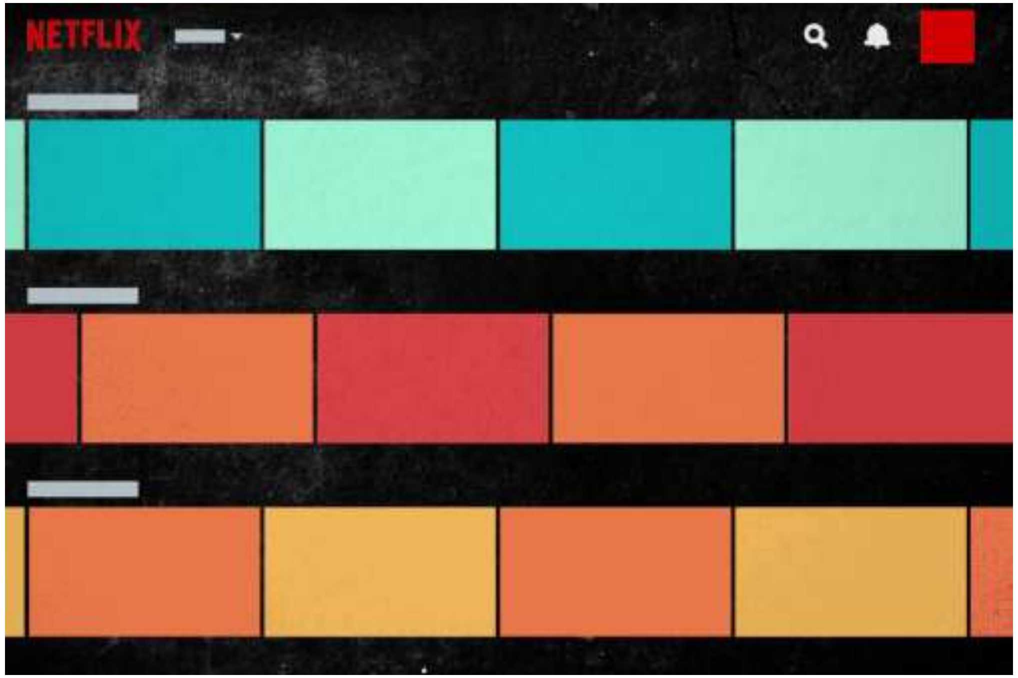


Illustrations from Colin Ware: Information Visualization, 3rd edition

Gestalt Law of Common Fate

- **Law of Common Fate**
  - Elements that move together are perceived as belonging together
  - *Example:* Scrolling / panning in user interfaces affects all items belonging to the same group

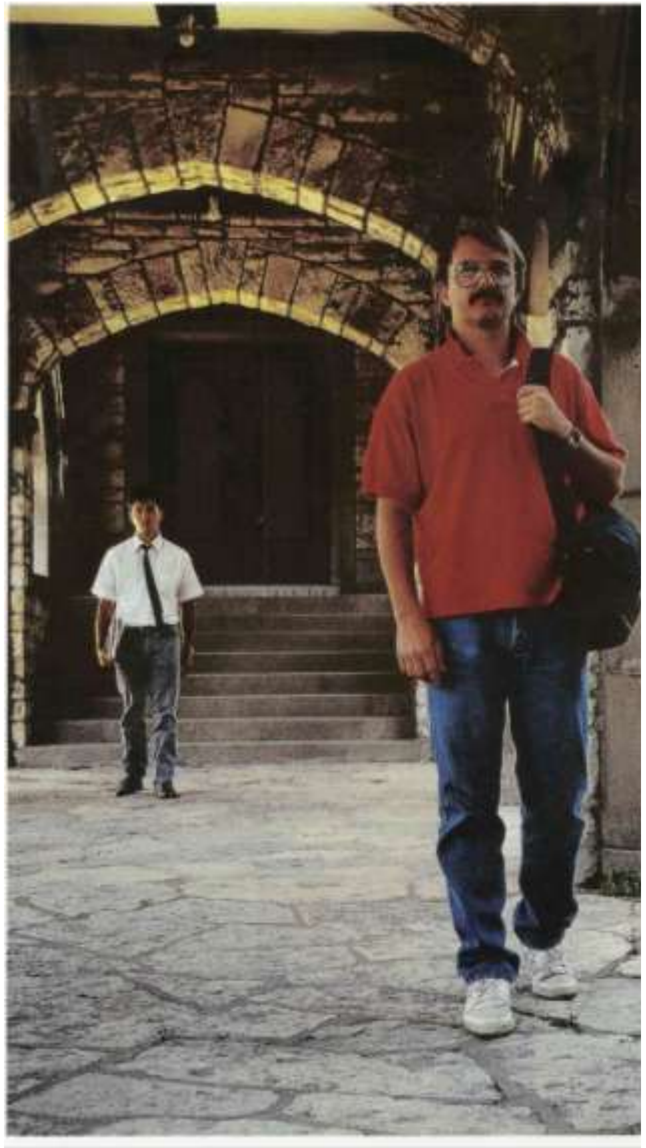




Animations from Jon Hensley / Gizmodo / ringer

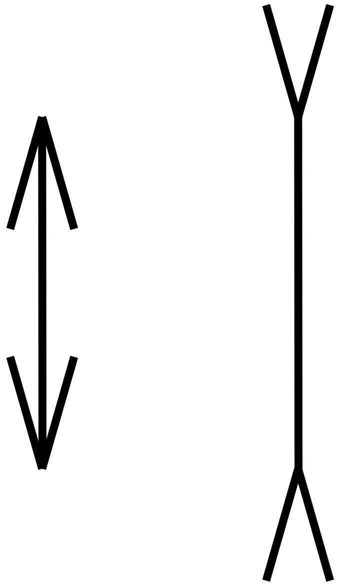
Size Constancy

- Similar to lightness and color constancy, our perception of perspective affords **size constancy**

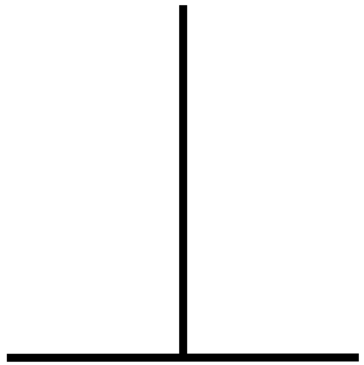


Illusions of Linear Extent

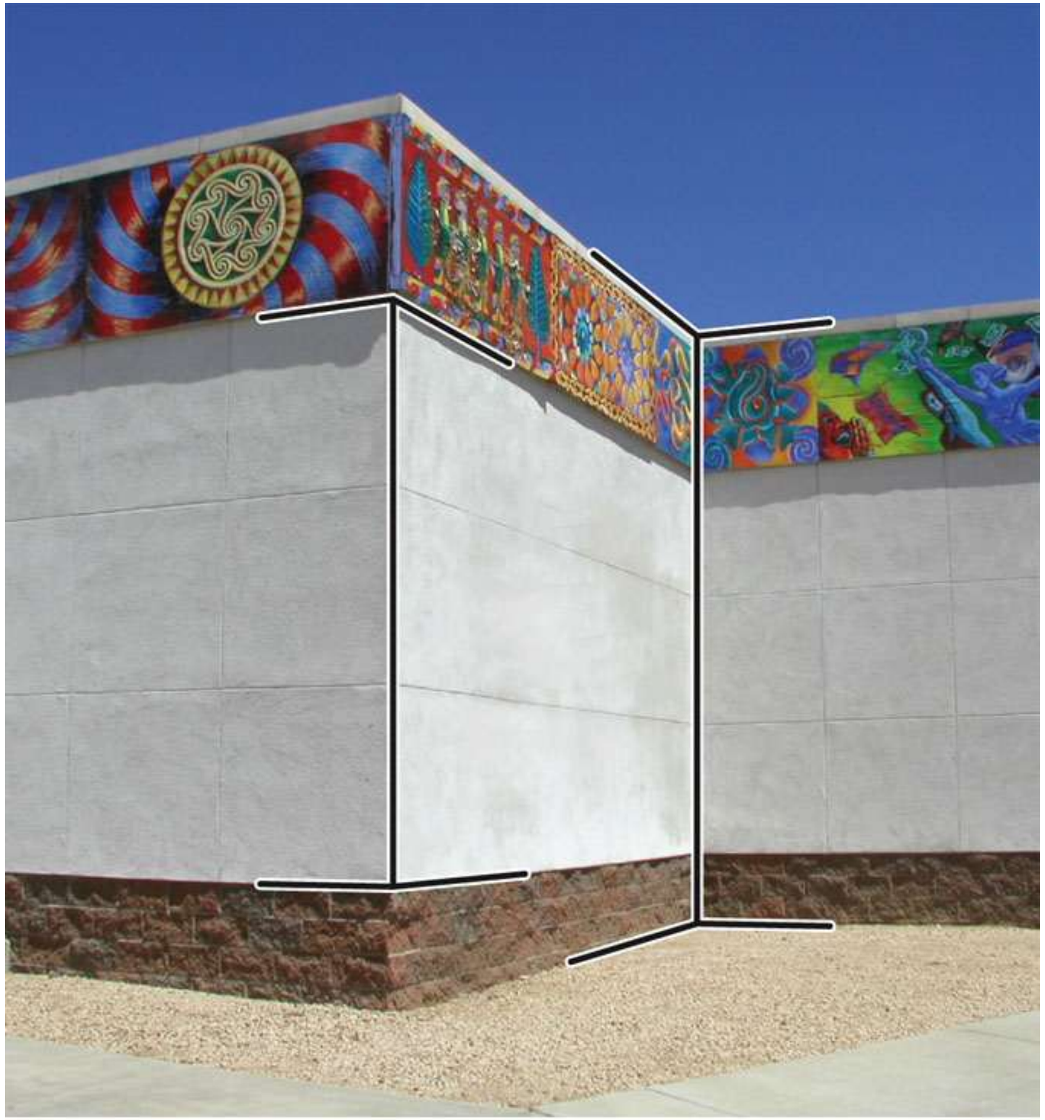
- Müller-Lyer (off by 25-30%)



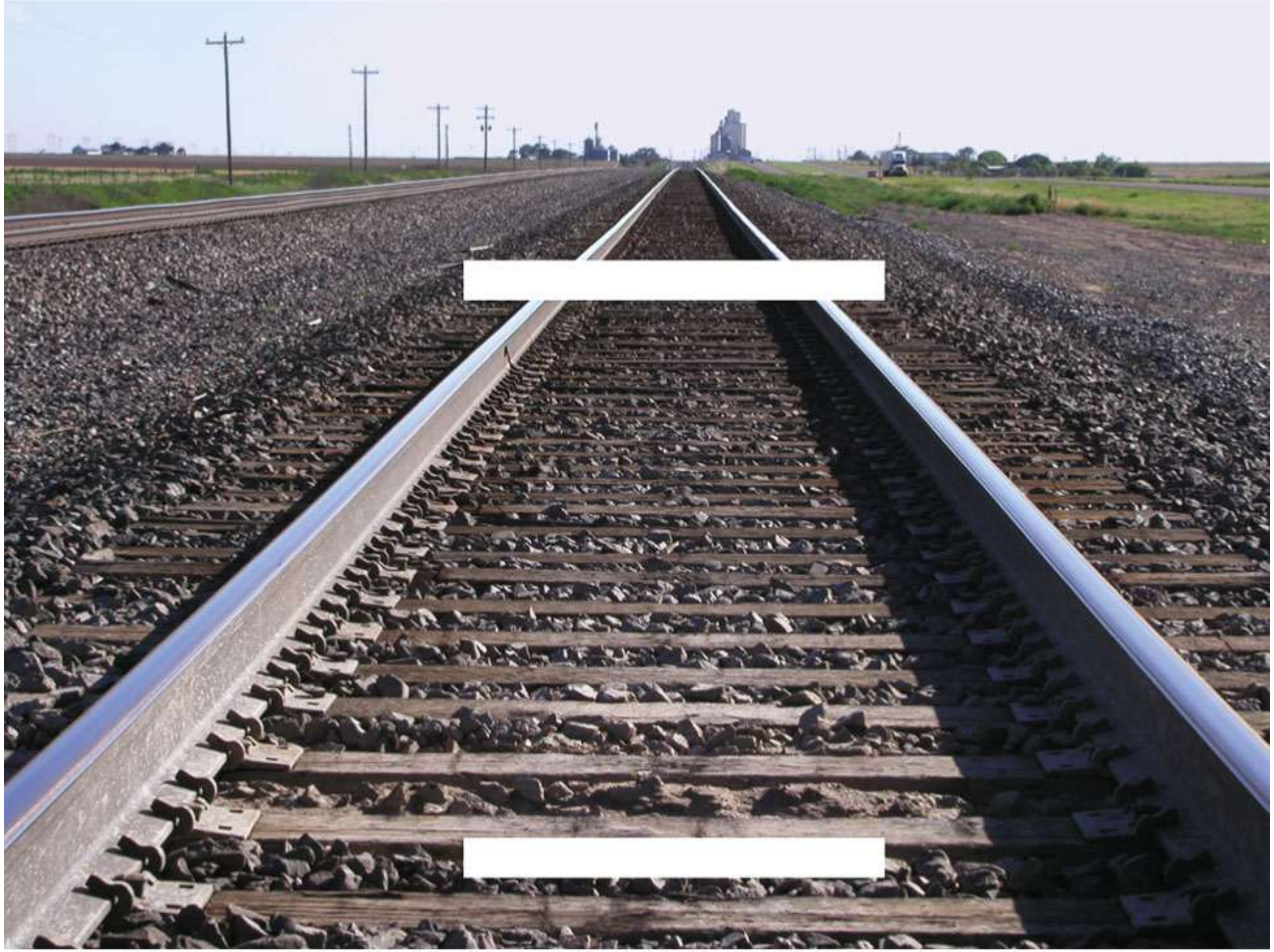
- Horizontal-Vertical



Müller-Lyer Explained

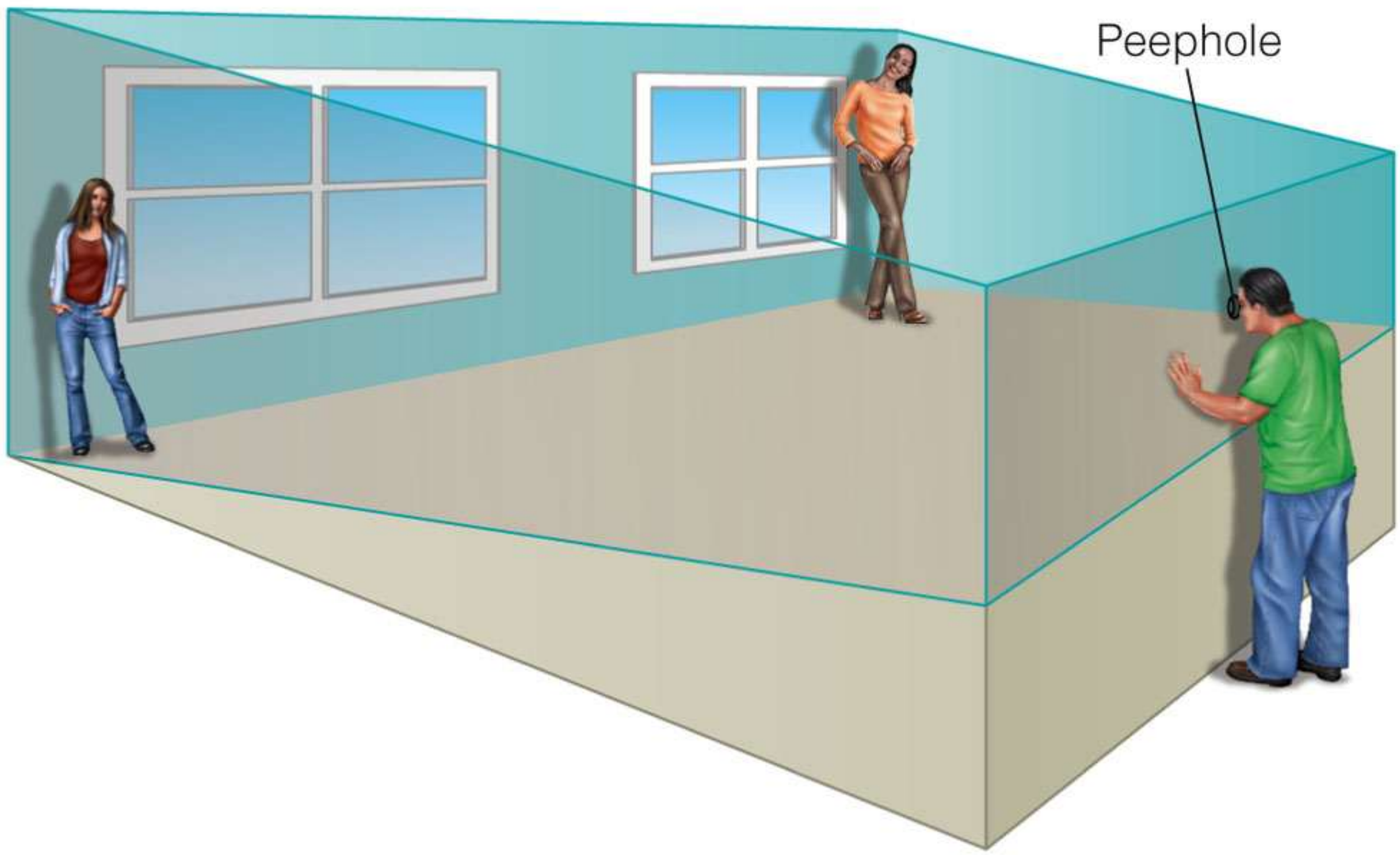


Railroad Track Illusion



Ames Room

Ames Room Explained



Binocular Depth Perception

- Binocular (retinal) disparity
  - Difference in images between the two eyes

(a) Left eye image

(b) Right eye image

Monocular Depth Cues

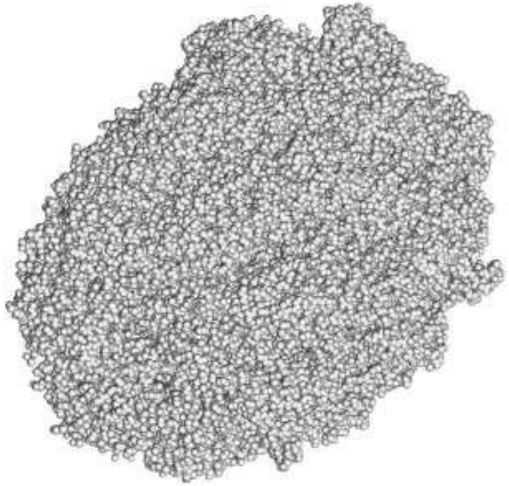
- **Occlusion:** Closer objects cover more distant ones
- **Shadows:** Indicate where objects are located
- **Perspective convergence** of parallel lines
- **Relative height:** More distant objects appear higher in the visual field
- **Relative area:** When objects are the same size, the one that takes up more area on the retina is closer

More Monocular Depth Cues

- **Atmospheric perspective:** Distant objects appear less saturated, less sharp, and may have a blue tint
- **Texture gradient:** Similarly sized elements are more closely packed as distance increases
- **Motion parallax:** Distant objects move more slowly than close ones

Monocular Depth Cues For Visualization

Standard lighting



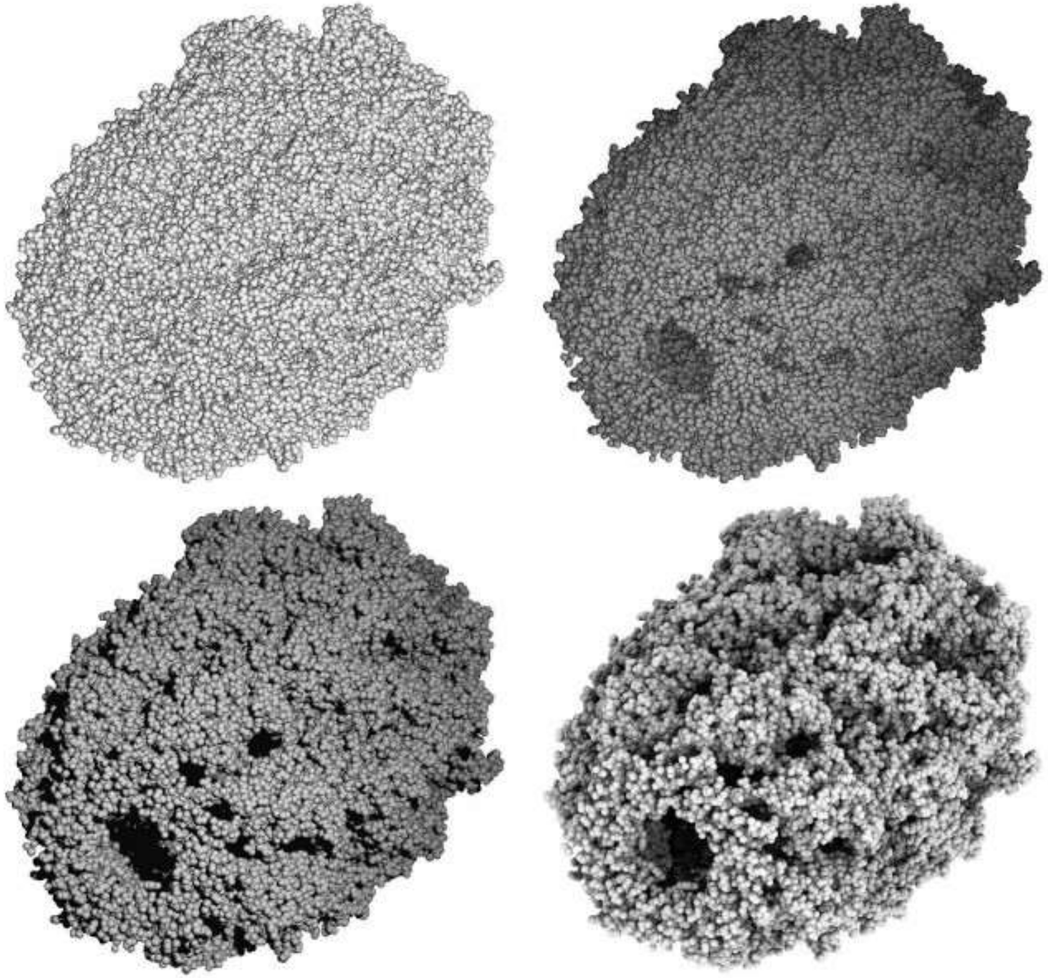
(b) Plain line rendering.

[Tarini et al. 2006]

Line Halos, [Everts et al. 2009]

Monocular Depth Cues For Visualization

Standard lighting Depth Cueing



Depth Cueing + Shadows Ambient Occlusion

[Tarini et al. 2006]

(b) Plain line rendering.

(d) Lines with illustrative halos.

Line Halos, [Everts et al. 2009]

Summary: Gestalt, Perspective, Depth

- We should be aware of **Gestalt laws**
  - Perceptual grouping and emergent patterns
  - Proximity, similarity, continuity, closure, figure/ground
- **Perspective projections** shape our perception
  - Enables size constancy
  - Can lead to misjudgment of lengths in 2D
- **Depth perception** relies not just on binocular, but also on monocular cues
  - Include occlusions, shadows, desaturation
  - Such cues can greatly improve 3D visualizations

Section 4.5:
Change Blindness

Change Blindness

- The capacity of our **visual working memory** is very limited
  - Experience of a rich and detailed world around us partly relies on information from long-term memory, and the fact that “the world is its own memory”
- Consequently, we often do not notice even substantial changes, unless we explicitly pay **attention** to what changes
  - Motion strongly draws our attention
  - Masking out motion makes us **blind to changes** in images
- Sequences courtesy of Ron Rensink


PERIGOU
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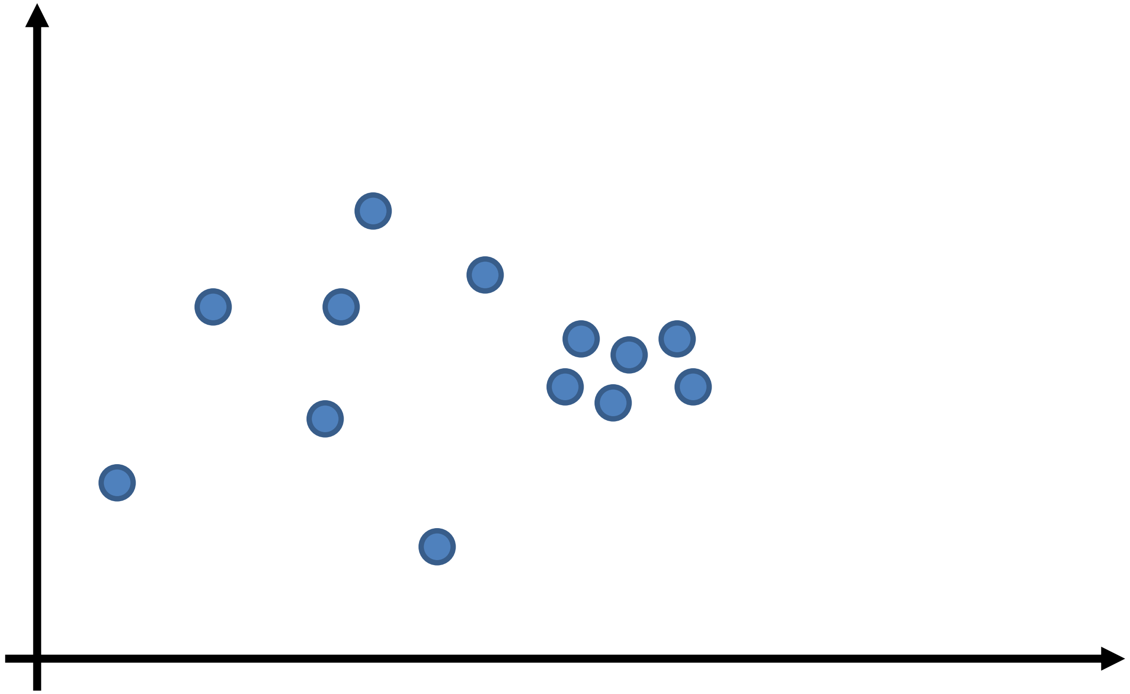
Parc
Pont Neuf
TABLEAUX MODERNES
Piano Bar
715 GCY 75

Change Blindness in Visualization

- Can you spot the difference in the data?

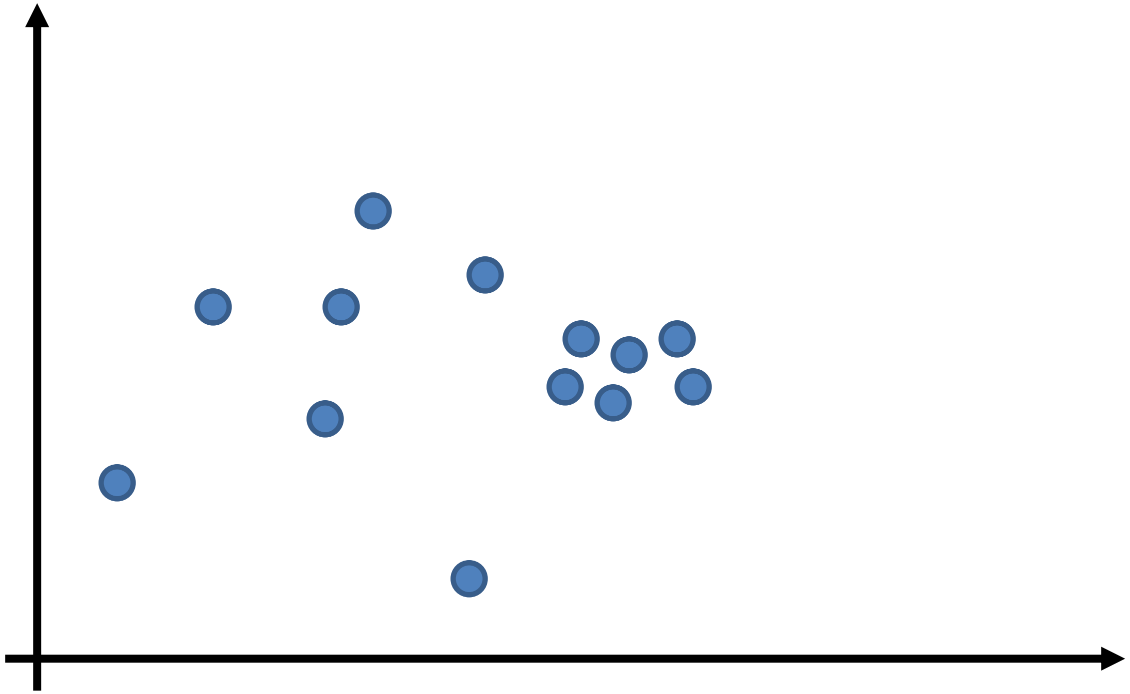


Change Blindness in Visualization

- Can you spot the difference in the data?

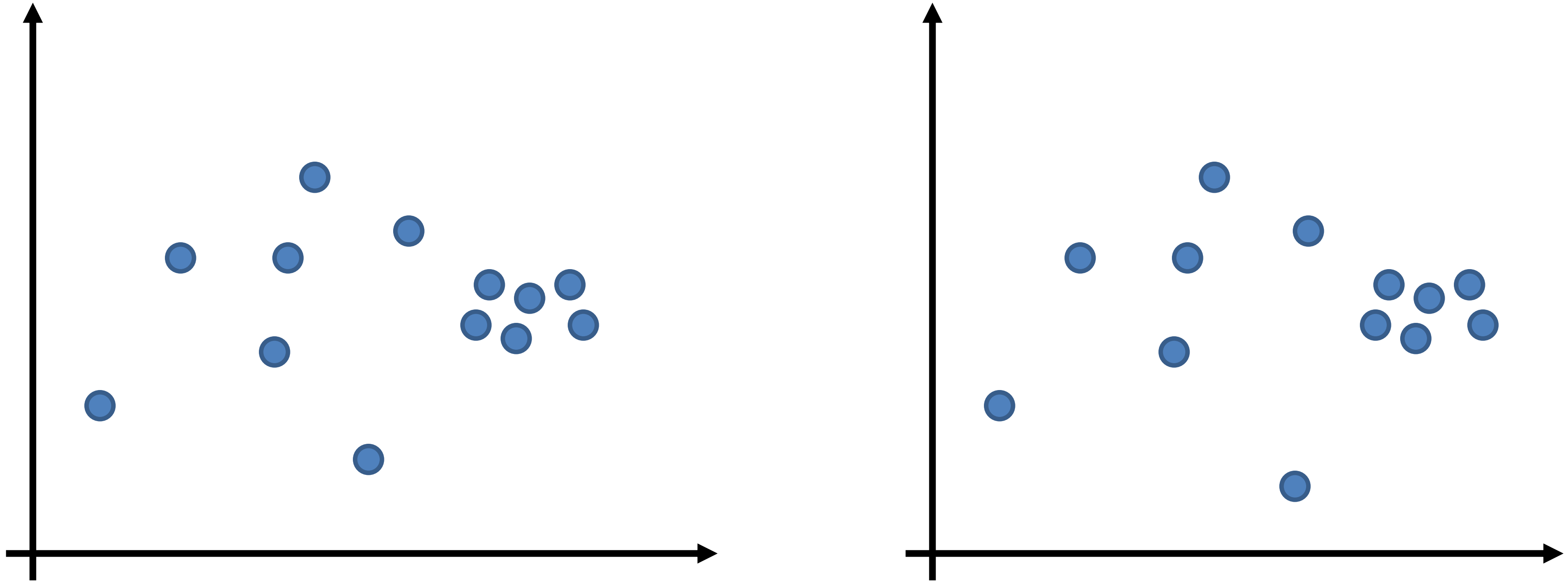
Change Blindness in Visualization

- Can you spot the difference in the data?



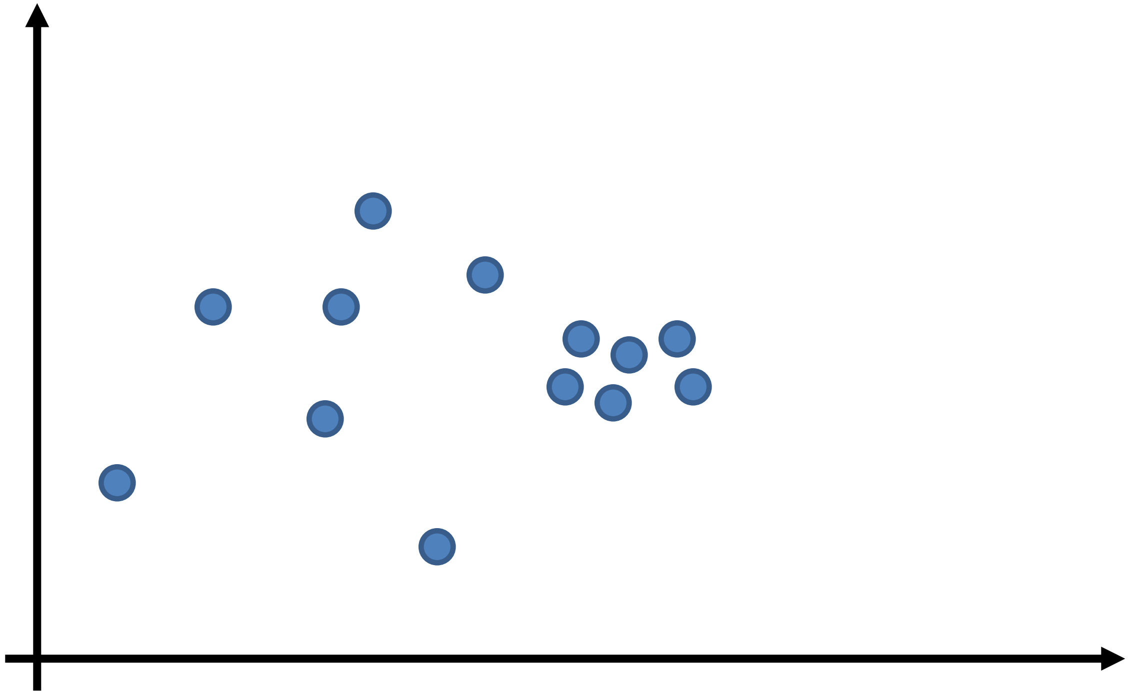
Change Blindness in Visualization

- Can you spot the difference in the data?



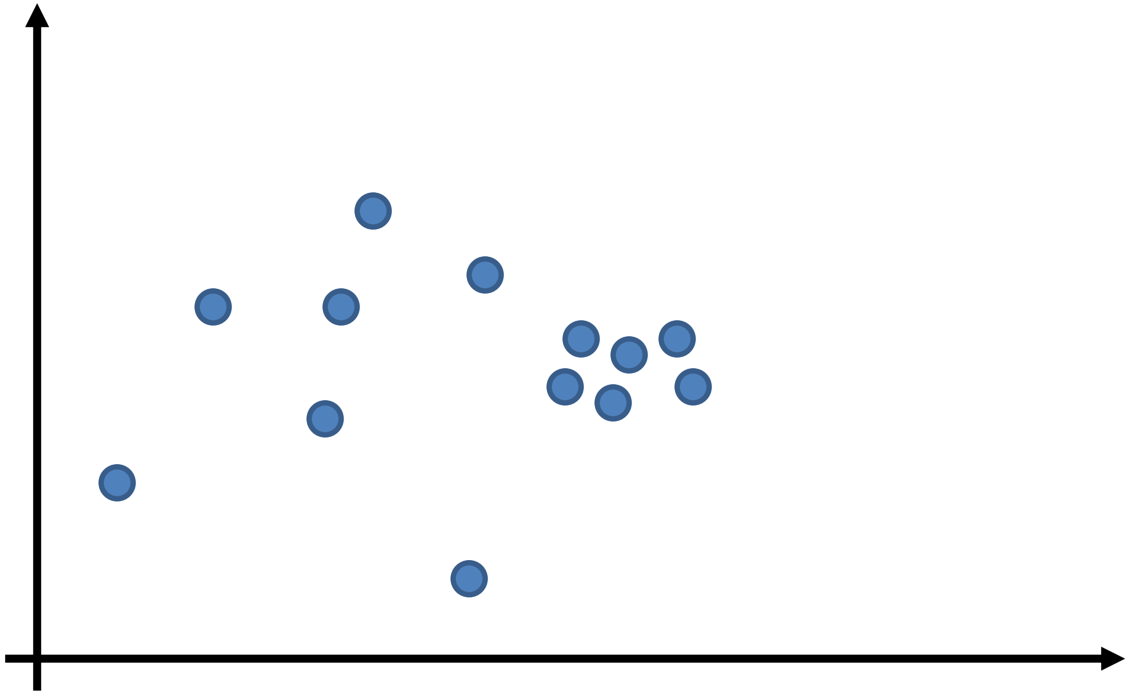
Change Blindness in Visualization

- Can you spot the difference in the data?



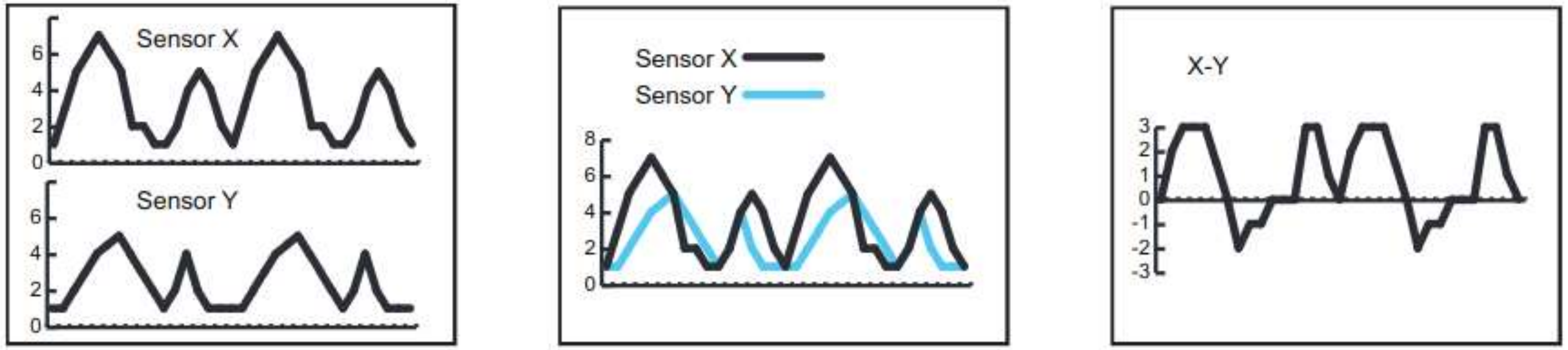
Change Blindness in Visualization

- Can you spot the difference in the data?



Summary: Change Blindness

- Limitations of human visual working memory can cause **change blindness**
- *Consequence for visualization design:* We should not rely on viewers' memory to notice differences
  – When switching between visualizations, changes are perceived as motion, which draws the viewers' attention to them
  – Other options are *juxtaposition*, *superposition*, or *explicit encoding*



(a) Juxtaposition (b) Superposition (c) Explicit Encoding: Difference

Figures from [Gleicher 2018]

Acknowledgement

- Sources of this lecture include slides by:
  - Robert Winningham, Western Oregon University
  - Christopher Healey, North Carolina State University
  - Liz Marai, University of Illinois at Chicago
  - Gordon Kindlmann, University of Chicago
  - Kristen Grauman, UT-Austin
- …as well as much material from Colin Ware’s *Information Visualization* book

Further Reading

- Colin Ware: *Information Visualization: Perception for Design.* Morgan Kaufmann, 2013
- Gunter Wyszecki, W. S. Stiles: *Color Science: Concepts and Methods, Quantitative Data and Formulae*. Second edition, Wiley, 2000
- Matthew Ward, Georges Grinstein, Daniel Keim: *Interactive Data Visualization: Foundations, Techniques, and Applications.* A. K. Peters, 2010

Exam-Like Questions

1. State the difference between luminance, brightness, and lightness.
2. Suggest a suitable color space for letting users select a color. Motivate your choice.
3. Name a scale of measurement, and describe how it would affect your choice of color map.
4. Name two Gestalt laws of perception. For each of them, give an example of how it might be used in visualization.